



WHO INFLUENZA CENTRE LONDON

**Report prepared for the WHO annual
consultation on the composition of influenza
vaccine for the Southern Hemisphere 2013**

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Southern Hemisphere Vaccine Recommendation Meeting for 2013 influenza season.
Summary of NIMR Results.
Final Report for WHO meeting: 17-19 September 2012.

Influenza Activity in the European region, October 2011 – August 2012

For the 2011-2012 influenza season, activity started late compared to the 2010-2011 season, but increased slowly over consecutive weeks after week 52/2011 to peak sharply in weeks 8-9/2012; this contrasts with the broad peak (weeks 52/10-6/11) seen in the 2010-2011 season (**Figure 1**). Approximately 46000 influenza detections have been reported across the whole WHO Euro region, compared to over 89000 in the 2010-11 season, but, again, type A detections (90.5%) predominated over type B (9.5%: **Figure 2**). The pattern of Influenza-like illness (ILI) in Europe has been inconsistent with some countries showing low levels of ILI (e.g. England, Romania, Switzerland, Russian Federation, Germany) and others reporting activity at levels similar to (e.g. Italy, Spain) or even greater than (e.g. Sweden) those of last year with most countries reporting later starts, and subsequent peaks, in influenza activity than usual (**Figure 3**).

Within type A, the H3 subtype (96.0%) predominated over H1pdm09 viruses (4.0%) by a 24-fold margin across the WHO Euro region but there was disparity between Western (35-fold) and Eastern (14-fold) parts of the region (**Figures 1 & 2**): this represents a total reversal of the situation in 2010-11 despite the late start and generally low intensity of the 2011-2012 season (**Figure 2**).

Influenza B detections have been low in number, accounting for 9.5% of total influenza detections compared to 31.5% in the previous season. Furthermore, type B detections as a percentage of total detections in 2011-2012 have been higher in the Eastern countries (17.4%) compared with those in the West (7.2%) with differing proportions of viruses of the Victoria and Yamagata lineages: approximately 9 to 1 in Eastern countries and 1 to 1 in the West (**Figures 1 & 2**).

Influenza positive specimens received at NIMR

Specimens, viruses or clinical samples, with collection dates after 01 February 2012 have been received at the WHO CC in London from 30 countries in Europe, Africa, South America and the Far East (**Table 1**) with the majority (23/30) coming from countries falling within the WHO-Euro Region (**Table 2**).

The majority were A(H3N2) viruses (67%) with smaller numbers of influenza B viruses (25%) and H1N1pdm09 viruses (8%). While a small number of type A and type B viruses have yet to be subtyped, B/Victoria lineage viruses (13.5%) and B/Yamagata lineage viruses (9.9%) were received at similar levels (**Table 2**). The proportions received are representative of the types/subtypes/lineages in circulation in Europe. **Table 3** shows the frequency of successful propagation of samples received with collection dates since 01 February 2012. **Table 4** summarises the number of viruses of the A(H1N1), A(H3N2) subtypes, and the two B-lineages that showed reductions in HI titres with post-infection ferret antisera compared with the titres against the homologous vaccine/reference viruses.

Antigenic and genetic characteristics of influenza viruses collected between 01 February 2012 and 31 August 2012

Influenza A(H1N1)pdm09 virus analysis.

Tables 5 & 6 show results of HI analyses of A(H1N1)pdm09 viruses received, with collection dates since 01 February 2012, and a summary is shown in **Table 7**.

Influenza A(H1N1)pdm09 viruses were the minority in most countries with the exception of Georgia and Norway; the majority of viruses collected in January 2012 in Kazakhstan were A(H1N1)pdm09 viruses. Of the viruses isolated, 28 were propagated and analysed by HI. All of the H1N1 viruses were antigenically similar to the vaccine virus but low HI titres, compared with the homologous titre, with ferret antiserum raised against the vaccine virus were seen for four viruses, three of which carried the amino acid substitution G155E, one in combination with A48D and G225A. The fourth virus that showed reduced reactivity carried a polymorphism at residue 142 and a substitution at 211 (K142K/E, K211R). Such amino acid substitutions have been associated with selection during culture. Virus propagation from clinical samples sent by Kazakhstan, collected in January 2012, and confirmed by rtRT-PCR at NIMR, was not successful and only 1 of 10 clinical samples from Georgia yielded a virus isolate.

Twenty-five of the 28 viruses analysed in HI assays have been subjected to HA and NA gene sequence analysis (**Figures 4 & 7** respectively). The viruses for which HI results are presented (**Tables 5 & 6**) are boxed, the four that were observed to be low reactors are indicated (@) as are those viruses proposed as serology antigens (#). Node defining amino acid substitutions for HA1 (black type) and HA2 (purple type) are shown (**Figure 4**). The HA genes of H1N1 viruses cluster into eight genetic groups, previously described, with groups 2 to 8 defined by the following amino acid substitutions in **HA1 & HA2**:

- (2) **N31D, S162N** (resulting in the gain of a glycosylation site), **A186T, E47K & S124N**, e.g. A/Czech Republic/32/2011;
- (3) **A134T & S183P**, e.g. A/Hong Kong/3934/2011;
- (4) **N125D & E47K**, e.g. A/Christchurch/16/2010;
- (5) **D97N, R205K, I216V, V249L & E47K**, e.g. A/Astrakhan/1/2011;
- (6) **D97N, S185T, E47K & S124N**, e.g. A/St Petersburg/27/2011;
- (7) **S143G, S185T, A197T, E47K & S124N**, e.g. A/St Petersburg/100/2011;
- (8) **A186T, V272A, E47K, N146D, T147K & V193A**, e.g. A/Ghana/FS1803/2011.

Specimens received at the London WHO CC, with collection dates since 01 February 2012 fell into genetic groups 6 and 7, and the four viruses showing low reactivity with ferret antiserum raised against the vaccine virus did not form a distinct group (**Figure 4**). Of the 10 clinical samples received from Georgia, eight fell into genetic group 6 and two into genetic group 7, with the isolate being one of the group 7 viruses. The Kazakhstan viruses, collected

in January 2012, fell into group 6. The genetic groups of test viruses are indicated in **Tables 5 & 6**.

Amino acid alignment of the HAs for a selected group of viruses is shown in **Figure 5**; the vaccine virus is shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue, the genetic group is indicated and the HA1/HA2 processing site is highlighted.

Figure 6 illustrates the location of amino acid residues defining genetic groups 6 to 8, representing the most recently circulating viruses analysed (April to July), on the structure of the H1 HA.

Figure 7 shows a phylogenetic tree for NA genes of representative H1N1 viruses and **Figure 8** shows an amino acid alignment for a selected group of these viruses. The HA and NA phylogenetic trees are generally congruent but some viruses are HA/NA reassortants, e.g. A/Ethiopia/198/2012, A/Florida/27/2011, A/Stockholm/36/2011.

Antiviral analysis.

All viruses analysed at the London WHO CC, 58 in total with 24 collected since 01 February 2012, showed normal inhibition (sensitive phenotype) with the neuraminidase inhibitors zanamivir and oseltamivir (**Table 40**). The subset assessed by M-gene sequencing all show resistance to adamantanes due to possession of the S31N substitution in the M2 ion-channel protein. Two cases of oseltamivir-resistant A(H1N1)pdm09 in Dutch travellers returning from separate holidays on the Catalanian (Spanish) coast, in samples collected in mid-August, have been reported. Both patients experienced mild disease and partial sequencing of the viruses in the clinical samples has shown them to be virtually identical, falling within genetic group 6.

Conclusion.

Of the small number of viruses received all but four reacted well, showing no more than a 2-fold reduction compared to the homologous titre, with antiserum raised against A/California/7/2009 indicating antigenic similarity to the vaccine virus.

Influenza A(H3N2) virus analysis.

Influenza A(H3N2) viruses have been received from Europe, Africa, South America and the Far-East. As described previously, these viruses have continued to be difficult to characterise antigenically by HI assay due to variable agglutination of red blood cells from guinea pig, turkey and humans. Those viruses with sufficient titre in HA assays using guinea pig red blood cells in the presence of 20 nM oseltamivir, to circumvent the NA-mediated binding of H3N2 viruses to the red blood cells, were analysed by HI assay. Approximately 75% of viruses propagated retained sufficient HA titre in the presence of oseltamivir on guinea pig red blood cells to allow HI analysis (**Table 1**). The results of HI analyses of H3N2 viruses received, with collection dates since 01 February 2012, are shown in **Tables 8 to 20**, and the results are summarised in **Table 21**.

HI analyses of 376 test viruses show that 76% of the viruses display HI titres ≥ 8 -fold reduced compared with the titre of the homologous virus with post-infection ferret antisera raised against the prototype virus A/Perth/16/2009; reductions for the two ferret antisera employed, F30/09 and F35/11, were 51% and 90% respectively (**Tables 8 to 20**). Titres for test viruses were also low (≥ 8 -fold reduced), in comparison with the homologous virus, for post-infection ferret sera raised against other egg-propagated reference and vaccine viruses: 97% of viruses showed reduced titres with ferret antiserum raised against A/Victoria/208/2009; 92% with antiserum raised against A/Iowa/19/2010; and 96% with antiserum raised against exclusively egg-propagated vaccine virus, A/Victoria/361/2011. By contrast, in HI assays using post-infection ferret antisera raised against exclusively tissue-culture propagated reference viruses (e.g. A/Hong Kong/3969/2011, A/Stockholm/18/2011, A/Norway/1789/2011, A/Berlin/93/2011, A/Athens/GR112/2012 and the MDCK cell propagated A/Victoria/361/2011) the vast majority of test viruses (range 96.3 - 99.7%) had titres within 4-fold of that of the corresponding homologous reference virus. This pattern of reactivity was consistent for all currently circulating viruses, independent of the HA genetic group into which they fell as indicated in **Tables 8 to 20**.

These antigenic analyses indicate that egg-adaptation affects the antigenic characteristics of egg-propagated H3N2 viruses making them antigenically distinct from viruses propagated exclusively in mammalian cells. There is no evidence of significant antigenic diversity within circulating H3N2 viruses when assessed with antisera raised against MDCK or MDCK-SIAT1 cell-propagated reference viruses.

Virus neutralisation assays on MDCK-SIAT1 cells were used to complement HI and the results are shown in **Tables 22 to 25**. The HA genetic group of all but one (34/35) of the test viruses is known. As observed previously, when using an endpoint of 80% plaque reduction, post-infection ferret antiserum raised against A/Perth/16/2009 (F35/11) reacted with a titre of at least 4-fold lower against recently circulating test viruses, compared with the titre against the homologous virus (**Table 22**). As seen in the HI assays, homologous titres for ferret antisera raised against egg-propagated viruses were considerably higher than for those raised against mammalian cell-propagated viruses, and all cell-propagated test viruses were poorly neutralised by antisera raised against egg-propagated viruses regardless of the HA genetic group (A/Iowa/19/2010 [group 6], A/Victoria/361/2011 [group 3C]). In contrast, post-infection ferret antisera raised against cell-propagated viruses (A/Hong Kong/3969/2011 [group 3C], A/Stockholm/18/2011 [group 3A], A/Athens/GR112/2012 and A/Victoria/361/2011

[group 3C]) generally neutralised test viruses well, regardless of the HA genetic group into which they fell (**Tables 22 to 25**).

Phylogenetic analysis of the HA genes of representative H3N2 viruses is shown in **Figure 9**. Viruses examined by HI analysis and virus neutralisation tests and presented in **Tables 8 to 25** are boxed. Two clades of viruses have been defined, designated as the A/Perth/16/2009 and A/Victoria/208/2009 clades. Within these clades, seven genetic groups have been assigned (1-7) and the group 3 viruses have been split into three subgroups (A-C). The seven genetic groups and three group 3 subgroups are defined by the following amino acid substitutions in **HA1** and **HA2**.

In the Perth/16 genetic clade:

- (1) **P162S, I260M, R261Q**, e.g. A/Norway/1186/2011;
- (2) **N133D** (resulting in the loss of a glycosylation site), **R142G, T212A, V213A, N169K & V176I**, e.g. A/Norway/1330/2010.

In the Victoria/208 genetic clade which all carry the substitutions: **K62E, K144N** (resulting in the gain of a glycosylation site, and **T212A** with respect to viruses of the Perth/16 genetic clade):

- (3A) **N144D** (resulting in the loss of a glycosylation site), **N145S, V223I & D158N**, e.g. A/Stockholm/18/2011;
- (3B) **N145S, A198S, V223I, N312S & D158N**, e.g. A/England/517/2012;
- (3C) **S45N** (resulting in the gain of a glycosylation site) **T48I, A198S, V223I & N312S**, e.g. A/Hong Kong/3969/2011, with many viruses also carrying the substitutions **Q33R & N278K**;
- (4) **N312S**, e.g. A/Serbia/71/2011;
- (5) **D53N, Y94H, I230V & E280A**, e.g. A/Perth/10/2011;
- (6) **D53N, Y94H, S199A, I230V & E280A**, e.g. A/Iowa/19/2010;
- (7) **S45N** (resulting in the gain of a glycosylation site), e.g. A/Maine/07/2012.

The HA genes of all recently circulating viruses received at the London WHO CC have fallen within the A/Victoria/208/2009 clade, specifically in genetic subgroups 3A, 3B, and 3C, and group 5 and group 6.

Figure 10 shows an HA amino acid alignment for a selection of the viruses represented in the HA phylogeny. The vaccine virus recommended for the 2012 Southern Hemisphere influenza season, A/Perth/16/2009, is shown in red, as is the A/Victoria/361/2011 virus recommended for the 2012-13 Northern Hemisphere season, while potential glycosylation motifs are highlighted and the reference viruses are coloured blue. The genetic group to which each of the sequences belongs is also shown.

Figures 11a & 11b show the locations on the H3 HA structure of amino acid substitutions that define genetic sub/groups 3A, 3B, 3C, 5 and 6. Several of the substitutions that have

accumulated are exposed on the surface of the HA and could potentially influence the antigenicity of the virus.

Figure 12 shows a NA gene phylogenetic tree for representative viruses with the HA clades and genetic groups indicated, together with indication of which viruses carry substitutions/polymorphisms in the NA 150-loop that can influence the NA contribution to agglutination of red blood cells. Viruses with these substitutions/polymorphisms are seen in all genetic groups.

Figure 13 shows a NA amino acid alignment for a selected group of these viruses. Vaccine viruses are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the genetic group to which the corresponding HA gene sequences belong is also shown as are viruses that are HA/NA genetic group reassortants.

The HA and NA phylogenetic trees are generally congruent but it is noteworthy that the viruses of the predominating genetic groups all carry the dual substitutions S367N and K369T in their NA which results in the addition of a potential glycosylation site at residue 367. Other potential glycosylation sites are frequently lost at position 402 and occasionally at position 329 in some genetic subgroups. **Figure 14** localises these variable sites on the structure of a N2 NA with sialic acid in the active site.

Antiviral resistance.

All viruses analysed at the London WHO CC, 702 in total with 318 collected since 01 February 2012, showed normal inhibition (sensitive phenotype) with the neuraminidase inhibitors zanamivir and oseltamivir (**Table 40**). The subset assessed by M-gene sequencing all show resistance to adamantanes due to possession of the S31N substitution in the M2 ion-channel protein.

Conclusion.

Antigenic characterisation of H3N2 viruses has shown there to be clear differences between antisera raised against egg-propagated and cell-propagated A/Victoria/361/2011. Viruses circulating recently have shown good reactivity in HI assays with post-infection ferret antisera raised against cell-propagated reference viruses, including A/Victoria/361/2011, but low reactivity with antisera raised against egg-propagated viruses, notably A/Perth/16/2009 (recommended vaccine virus for the Southern Hemisphere 2012 season) and A/Victoria/361/2011 (recommended vaccine virus for the Northern Hemisphere 2012-13 season). The same pattern has been observed in plaque-reduction neutralisation assays for a subset of the viruses analysed by HI assay and across all HA genetic sub/groups (3A, 3B, 3C, 5 and 6) currently in circulation.

Influenza B viruses.

Approximately 200 influenza B viruses with collection dates from 01 February 2012 have been received, in the proportions of 58% B/Victoria and 42% B/Yamagata lineage viruses (**Table 1**). This represents a continuation of the increase in the prevalence of viruses of the B/Yamagata lineage over the past 12 months and reflects the proportions seen in WHO-EURO region countries that report to ECDC (**Figure 1**).

B/Victoria lineage.

Influenza B viruses of the B/Victoria lineage were received from countries in Europe, Africa (Madagascar, Senegal and South Africa) and from China, Hong Kong SAR.

Tables 26 to 32 show the results of HI analysis of viruses of the B/Victoria lineage. A summary of the results is shown in **Table 33**.

The vast majority of test viruses showed low reactivity with antisera raised against the egg-propagated, currently recommended vaccine virus, B/Brisbane/60/2008 compared to the titre with the homologous virus. The test viruses also showed similar reductions in titre with antiserum raised against the egg-propagated virus B/Malta/636714/2011. The viruses reacted better with antisera raised against surrogates for B/Brisbane/60/2008, reference viruses genetically closely related to B/Brisbane/60/2008 but propagated in cells: these antisera were raised against B/Paris/1762/2008, B/Hong Kong/514/2009 and B/Odessa/3886/2010. In HI analyses using these antisera all test viruses reacted with titres within 4-fold of the homologous titre with antiserum raised against B/Paris/1762/2008, all but three reacted with titres within 4-fold of the homologous titre with antiserum raised against B/Hong Kong/514/2009, but 13% of test viruses (12/90) reacted with titres reduced ≥ 8 -fold compared to the homologous titre with antiserum raised against B/Odessa/3886/2010.

Phylogenetic analysis of the HA gene of representative B/Victoria lineage viruses is shown in **Figure 15**, with viruses for which antigenic results are presented in **Tables 26 to 32** boxed, and an alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 16**. The HA genes of all recently collected viruses fall into clade 1, with only a small number of amino acid substitutions compared with the HA of the currently recommended vaccine virus, B/Brisbane/60/2008.

Figure 17 shows a phylogenetic tree for the NA gene and **Figure 18** shows a corresponding alignment for a selection of NA glycoproteins. While the majority of NA genes also clustered within genetic clade 1, a few fell into genetic clade 4 and a small minority into clade 3. Viruses showing HA clade 1 reassortment with non-clade 1 NA genes are indicated on the HA phylogenetic tree (**Figure 15**), e.g. HA clade1/NA clade 3, B/Dakar/10/2012, and HA clade 1/NA clade 4, B/Hong Kong/3534/2012 and B/St. Petersburg/27/2012.

B/Yamagata lineage.

Influenza B viruses of the B/Yamagata lineage were received from countries in Europe, Africa (Ghana, Madagascar and South Africa), Guyane (South America) and from China, Hong Kong SAR.

Tables 34 to 38 show the results of HI analysis of B/Yamagata lineage viruses and the results are summarised in **Table 39**.

The reactivity of the test viruses with post-infection ferret antiserum raised against the currently recommended vaccine virus B/Wisconsin/1/2010 was variable, with a group of viruses showing reduced reactivity with the antiserum, when compared with the homologous titre of B/Wisconsin/1/2010. Reactivity with antiserum raised against B/Serbia/1894/2011 was also heterogeneous but reactivity with antiserum raised against B/Stockholm/12/2011 was much more homogenous. Test viruses analysed with antiserum raised against B/Estonia/55669/2011 fell into two antigenic groupings, one that reacted within 4-fold of the titre with the homologous virus (43 of 77 [56%] test viruses) and a group (44% of test viruses) that reacted with ≥ 8 -fold reduced titres. For the post-infection ferret antiserum raised against B/Wisconsin/1/2010 23 of 77 (30%) test viruses reacted with titres reduced by ≥ 8 -fold compared with the homologous titre.

Figure 19 shows a phylogenetic analysis of the HA gene of representative B/Yamagata lineage viruses and an alignment of a selection of the HA glycoproteins is shown in **Figure 20**. Viruses with collection dates since 01 February 2012 fell into two clades in similar proportions (26 in clade 2, 24 in clade 3), and numbers of countries with viruses in each clade were also similar (14 countries had viruses in clade 2, 13 countries/regions had viruses in clade 3). Of the test viruses, for which clade was determined, 25/47 from 15 countries were in clade 2 and 22/47 from 10 countries were in clade 3 (**Tables 34 to 38**). The viruses that could be distinguished antigenically based on their HI reactions with post-infection ferret antisera raised against B/Wisconsin/1/2010 and B/Estonia/55669/2011 matched their genetic clade designations.

Clade 3 viruses are represented by the reference virus B/Bangladesh/3333/2007 and the vaccine virus B/Wisconsin/1/2010, while clade 2 viruses are represented by the reference viruses B/Brisbane/3/2007 and B/Estonia/55669/2011. For the HA gene, clade 3 can be subdivided into two to four genetic groups: a group similar to B/Wisconsin/1/2010, defined by the amino acid substitution **N202S**, a group defined by the substitution **T181K** (e.g. B/Ireland/M1522/2012), and two groups defined by the substitution **M251V** with either the substitutions **T181A & K253R** (e.g. B/Serbia/1894/2011) or with the substitutions **V29A & L172Q** (e.g. B/Stockholm/12/2011) (**Figure 19**). The locations of HA amino acid substitutions that distinguish the vaccine virus, B/Wisconsin/1/2012 (clade 3), from B/Estonia/55669/2011 (clade 2) are shown on the influenza B HA structure (**Figure 21**).

Figure 22 shows a phylogenetic tree of the NA gene and **Figure 23** shows a corresponding alignment for selected NA glycoproteins. As for the HA gene, the NA genes fall into clades 2 and 3 with similar groupings within a clade; there is no evidence of viral inter-clade gene reassortment.

Antiviral Analysis.

All influenza B viruses subjected to phenotypic analysis, 112 in number with collection dates since 01 February 2012, showed normal inhibition (sensitive phenotype) for the neuraminidase inhibitors zanamivir and oseltamivir (**Table 40**).

Conclusion.

Of the 167 influenza B viruses analysed approximately 60% and 40% fell into the B/Victoria and B/Yamagata lineages respectively. This represents a sustained rise in the proportion of B/Yamagata viruses. For viruses of the B/Victoria lineage, the great majority were antigenically similar to the currently recommended vaccine virus, B/Brisbane/60/2008, or viruses closely related to it. By HI analysis, B/Yamagata lineage viruses circulating since 01 February 2012 have continued to fall into two antigenically distinguishable groups in approximately equal numbers. At the HA gene level these groups correspond to distinct clades: clade 3 represented by the vaccine virus B/Wisconsin/1/2010 and clade 2 represented by B/Estonia/55669/2011.

Figure 1. Influenza Activity in Europe: 2011-12 Season (weeks 40/11-34/12)

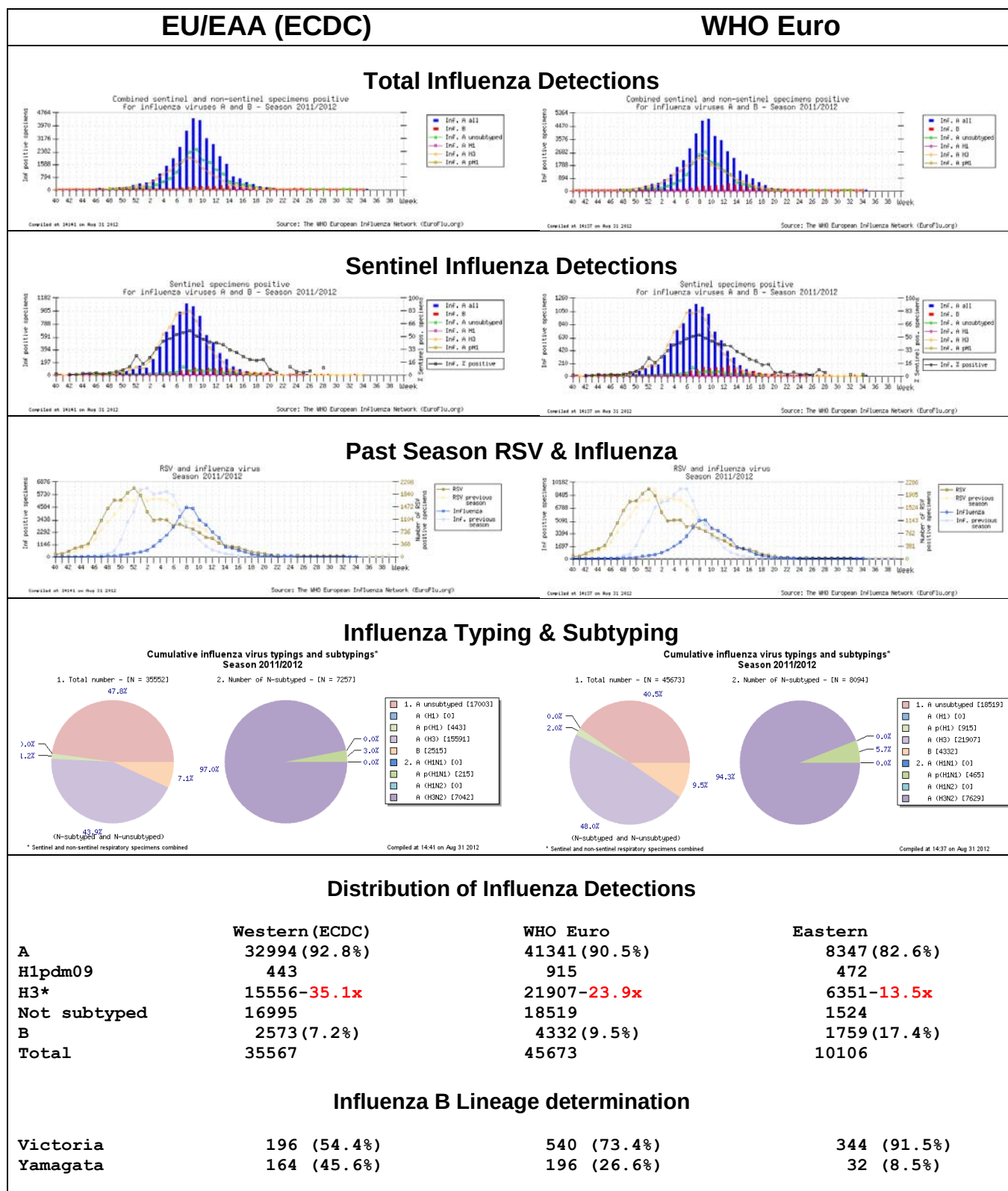


Figure 2. Comparison of 2010-11 & 2011-12 Influenza Seasons in Europe

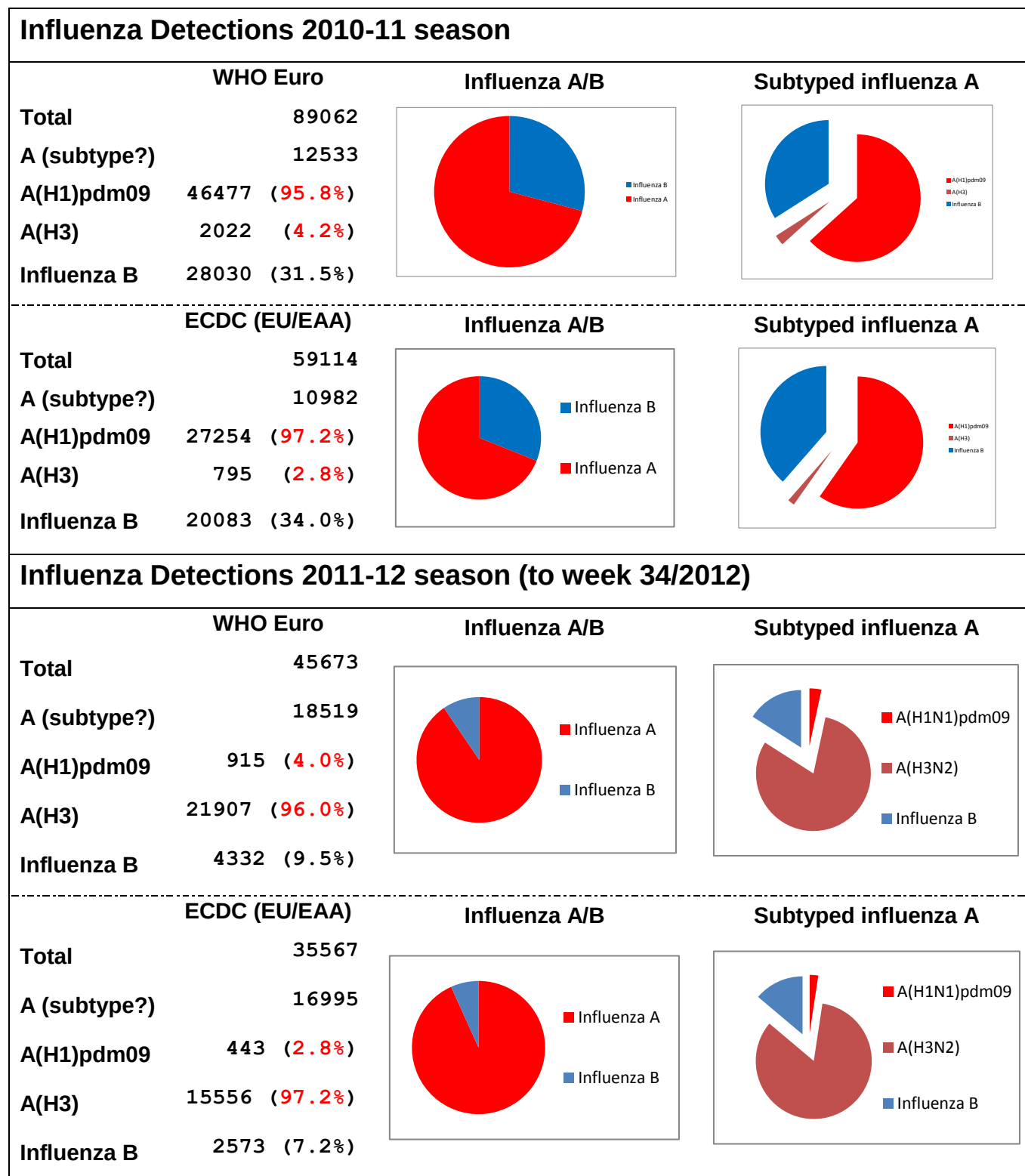


Figure 3. Country Variation in the Intensity of the 2011-12 Season

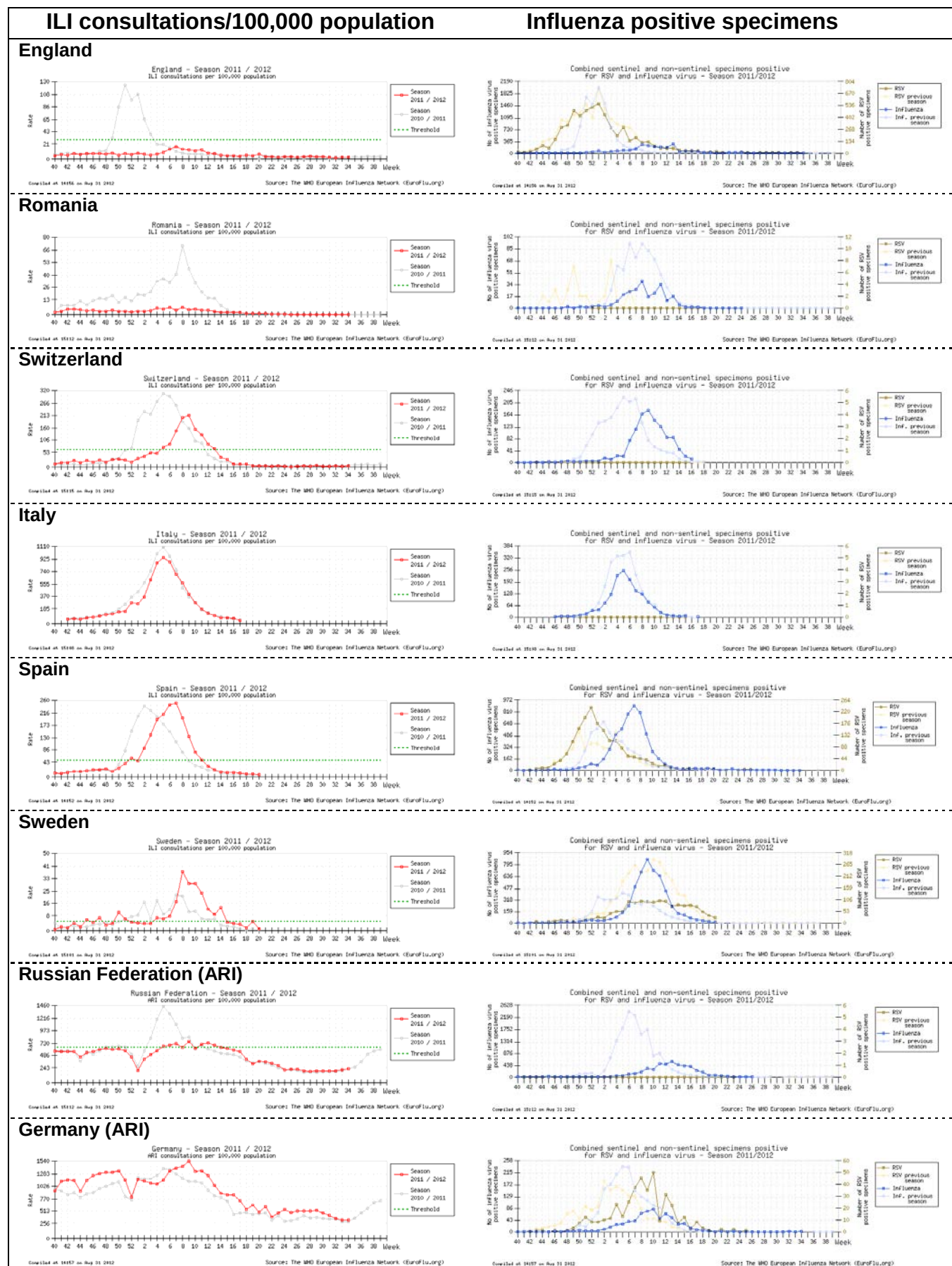


Table 1. Summary of clinical samples and isolates received, with collection dates since 2012-02-01, by country

MONTH	TOTAL RECEIVED	A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage	
		Not subtyped		Number received	Number propagated ¹	Number received	Number propagated ²	Lineage unknown		Number received	Number propagated ¹	Number received	Number propagated ¹
Country													
UNKNOWN													
Czech Republic ³	26					10	10			14	10	2	2
FEBRUARY													
Albania	2					2	1						
Bulgaria	8					8	8						
Denmark	11					6	4			2	2	3	3
Estonia	18					18	2						
Finland	5					4	2			1	1		
France	5			2	2	3	3						
Georgia	6			2	0	4	4						
Ghana	1							1					
Greece	24					14	13	4		2	2	4	4
Iceland	11					11	11						
Ireland	3			1	1	1	1			1	1		
Italy	12					6	5			1	1	5	5
Macedonia	6					6	0						
Madagascar	11	2		5	1					4	4		
Montenegro	2					2	0						
Norway	29			10	6	15	15			1	1	3	3
Portugal	3					1	1					2	1
Russia	6					3	3					3	3
Serbia	1					1	1						
Slovenia	12			1	1	9	7			1	0	1	0
Sweden	5			3	3					1	1	1	1
Ukraine	14	1				12	12	1					
United Kingdom	7					5	4			1	1	1	1
MARCH													
Belarus	7					7	5						
Denmark	13			3	0	8	7			1	0	1	0
Djibouti	1									1	1		
Estonia	11					11	7						
Finland	1					1	1						
France	19			2	2	11	10			4	4	2	2
Georgia	19			10	1	6	3			3	2		
Ghana	1					1	0						
Guyane	3					2	2					1	1
Iceland	3					3	2						
Ireland	4					4	4						
Italy	11					4	1			1	0	6	6
Macedonia	9					9	0						
Madagascar	5									5	5		
Montenegro	35					35	7						
Norway	14			3	0	6	6			2	1	3	3
Portugal	10					6	4			1	1	3	2
Russia	6					6	6						
Senegal	2			1	1					1	1		
Serbia	14					14	13						
Slovenia	16			1	0	9	7			4	1	2	0
Sweden	2					1	1					1	1
Ukraine	91	2				86	79			3	3		
United Kingdom	10			1	1	7	4			1	1	1	1
APRIL													
Denmark	6					4	3			1	1	1	0
Estonia	8					7	3					1	1
France	6									3	3	3	3
Georgia	10			6	0	2	1			2	1		
Ghana	3					3	0						
Hong Kong	1									1	1		
Iceland	1					1	1						
Ireland	10					10	2						
Madagascar	8			3	0					5	2		
Montenegro	9					9	0						
Norway	11					8	8			1	1	2	2
Russia	13					9	9			2	2	2	2
Senegal	15									15	15		
Serbia	5					5	5						
Slovenia	2					1	1	1					
South Africa	1											1	1
Sweden	5			1	1	3	2					1	1
Ukraine	20					20	17						
United Kingdom	10					7	6			2	2	1	1
MAY													
Finland	1									1	1		
Georgia	1									1	1		
Ghana	4					4	1						
Hong Kong	7			2	2					5	5		
Madagascar	13					9	9	1		3	3		
Norway	2					2	2						
Senegal	1									1	1		
South Africa	3									1	1	2	2
Ukraine	1					1	1						
United Kingdom	3					2	in progress			1	in progress		
JUNE													
Ghana	9					2	0	2				5	5
Hong Kong	13			1	1	6	4					6	6
Madagascar	9	1				8	2						
South Africa	19					14	in progress					5	in progress
United Kingdom	1											1	in progress
JULY													
Ghana	1												
South Africa	3					2	2			1	in progress	1	1
AUGUST													
Ireland	1			1	0								
South Africa	3									3	in progress		
United Kingdom	2					2	in progress						
	776	6		59	23	519	355	10		105	84	77	64
				7.6%		66.9%				13.5%		9.9%	

1. Propagated to sufficient titre to perform HI assay

2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir

3. Package received 2012-05-10

Table 2. Summary of clinical samples and isolates received, with collection dates since 2012-02-01, by WHO region

MONTH	TOTAL RECEIVED	A	H1N1pdm09		H3N2		B	B Victoria lineage		B Yamagata lineage	
WHO Region		Not subtyped	Number received	Number propagated	Number received	Number propagated ¹	Lineage unknown	Number received	Number propagated	Number received	Number propagated
UNKNOWN											
EURO ²	26				10	10		14	10	2	2
FEBRUARY											
AFRO	12	2	5	1			1	4	4		
EURO	190	1	19	13	131	97	5	11	10	23	21
MARCH											
AFRO	9		1	1	1	0		7	7		
AMRO	3				2	2				1	1
EURO	295	2	20	4	234	167		20	13	19	15
APRIL											
EURO	116		7	1	86	58	1	11	10	11	10
AFRO	27		3	0	3	0		20	17	1	1
WPRO	1							1	1		
MAY											
AFRO	21				13	10	1	7	7	2	2
EURO	8				5	3		3	2		
WPRO	7		2	2				3	3		
JUNE											
AFRO	37	1			24	2	2	0	0	10	5
EURO	1									1	in progress
WPRO	13		1	1	6	4				6	6
JULY											
AFRO	4				2	2		1	in progress	1	1
AUGUST											
AFRO	3							3	in progress		
EURO	3		1	0	2	in progress					
	776	6	59	23	519	355	10	105	84	77	64
				7.6%		66.9%			13.5%		9.9%

1. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir

2. Package received 2012-05-10

Table 3. Isolation rates for samples collected after 2012-02-01

Virus subtype lineage	Clinical sample			Virus isolate		
	number received	number propagated ¹	percent recovered	number received	number propagated ¹	percent recovered
H1	30	1	3.3	33	26	78.8
H3	132	44	33.3	374	314	84.0
B Victoria	13	7	53.8	91	79	86.8
B Yamagata	7	3	42.9	76	72	94.7

1. Propagated to sufficient titre to perform HI assay

Table 4. Summary of isolates collected since 2012-02-01 showing reduction in reactivity with antisera raised against reference viruses

Virus subtype/lineage	Number tested	≤2 fold^a	4 fold^a	≥8 fold^a
H1¹	23	19 (82.6%)	1 (4.3%)	3 (13.1%)
H3²	355	331 (93.2%)	20 (5.6%)	4 (1.2%)
B/Victoria³	84	56 (66.6%)	14 (16.7%)	14 (16.7%)
B/Yamagata⁴	64	51 (79.7%)	11 (17.2%)	2 (3.1%)

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

1. Compared to homologous titre for ferret antiserum raised against vaccine virus A/California/7/2009
2. Compared to homologous titres for ferret antisera raised against tissue-culture grown A/Victoria/361/2011 (Tables 11-20) or A/Norway/1789/2011 (Tables 8-10)
3. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents
4. Compared to homologous titre for ferret antiserum raised against prototype virus B/Stockholm/12/2011

Table 5. Antigenic analyses of influenza A(H1N1)pdm09 viruses

Haemagglutination inhibition titre ¹											
Post infection ferret sera											
Viruses		Collection date	Passage History	A/Cal 7/09 F29/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/C'church 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P'burg 27/11 F23/11	A/St. P'burg 100/11 F24/11
	Genetic Group						Group 4	Group 3	Group 5	Group 6	Group 7
REFERENCE VIRUSES											
A/California/7/2009		2009-04-09	E1/E2	1280	2560	1280	1280	1280	1280	1280	2560
A/Bayern/69/2009		2009-07-01	MDCK5	160	640	320	160	80	160	160	80
A/Lviv/N6/2009		2009-10-27	M4/S1/M2	1280	2560	1280	320	160	320	320	320
A/Christchurch/16/2010		2010-07-12	E2/E1	5120	5120	5120	5120	5120	5120	5120	5120
A/Hong Kong/3934/2011		2011-03-29	M2/M2	2560	640	640	1280	2560	2560	2560	5120
A/Astrakhan/1/2011		2011-02-28	M1/M5	2560	1280	1280	1280	2560	2560	2560	5120
A/St. Petersburg/27/2011		2011-02-14	E1/E2	2560	2560	2560	1280	2560	2560	2560	5120
A/St. Petersburg/100/2011		2011-03-14	E1/E2/E1	2560	1280	2560	2560	2560	2560	2560	5120
TEST VIRUSES											
A/Norway/421/2012	7	2012-02-01	MDCK1/MDCK2	5120	2560	2560	2560	5120	5120	5120	5120
A/Norway/418/2012	7	2012-02-02	MDCK1/MDCK1	2560	1280	1280	2560	2560	2560	5120	5120
A/Norway/446/2012	7	2012-02-02	MDCK2/MDCK2	5120	2560	5120	2560	5120	5120	5120	5120
A/Norway/560/2012	6	2012-02-02	MDCK2/MDCK2	640	320	640	640	640	1280	1280	2560
A/Norway/716/2012	7	2012-02-17	MDCK1/MDCK2	1280	640	1280	1280	2560	2560	2560	5120
A/Stockholm/17/2012	6	2012-02-17	C2/MDCK2	160	160	160	160	320	160	160	320
A/Norway/717/2012	7	2012-02-19	MDCK1/MDCK2	640	320	640	640	1280	1280	1280	2560
A/Slovenia/421/2012	7	2012-02-20	Mx/MDCK3	1280	640	640	640	1280	1280	640	2560
A/Stockholm/16/2012	6	2012-02-21	C2/MDCK2	640	320	640	640	640	640	640	2560
A/Stockholm/18/2012	7	2012-02-27	C3/MDCK3	80	640	160	80	40	40	80	80
A/Ireland/M17525/2012	7	2012-02-29	C1/SIAT1	1280	1280	1280	1280	1280	1280	1280	2560
A/England/344/2012		2012-03-01	SIAT2/MDCK2	1280	640	640	640	1280	1280	1280	2560
A/Stockholm/25/2012	6	2012-04-07	C2/MDCK2	40	160	80	40	<	<	<	<

1. < = <40

Vaccine

Sequences in phylogenetic trees

K142E/K, K211R

G155E

A48D, G155E, G225A

Table 6. Antigenic analyses of influenza A(H1N1)pdm09 viruses

Haemagglutination inhibition titre ¹											
Post infection ferret sera											
Viruses	Collection date	Passage History	A/Cal 7/09 F29/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 C4/34/09	A/C'church 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/11	A/St. P'burg 27/11 F23/11	A/St. P'burg 100/11 F24/11	
	Genetic Group					Group 4	Group 3	Group 5	Group 6	Group 7	
REFERENCE VIRUSES											
A/California/7/2009		2009-04-09	E1/E2	1280	1280	1280	640	640	640	320	640
A/Bayern/69/2009		2009-07-01	MDCK4/MDCK1	80	320	160	80	40	80	80	40
A/Lviv/N6/2009		2009-10-27	M4/S1/M2	640	1280	640	160	80	160	160	160
A/Christchurch/16/2010		2010-07-12	E2/E2	2560	1280	2560	5120	1280	2560	1280	5120
A/Hong Kong/3934/2011		2011-03-29	M2/M2	640	160	320	320	1280	1280	640	1280
A/Astrakhan/1/2011		2011-02-28	M1/M5	1280	640	1280	1280	2560	2560	2560	5120
A/St. Petersburg/27/2011		2011-02-14	E1/E2	2560	1280	1280	1280	2560	2560	2560	5120
A/St. Petersburg/100/2011		2011-03-14	E1/E1	320	320	320	320	320	640	640	2560
TEST VIRUSES											
A/Caen/635/2012	6	2012-02-20	MDCK2/MDCK1	1280	320	1280	1280	2560	2560	1280	5120
A/Madagascar/01586/2012		2012-02-21	MDCK2	640	320	640	640	1280	640	640	2560
A/Paris/1054/2012	7	2012-02-21	MDCK1/MDCK1	1280	320	640	640	1280	1280	1280	2560
A/Georgia/427/2012	7	2012-03-19	MDCK3	1280	320	640	640	1280	1280	1280	2560
A/Paris/1119/2012	7	2012-03-21	MDCK2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120
A/Dakar/03/2012		2012-03-23	MDCK4/SIAT2	1280	320	640	640	1280	1280	1280	2560
A/Pays de Loire/1224/2012	7	2012-03-31	MDCK2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120
A/Maryland/13/2012	6	2012-04-01	M1/C1/MDCK1	2560	640	1280	1280	2560	2560	2560	5120
A/Hong Kong/5659/2012	6	2012-05-21	MDCK3/MDCK1	1280	320	640	640	1280	1280	1280	2560
A/Victoria/514/2012	7	2012-05-28	MDCK2/MDCK1	2560	640	1280	1280	2560	2560	2560	5120
A/Hong Kong/6609/2012	6	2012-05-30	MDCK3/MDCK1	1280	320	1280	1280	2560	2560	2560	5120
A/Hong Kong/6747/2012	6	2012-06-01	MDCK2/MDCK1	320	160	160	160	80	160	80	160
A/Victoria/523/2012	6	2012-06-11	E5/E1	640	320	640	320	1280	1280	640	1280
A/Washington/24/2012	7	2012-06-17	C2/MDCK1	1280	320	640	640	1280	1280	1280	5120
A/Washington/24/2012	7	2012-06-17	E4/E1	1280	640	640	640	1280	1280	1280	2560

G155E

1. < = <40

Vaccine

Sequences in phylogenetic trees

G155E

Table 7. Summary of influenza A (H1N1)pdm09 samples analysed, collected between 2012-02-01 and 2012-08-31

MONTH	H1N1			
	Country	Number propagated ¹	Fold reduction in HI titre	
			≤2 fold ²	≥8 fold ²
FEBRUARY				
France	2	2		
Ireland	1	1		
Madagascar	1	1		
Norway	6	6		
Slovenia	1	1		
Sweden	3	1		2
MARCH				
France	2	2		
Georgia	1	1		
Senegal	1	1		
United Kingdom	1	1		
APRIL				
Sweden	1			1
MAY				
Hong Kong	2	2		
JUNE				
Hong Kong	1		1	
Total	23	19	1	3

* Reduction in HI titre with ferret antiserum raised against A/California/7/2009 (egg grown vaccine virus)

All viruses with >2 fold reduction in HI titre contained HA substitutions as a result of cell propagation that affect receptor binding/antigenicity

Figure 4. Phylogenetic comparison of influenza A(H1N1)pdm09 HA genes

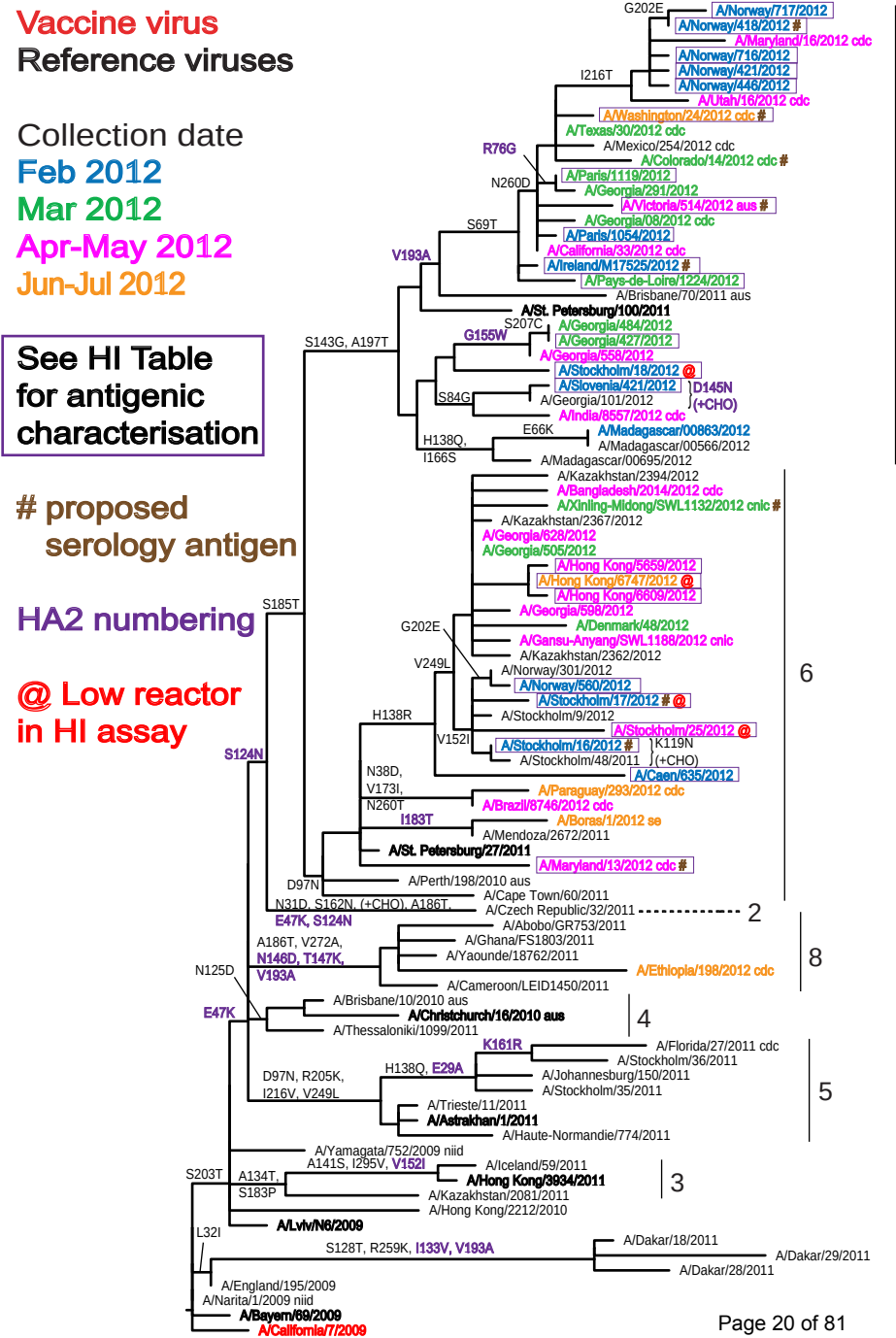
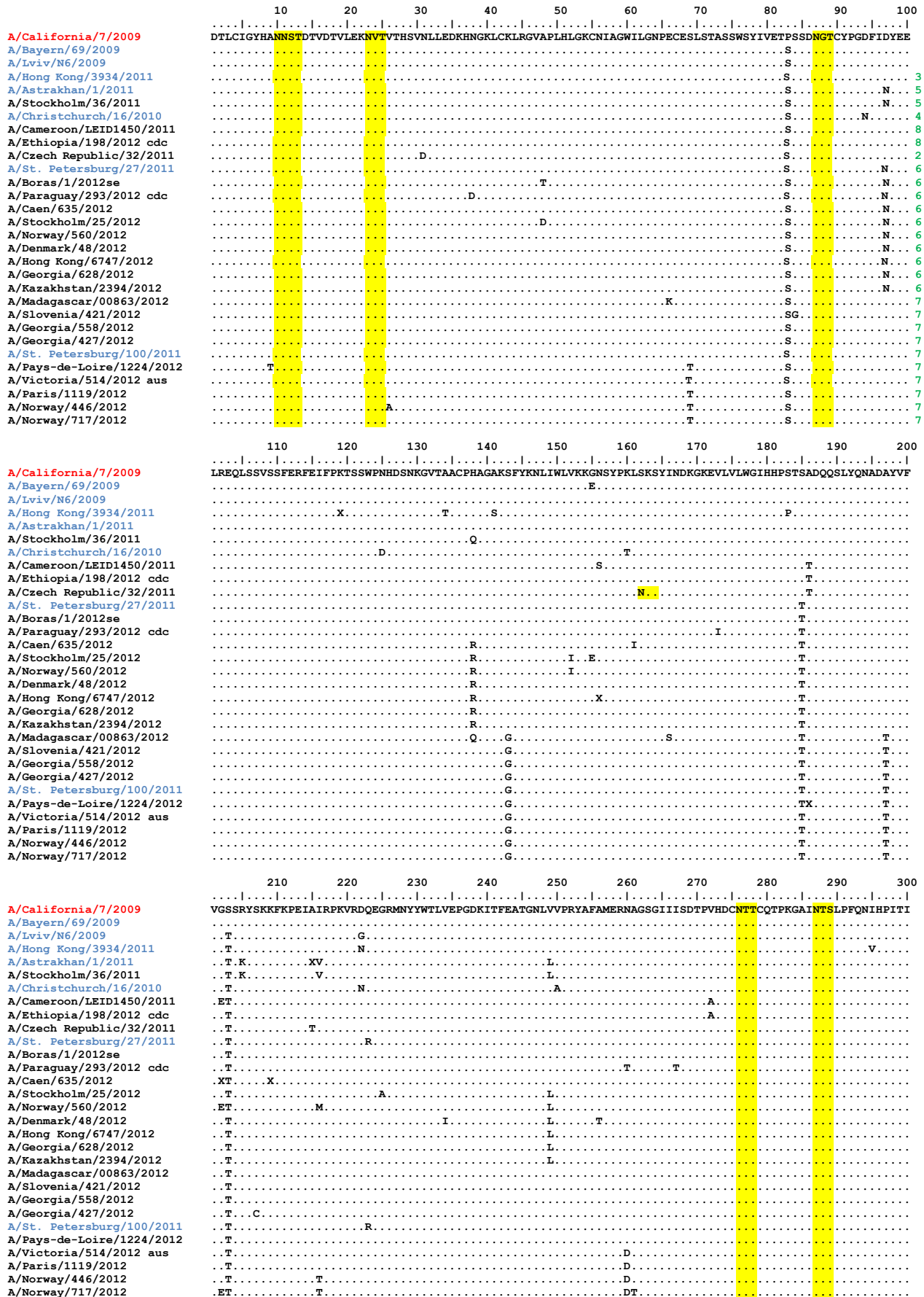


Figure 5. HA protein alignment for A(H1N1)pdm09 viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400
A/California/7/2009	GKCPKYVKSTKLRLATGLRNIPSIQSR	GLPGAIA	GFIEGGWTGMVDGWYGYHHQNEQ	SGSYAADLKSTQNAIDEITNKVNSVIERMNTQPTAVGKEFNHL						
A/Bayern/69/2009		V								
A/Lviv/N6/2009		V								
A/Hong Kong/3934/2011		V								
A/Astrakhan/1/2011		V						K		
A/Stockholm/36/2011		V				A		K		
A/Christchurch/16/2010		V						K		
A/Cameroon/LEID1450/2011		V						K		
A/Ethiopia/198/2012 cdc		V					Q	K		
A/Czech Republic/32/2011		V						K		
A/St. Petersburg/27/2011		V						K		
A/Boras/1/2012se		V						K		
A/Paraguay/293/2012 cdc		V						K		
A/Caen/635/2012		V						K		
A/Stockholm/25/2012		V						K		
A/Norway/560/2012		V						K		
A/Denmark/48/2012		V						K		
A/Hong Kong/6747/2012		V						K		
A/Georgia/628/2012		V						K		
A/Kazakhstan/2394/2012		V						K		
A/Madagascar/00863/2012		V						K		
A/Slovenia/421/2012		V						K		
A/Georgia/558/2012		V						K		
A/Georgia/427/2012		V						K		
A/St. Petersburg/100/2011		V						K		
A/Pays-de-Loire/1224/2012		V						K		
A/Victoria/514/2012 aus		V						K		
A/Paris/1119/2012		V						K		
A/Norway/446/2012		V						K		
A/Norway/717/2012		V						K		
	410	420	430	440	450	460	470	480	490	500
A/California/7/2009	EKRIENLNKKVDDGFLDIWTYNAELLVLL	ENERTLDYHDSNVKNLYEKVRSQ	LKNNAKEIGNGCFEYHKCDNTCMESV	KNGT	YDYPKYSEEAKLNREEI					
A/Bayern/69/2009										
A/Lviv/N6/2009										
A/Hong Kong/3934/2011								I		
A/Astrakhan/1/2011										
A/Stockholm/36/2011									R	
A/Christchurch/16/2010										
A/Cameroon/LEID1450/2011								DK		
A/Ethiopia/198/2012 cdc								DK		
A/Czech Republic/32/2011				N						
A/St. Petersburg/27/2011				N						
A/Boras/1/2012se				N						K
A/Paraguay/293/2012 cdc				N						
A/Caen/635/2012				N						
A/Stockholm/25/2012				N						
A/Norway/560/2012				N						
A/Denmark/48/2012				N						
A/Hong Kong/6747/2012				N						
A/Georgia/628/2012				N						
A/Kazakhstan/2394/2012				N						
A/Madagascar/00863/2012				N						
A/Slovenia/421/2012			N				N			
A/Georgia/558/2012				N				W		
A/Georgia/427/2012				N				W		
A/St. Petersburg/100/2011				N						
A/Pays-de-Loire/1224/2012				N						
A/Victoria/514/2012 aus	X			N						
A/Paris/1119/2012	G			N						
A/Norway/446/2012				N						
A/Norway/717/2012			E		N					
	510	520	530	540	550					
A/California/7/2009	DGVKLESTRIYQILAIYSTVASSLVLVSL	GAISFWMCNS	NGSLQCRICI*							
A/Bayern/69/2009					*					
A/Lviv/N6/2009					*					
A/Hong Kong/3934/2011					*					
A/Astrakhan/1/2011					*					
A/Stockholm/36/2011			A		*					
A/Christchurch/16/2010					*					
A/Cameroon/LEID1450/2011			A		*					
A/Ethiopia/198/2012 cdc			A		*					
A/Czech Republic/32/2011					*					
A/St. Petersburg/27/2011					*					
A/Boras/1/2012se		T			*					
A/Paraguay/293/2012 cdc					*					
A/Caen/635/2012					*					
A/Stockholm/25/2012					*					
A/Norway/560/2012					*					
A/Denmark/48/2012					*					
A/Hong Kong/6747/2012					*					
A/Georgia/628/2012					*					
A/Kazakhstan/2394/2012					*					
A/Madagascar/00863/2012					*					
A/Slovenia/421/2012					*					
A/Georgia/558/2012					*					
A/Georgia/427/2012					*					
A/St. Petersburg/100/2011					*					
A/Pays-de-Loire/1224/2012			A		*					
A/Victoria/514/2012 aus			A		*					
A/Paris/1119/2012			A		*					
A/Norway/446/2012			A		*					
A/Norway/717/2012			A		*					

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 6. H1 HA locations of genetic group defining amino acid substitutions.

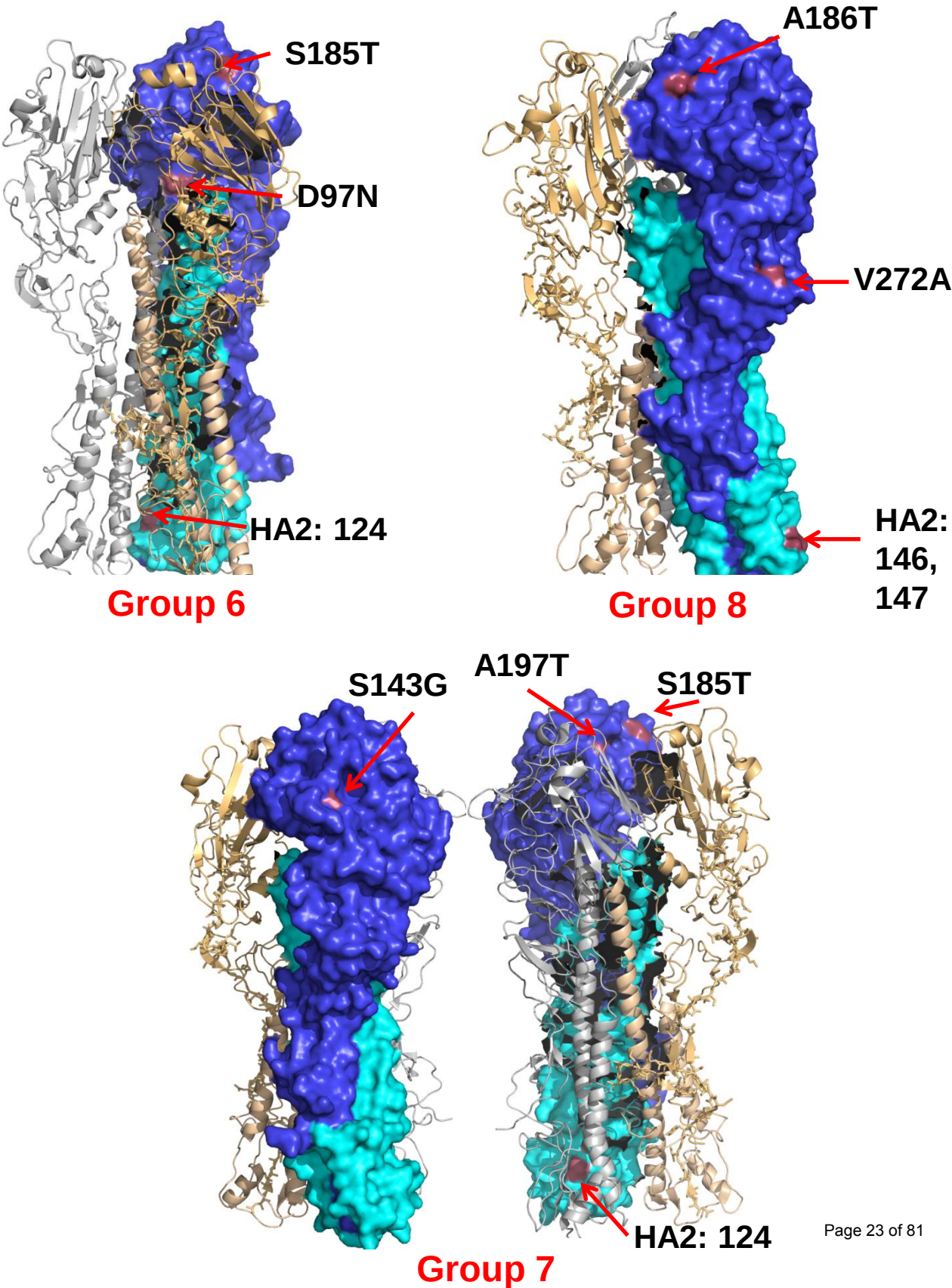


Figure 7. Phylogenetic comparison of influenza A(H1N1)pdm09 NA genes

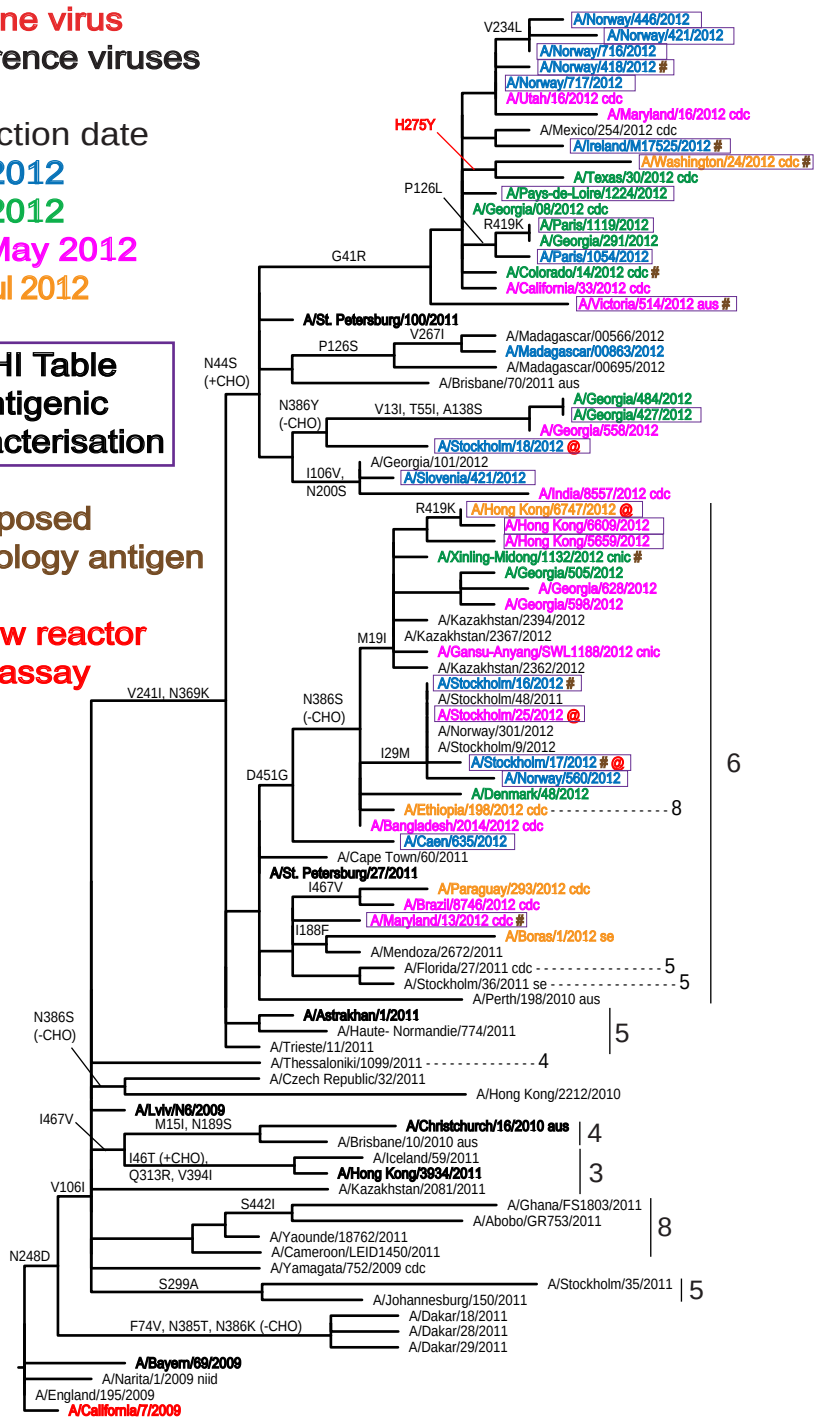
Vaccine virus
Reference viruses

Collection date
Feb 2012
Mar 2012
Apr-May 2012
Jun-Jul 2012

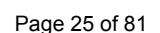
See HI Table
for antigenic
characterisation

proposed
serology antigen

@ Low reactor
in HI assay



Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site: NA/HA reassortants



	310	320	330	340	350	360	370	380	390	400
A/California/7/2009	RPWVSFNQ	LEYQIGYICSGIFG	DNPRPNDKTGSCGPVSS	NGANGVKGFSFR	YGNVWIGRTKSI	SSRNGFEMIWD	PNGWTGTDN	NFSIKQDIVG	IN	NEWS
A/Bayern/69/2009										
A/Lviv/N6/2009										
A/Hong Kong/3934/2011		R.....							I.....	
A/Astrakhan/1/2011						K.....				
A/Stockholm/36/2011 se						K.....				
A/Christchurch/16/2010							I.....			
A/Cameroon/LEID1450/2011										
A/Ethiopia/198/2012 cdc						K.....		S.....		
A/Czech Republic/32/2011								S.....		
A/St. Petersburg/27/2011						K.....				
A/Boras/1/2012 se						K.....				
A/Paraguay/293/2012 cdc						K.....				
A/Caen/635/2012						K.....				
A/Stockholm/25/2012						K.....		S.....		
A/Norway/560/2012			R.....			K.....		S.....		
A/Denmark/48/2012						K.....		S.....		
A/Hong Kong/6747/2012						K.....		S.....		
A/Georgia/628/2012						K.....		S.....		
A/Kazakhstan/2394/2012						K.....		S.....		
A/Madagascar/00863/2012						K.....				
A/Slovenia/421/2012						K.....				
A/Georgia/558/2012						K.....		Y.....		
A/Georgia/427/2012						K.....		Y.....		
A/St. Petersburg/100/2011					I.....	K.....				
A/Pays-de-Loire/1224/2012						K.....				
A/Victoria/514/2012 aus						K.....				
A/Paris/1119/2012						K.....				
A/Norway/446/2012						K.....				
A/Norway/717/2012						K.....				

	410	420	430	440	450	460	470
A/California/7/2009	GYSGSFVQHP	ELTGLDCIRPCF	WVELIRGRPK	ENTIWTSGSSI	SFCGVNSDT	VGWSWPDGA	ELPFTIDK*
A/Bayern/69/2009							
A/Lviv/N6/2009							
A/Hong Kong/3934/2011						V.....	
A/Astrakhan/1/2011							
A/Stockholm/36/2011 se							
A/Christchurch/16/2010							
A/Cameroon/LEID1450/2011							
A/Ethiopia/198/2012 cdc			G.....				
A/Czech Republic/32/2011							
A/St. Petersburg/27/2011							
A/Boras/1/2012 se							
A/Paraguay/293/2012 cdc					V.....		
A/Caen/635/2012		G.....					
A/Stockholm/25/2012		G.....					
A/Norway/560/2012		G.....					
A/Denmark/48/2012		G.....					
A/Hong Kong/6747/2012		K.....			G.....		
A/Georgia/628/2012					G.....		
A/Kazakhstan/2394/2012					G.....		
A/Madagascar/00863/2012							
A/Slovenia/421/2012							
A/Georgia/558/2012							
A/Georgia/427/2012							
A/St. Petersburg/100/2011							
A/Pays-de-Loire/1224/2012							
A/Victoria/514/2012 aus							
A/Paris/1119/2012		K.....					
A/Norway/446/2012							
A/Norway/717/2012							

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site: NA/HA reassortants

Table 8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir)

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹								
			Post infection ferret sera								
			A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/HK 3969/11	A/Stock 18/11	A/Iowa 19/10	A/Fin 190/11	A/Norway 1789/11	A/Vic 361/11
			F35/11	F7/10	F27/10	F27/11	F28/11	F15/11	F01/12	F03/12	F05/12
Genetic group					Group 5	Group 3C	Group 3A	Group 6	Group 3C	Group 3B	Group 3C
REFERENCE VIRUSES											
A/Perth/16/2009	2009-07-04	E3/E1	1280	80	160	640	160	160	320	640	160
A/Victoria/208/2009	2009-06-02	E3/E1	1280	5120	1280	2560	2560	5120	5120	5120	2560
A/Alabama/5/2010	2010-07-13	MK1/M2/SIAT1	40	40	160	320	160	160	160	320	80
A/Hong Kong/3969/2011	2011-05-19	MDCK3	160	320	320	1280	640	640	1280	1280	640
A/Stockholm/18/2011	2011-03-28	MDCK2/SIAT2	80	160	160	640	640	160	320	640	640
A/Iowa/19/2010	2010-12-30	E3/E1	640	5120	1280	2560	2560	5120	2560	5120	2560
A/Finland/190/2011	2011-11-25	Cx/SIAT3	160	320	320	640	640	640	1280	1280	640
A/Norway/1789/2011	2011-08-02	Cx/SIAT3	160	320	320	640	640	640	1280	1280	640
A/Victoria/361/2011	2011-10-24	E3/E1	320	640	640	1280	160	1280	1280	1280	5120
TEST VIRUSES											
A/Ioannina/GR148/2012	2012-02-01	MDCK2	80	160	160	640	640	320	640	640	ND
A/Athens/GR112/2012	2012-02-01	C1/MDCK2	80	160	80	320	320	160	320	640	ND
A/Athens/GR125/2012	2012-02-01	C1/MDCK2	40	80	80	320	320	160	320	320	ND
A/Astrakhan/65/2012	2012-02-02	C3/SIAT1	<	40	40	160	80	80	80	80	40
A/Shkoder/ALB-4101/2012	2012-02-02	SIAT2	80	160	160	320	320	160	640	320	320
A/Athens/GR131/2012	2012-02-02	C1/MDCK2	80	160	160	320	320	160	640	1280	ND
A/Athens/GR134/2012	2012-02-02	C1/MDCK2	80	160	160	640	640	160	640	640	ND
A/Athens/GR165/2012	2012-02-04	MDCK2	80	160	160	640	640	320	640	640	ND
A/Athens/GR166/2012	2012-02-04	MDCK2	80	160	160	640	640	320	640	640	ND
A/Athens/GR208/2012	2012-02-04	SIAT2	80	160	160	640	320	320	1280	640	160
A/Athens/GR173/2012	2012-02-06	MDCK2	80	160	160	640	320	160	640	640	ND
A/Athens/GR237/2012	2012-02-06	SIAT3	160	320	160	1280	2560	640	1280	1280	640
A/Athens/GR193/2012	2012-02-07	SIAT3	40	40	80	320	160	80	320	1280	80
A/Athens/GR209/2012	2012-02-08	SIAT4	80	80	160	320	320	320	640	1280	160
A/Athens/GR248/2012	2012-02-09	SIAT3	80	80	160	320	320	160	640	640	160

1. < = <40; ND = not determined

Sequences in phylogenetic trees

Vaccine
2011-12
2012

Vaccine

Table 9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir)

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
			Post infection ferret sera									
			A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/HK 3969/11	A/Stock 18/11	A/Iowa 19/10	A/Fin 190/11	A/Norway 1789/11	A/Vic 361/11	A/Berlin 93/11
			F35/11	F7/10	F27/10	F27/11	F28/11	F15/11	F01/12	F03/12	Egg F05/12	T/C F11/12
Genetic group					group 5	group 3C	group 3A	group 6	group 3C	group 3C	group 3C	group 3C
REFERENCE VIRUSES												
A/Perth/16/2009	2009-07-04	E3/E2	1280	160	320	640	320	320	160	640	160	640
A/Victoria/208/2009	2009-06-02	E3/E1	1280	5120	1280	5120	2560	5120	5120	5120	5120	5120
A/Alabama/5/2010	2010-07-13	MK1/C2/SIAT2	40	40	160	320	160	160	160	320	80	640
A/Hong Kong/3969/2011	2011-05-19	MDCK2/SIAT4	320	320	320	1280	320	320	1280	640	640	1280
A/Stockholm/18/2011	2011-03-28	MDCK2/SIAT3	40	80	80	320	320	160	160	320	160	640
A/Iowa/19/2010	2010-12-30	E3/E1	1280	5120	1280	2560	2560	5120	5120	5120	2560	5120
A/Finland/190/2011	2011-11-25	Cx/SIAT1	320	320	320	1280	640	320	2560	1280	640	2560
A/Norway/1789/2011	2011-08-02	Cx/SIAT1	320	320	320	1280	640	640	1280	1280	640	2560
A/Victoria/361/2011	2011-10-24	E3/E1	640	2560	320	1280	320	1280	1280	1280	5120	2560
A/Berlin/93/2011	2011-12-07	NVD3/S2	160	640	320	1280	640	640	2560	1280	640	2560
TEST VIRUSES												
A/Bulgaria/182/2012	6	2012-02-02	C3/SIAT1	80	640	320	640	320	640	640	160	1280
A/Bulgaria/217/2012	6	2012-02-09	C3/SIAT1	40	320	80	320	160	160	320	80	640
A/Bulgaria/218/2012	6	2012-02-09	C3/SIAT1	40	320	80	160	160	320	320	80	640
A/Slovenia/402/2012	3C	2012-02-20	Mx/SIAT1	40	160	80	320	160	160	640	160	640
A/Slovenia/435/2012	3C	2012-02-20	Mx/SIAT1	40	160	80	640	160	160	640	80	1280
A/Slovenia/459/2012	3C	2012-02-20	Mx/SIAT1	160	320	320	640	640	2560	1280	320	2560
A/Bulgaria/280/2012	6	2012-02-20	C3/SIAT1	80	160	320	320	320	640	640	160	1280
A/Bulgaria/311/2012	6	2012-02-24	C3/SIAT1	40	320	160	320	160	320	320	80	1280
A/Bulgaria/312/2012	6	2012-02-24	C3/SIAT1	80	320	160	320	160	640	640	160	1280
A/Bulgaria/313/2012	6	2012-02-24	C3/SIAT1	80	320	160	640	320	320	640	160	1280
A/Bulgaria/314/2012	6	2012-02-24	C3/SIAT1	80	320	320	640	320	640	640	160	1280
A/Slovenia/566/2012	3C	2012-02-27	Mx/SIAT1	40	160	80	320	160	320	640	80	640
A/Slovenia/599/2012	3C	2012-02-27	Mx/SIAT1	80	320	320	640	640	1280	1280	320	2560
A/St Petersburg/2/2012	3B	2012-02-27	C3/SIAT1	40	320	80	320	160	160	320	80	640
A/Petrozavodsk/1/2012	3B	2012-02-27	C3/SIAT1	80	320	80	640	160	320	320	80	1280
A/Slovenia/637/2012	3C	2012-03-05	Mx/SIAT1	160	640	320	1280	640	640	1280	320	2560
A/Slovenia/815/2012	3C	2012-03-12	Mx/SIAT1	40	160	80	320	160	160	640	160	1280
A/Slovenia/879/2012	3C	2012-03-19	Mx/SIAT1	160	320	160	1280	640	640	1280	160	2560
A/St Petersburg/4/2012	3B	2012-03-25	C2/SIAT1	80	80	80	320	160	320	640	80	1280
A/Novosibirsk/6/2012	3C	2012-03-25	C1/SIAT1	320	320	320	1280	640	640	1280	640	2560
A/Omsk/1/2012	3C	2012-03-25	C1/SIAT1	80	80	160	640	320	320	640	320	1280
A/St Petersburg/8/2012	3C	2012-03-25	C2/SIAT1	80	160	160	640	320	320	640	320	1280
A/St Petersburg/10/2012	3B	2012-03-25	C2/SIAT1	40	160	80	320	160	320	320	80	640
A/Novosibirsk/8/2012	3C	2012-03-25	C1/SIAT1	160	320	320	1280	320	640	1280	320	2560
A/Slovenia/1049/2012	3C	2012-04-02	Mx/SIAT1	160	320	160	640	320	320	640	160	1280

1. <= <40

Sequences in phylogenetic trees

Vaccine
2011-12
2012

Vaccine

Table 10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir)

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
			Post infection ferret sera									
			A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/HK 3969/11	A/Stock 18/11	A/Iowa 19/10	A/Fin 190/11	A/Norway 1789/11	A/Vic 361/11	A/Berlin 93/11
			F35/11	F7/10	F27/10	F27/11	F28/11	F15/11	F01/12	F03/12	Egg F05/12	T/C F11/12
Genetic group					group 5	group 3C	group 3A	group 6	group 3C	group 3C	group 3C	group 3C
REFERENCE VIRUSES												
A/Perth/16/2009	2009-07-04	E3/E2	1280	80	160	640	160	160	160	320	80	320
A/Victoria/208/2009	2009-06-02	E3/E1	1280	2560	1280	2560	1280	5120	2560	2560	2560	5120
A/Alabama/5/2010	2010-07-13	MK1/C2/SIAT2	40	<	160	320	80	160	80	160	40	160
A/Hong Kong/3969/2011	2011-05-19	MDCK2/SIAT4	160	160	160	640	320	320	640	640	160	1280
A/Stockholm/18/2011	2011-03-28	MDCK2/SIAT3	40	40	80	320	320	160	160	320	80	320
A/Iowa/19/2010	2010-12-30	E3/E1	640	2560	1280	2560	1280	5120	2560	2560	1280	5120
A/Finland/190/2011	2011-11-25	Cx/SIAT1	160	160	320	640	320	320	1280	640	160	1280
A/Norway/1789/2011	2011-08-02	Cx/SIAT1	160	160	160	640	160	320	640	640	160	1280
A/Victoria/361/2011	2011-10-24	E3/E1	320	1280	320	1280	160	1280	1280	1280	5120	1280
A/Berlin/93/2011	2011-12-07	NVD3/S2	160	160	320	1280	320	640	640	1280	320	1280
TEST VIRUSES												
A/Norway/410/2012	2012-02-01	SIAT2/SIAT2	40	80	80	320	160	80	320	320	80	640
A/Norway/433/2012	2012-02-05	SIAT1/SIAT1	160	160	320	1280	640	640	1280	1280	320	1280
A/Slovenia/274/2012	2012-02-06	P1/SIAT1	40	160	160	160	160	160	160	640	80	640
A/Norway/479/2012	2012-02-08	MDCK2/SIAT2	80	80	160	320	160	160	640	320	160	640
A/Norway/596/2012	2012-02-09	SIAT2/SIAT2	80	80	160	320	160	160	320	320	80	640
A/Norway/564/2012	2012-02-10	SIAT1/SIAT2	80	80	160	320	640	160	320	320	80	640
A/Estonia/65266/2012	2012-02-13	MDCK1/SIAT1	320	320	640	1280	640	640	1280	1280	320	2560
A/Norway/528/2012	2012-02-13	SIAT1/SIAT1	80	160	320	640	640	320	640	1280	160	1280
A/Norway/531/2012	2012-02-13	SIAT1/SIAT1	160	160	320	1280	320	640	1280	640	320	1280
A/Slovenia/326/2012	2012-02-13	MDCKx/SIAT2	80	160	160	320	320	160	320	320	80	640
A/Norway/722/2012	2012-02-15	SIAT1/SIAT2	80	80	160	320	160	160	320	320	160	640
A/Norway/648/2012	2012-02-15	SIAT1/SIAT1	160	320	640	1280	640	640	1280	1280	320	5120
A/Norway/606/2012	2012-02-20	SIAT1/SIAT1	60	160	320	640	320	320	640	640	160	1280
A/Norway/624/2012	2012-02-20	SIAT1/SIAT1	160	160	320	640	320	320	640	640	160	1280
A/Norway/625/2012	2012-02-20	SIAT2/SIAT2	80	80	320	320	160	320	320	320	160	640
A/Norway/657/2012	2012-02-22	SIAT2/SIAT2	80	80	320	320	160	320	320	320	160	640
A/Norway/708/2012	2012-02-27	MDCK1/SIAT1	160	320	640	640	640	640	1280	1280	320	2560
A/Estonia/65858/2012	2012-02-28	MDCK1/SIAT2	80	320	320	640	640	640	640	640	160	1280
A/Slovenia/663/2012	2012-03-05	MDCKx/SIAT2	80	160	160	640	320	320	640	640	160	640
A/Slovenia/646/2012	2012-03-05	MDCKx/SIAT3	40	80	160	640	160	160	320	160	40	640
A/Estonia/66073/2012	2012-03-06	MDCK2/SIAT1	40	80	40	80	160	80	320	320	160	640
A/Estonia/66103/2012	2012-03-07	MDCK2/SIAT2	80	160	160	1280	640	320	640	640	160	1280
A/Estonia/66234/2012	2012-03-09	MDCK2/SIAT1	160	160	320	640	320	640	1280	1280	640	1280
A/Estonia/66239/2012	2012-03-09	MDCK2/SIAT2	40	160	80	640	320	160	640	320	80	640
A/Estonia/66240/2012	2012-03-09	MDCK1/SIAT2	40	80	80	160	320	160	160	320	80	640
A/Pančevo/1148/2012	2012-03-13	SIAT3	640	1280	640	1280	640	640	2560	1280	640	1280
A/Cuprija/1214/2012	2012-03-16	SIAT3	80	160	160	640	160	160	640	640	160	1280
A/Slovenia/910/2012	2012-03-19	MDCKx/SIAT2	40	80	160	640	160	160	640	640	80	640
A/Slovenia/913/2012	2012-03-19	MDCKx/SIAT2	40	80	160	640	160	160	320	640	80	640
A/Sabac/1370/2012	2012-03-20	SIAT3	80	80	80	640	160	160	1280	640	160	1280
A/Sabac/1371/2012	2012-03-21	C1/SIAT3	160	320	320	1280	320	320	640	1280	320	1280
A/Estonia/66788/2012	2012-03-27	MDCK1/SIAT1	160	160	320	320	320	640	320	640	320	1280
A/Estonia/66837/2012	2012-03-28	MDCK1/SIAT1	40	40	80	160	80	320	160	320	160	320
A/Glasgow/407581/2012	2012-03-30	SIAT2	80	80	160	320	320	160	320	320	80	640
A/Glasgow/407664/2012	2012-04-03	SIAT2	80	160	160	640	320	320	640	640	80	1280
A/Glasgow/407665/2012	2012-04-03	SIAT2	160	160	320	640	320	320	640	640	160	1280
A/Estonia/67179/2012	2012-04-09	MDCK1/SIAT3	160	320	640	1280	640	640	1280	1280	160	2560
A/St. Petersburg/20/2012	2012-04-09	C3/SIAT1	80	160	80	640	320	160	320	320	160	640
A/Novosibirsk/26/2012	2012-04-09	C1/SIAT1	160	160	160	1280	320	320	640	640	160	1280
A/St. Petersburg/16/2012	2012-04-09	C2/M1/S1	160	320	320	1280	640	640	1280	1280	320	1280
A/Estonia/67238/2012	2012-04-10	MDCK1/SIAT3	160	160	640	1280	640	640	1280	1280	160	1280
A/Estonia/67221/2012	2012-04-11	MDCK1/SIAT2	160	320	640	1280	640	640	1280	640	160	1280
A/Ireland/12M28404/2012	2012-04-11	SIAT3	160	320	160	640	640	320	640	640	160	640
A/St. Petersburg/28/2012	2012-04-19	C2/SIAT1	320	640	640	2560	640	640	1280	1280	320	2560
A/Rostov-on-Don/1/2012	2012-04-19	C1/SIAT1	160	320	320	1280	640	640	1280	1280	320	1280
A/St. Petersburg/29/2012	2012-04-19	C2/SIAT1	160	320	640	2560	2560	640	1280	1280	320	2560
A/Kaliningrad/4/2012	2012-04-19	C1/SIAT1	160	160	320	1280	320	640	1280	1280	160	1280
A/Ekaterinburg/2/2012	2012-04-19	C1/SIAT1	40	80	160	320	160	160	320	320	80	640
A/Ekaterinburg/4/2012	2012-04-19	C2/SIAT1	40	80	80	320	160	160	320	320	160	640

1. <= <40

Vaccine

Vaccine

Sequences in phylogenetic trees

2011-12

2012

Table 11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir)

				Haemagglutination inhibition titre ¹									
Viruses		Collection Date	Passage History	Post infection ferret sera									
				A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/HK 3969/11	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens GR112/12
				F30/09	F7/10	F27/10	F27/11	F28/11	F15/11	Egg F05/12	T/C F17/12	T/C F14/12	F16/12
Genetic group				group 5	group 3C	group 3A	group 6	group 3C	group 3C	group 3C	group 3B		
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E2	640	80	160	640	160	160	160	640	640	
A/Victoria/208/2009		2009-06-02	E3/E1	2560	5120	1280	5120	2560	5120	5120	5120	5120	
A/Alabama/5/2010		2010-07-13	MK1/C2/SIAT2	40	40	160	640	160	160	80	640	640	
A/Hong Kong/3969/2011		2011-05-19	MDCK2/SIAT4	320	320	640	1280	640	640	640	2560	5120	
A/Stockholm/18/2011		2011-03-28	MDCK2/SIAT4	40	80	80	640	320	80	160	640	1280	
A/Iowa/19/2010		2010-12-30	E3/E1	1280	5120	2560	5120	5120	5120	5120	5120	5120	
A/Victoria/361/2011		2011-10-24	E3/E1	160	640	320	640	160	640	5120	1280	640	
A/Berlin/93/2011		2011-12-07	NVD3/S2	160	160	320	1280	640	640	640	1280	2560	
A/Victoria/361/2011		2011-10-24	M2/S2	80	160	160	640	160	320	160	640	1280	
A/Athens/GR112/2012		2012-02-01	SIAT3	80	160	160	640	160	160	160	640	1280	
TEST VIRUSES													
A/Pardubice/7/2012		unknown	MDCK5/SIAT2	160	160	640	1280	320	640	320	2560	2560	
A/Ceska Lipa/81/2012	6	unknown	MDCK3/SIAT2	160	160	640	1280	320	640	320	1280	2560	
A/Jihlava/83/2012	6	unknown	MDCK3/SIAT2	160	160	640	1280	320	640	320	1280	2560	
A/Vsetin/85/2012		unknown	MDCK3/SIAT2	160	320	640	1280	640	640	320	2560	2560	
A/Prerov/87/2012	6	unknown	MDCK3/SIAT2	160	160	320	640	320	320	160	1280	1280	
A/Praha/130/2012	6	unknown	MDCK3/SIAT2	160	320	640	2560	640	1280	640	2560	2560	
A/Ukraine/65/2012		2012-02-16	C1/SIAT1	80	320	320	1280	320	640	320	1280	1280	
A/Ukraine/78/2012	6	2012-02-16	C1/SIAT1	160	640	640	2560	640	1280	640	2560	5120	
A/England/294/2012	3A	2012-02-20	SIAT2/SIAT1	80	160	160	640	160	160	160	640	640	
A/Ukraine/74/2012		2012-02-21	C1/SIAT1	160	160	320	1280	320	640	320	1280	2560	
A/Ukraine/82/2012		2012-02-21	C1/SIAT1	160	160	320	1280	320	640	320	1280	2560	
A/Ukraine/87/2012		2012-02-22	C1/SIAT1	320	640	640	2560	640	1280	640	5120	5120	
A/Ukraine/92/2012		2012-02-24	C1/SIAT1	160	640	640	2560	640	1280	640	2560	5120	
A/England/49/2012	6	2012-02-26	SIAT1/SIAT1	160	320	640	1280	640	640	320	2560	2560	
A/Ukraine/96/2012	6	2012-02-26	C1/SIAT1	320	640	640	2560	640	1280	640	2560	5120	
A/Ukraine/131/2012	5	2012-02-27	C1/SIAT1	160	160	640	2560	640	1280	640	2560	5120	
A/England/316/2012	3C	2012-02-28	SIAT1/SIAT1	160	320	320	1280	640	640	320	2560	2560	
A/Ukraine/116/2012		2012-03-05	C1/SIAT1	160	640	640	2560	640	1280	640	2560	5120	
A/Ukraine/175/2012		2012-03-10	C1/SIAT1	320	1280	640	2560	1280	1280	640	2560	2560	
A/Ukraine/154/2012		2012-03-11	C1/SIAT1	320	1280	640	2560	1280	1280	640	1280	2560	
A/Ukraine/155/2012	6	2012-03-11	C1/SIAT1	320	1280	1280	2560	1280	1280	640	2560	2560	
A/England/392/2012		2012-03-13	SIAT1/SIAT1	80	320	160	640	640	320	160	640	1280	
A/England/476/2012	3B	2012-03-19	SIAT1/SIAT1	160	320	320	1280	640	640	640	2560	2560	
A/England/517/2012	3B	2012-04-05	SIAT1/SIAT1	80	160	160	640	320	320	160	1280	1280	
A/England/531/2012	3C	2012-04-16	MDCK1/SIAT1	40	40	40	320	80	80	80	320	640	
A/Ukraine/1/2012		2012-05-01	C1/SIAT1	80	160	160	640	320	640	160	1280	1280	

1. <= <40

Sequences in phylogenetic trees

Vaccine
2011-12
2012

Vaccine

Vaccine

Table 12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir)

Haemagglutination inhibition titre ¹												
Viruses	Collection Date	Passage History	Post infection ferret sera									
			A/Perth	A/Vic	A/Ala	A/HK	A/Stock	A/Iowa	A/Vic	A/Berlin	A/Vic	A/Athens
			16/09	208/09	5/10	3969/11	18/11	19/10	361/11	93/11	361/11	GR112/12
Genetic group			F35/11	F7/10	F27/10	F27/11	F28/11	F15/11	Egg F05/12	T/C F17/12	T/C F14/12	F16/12
					group 5	group 3C	group 3A	group 6	group 3C	group 3C	group 3C	group 3B
REFERENCE VIRUSES												
A/Perth/16/2009	2009-07-04	E3/E2	640	40	80	320	80	160	80	320	320	320
A/Victoria/208/2009	2009-06-02	E3/E1	640	2560	1280	1280	1280	2560	1280	5120	2560	2560
A/Alabama/5/2010	2010-07-13	MK1/C2/SIAT2	40	<	160	320	80	160	40	640	160	320
A/Hong Kong/3969/2011	2011-05-19	MDCK2/SIAT4	80	80	160	640	160	160	160	1280	640	640
A/Stockholm/18/2011	2011-03-28	MDCK2/SIAT4	40	40	40	320	160	80	80	640	160	320
A/Iowa/19/2010	2010-12-30	E3/E1	640	2560	1280	2560	1280	5120	1280	5120	2560	2560
A/Victoria/361/2011	2011-10-24	E3/E1	160	640	320	640	80	640	2560	1280	320	320
A/Berlin/93/2011	2011-12-07	NVD3/S2	80	160	160	640	160	320	320	1280	640	640
A/Victoria/361/2011	2011-10-24	M2/S2	80	160	160	640	160	320	320	1280	640	640
A/Athens/GR112/2012	2012-02-01	SIAT3	40	80	80	320	160	160	160	640	320	640
TEST VIRUSES												
A/Karlovy Vary/46/2012	unknown	MDCK5/SIAT3	40	80	80	320	80	160	80	640	320	320
A/Decin/42/2012	unknown	MDCK5/SIAT3	40	80	80	320	80	160	80	640	320	320
A/Cercany/96/2012	unknown	MDCK3/SIAT3	40	80	160	320	160	160	160	640	640	320
A/Praha/98/2012	unknown	MDCK3/SIAT3	40	80	80	320	160	160	160	640	320	320
A/Ireland/V1160/2012	3A 2012-02-01	C2/SIAT1	80	80	80	320	320	160	160	640	320	640
A/Belgrade/449/2012	3B 2012-02-02	C1/SIAT1	80	160	160	640	320	160	160	1280	640	640
A/Ukraine/38/2012	3A 2012-02-10	C1/SIAT3	160	320	160	640	640	320	320	2560	640	1280
A/Ukraine/134/2012	2012-02-25	C1/SIAT3	160	160	320	640	320	320	160	1280	640	640
A/England/354/2012	2012-02-28	SIAT2/SIAT2	40	80	80	320	160	160	80	320	160	320
A/Ukraine/114/2012	3B 2012-03-02	C1/SIAT3	80	160	160	320	320	320	320	2560	640	1280
A/Belgrade/898/2012	3B 2012-03-06	C1/SIAT1	40	80	80	320	160	160	80	640	320	640
A/Cuprija/1212/2012	3B 2012-03-15	C1/SIAT1	40	80	80	640	160	160	160	640	320	640
A/Ireland/12V2608/2012	2012-03-22	C1/SIAT1	40	80	80	320	320	160	160	640	320	640
A/Ireland/12V2628/2012	3C 2012-03-22	C1/SIAT1	160	160	320	1280	320	640	320	2560	1280	1280
A/Ireland/12V2734/2012	2012-03-26	C1/SIAT1	40	80	80	640	320	160	160	640	320	640
A/Nis/1471/2012	3A 2012-03-26	C1/SIAT1	80	80	160	640	320	320	160	1280	640	640
A/Nis/1472/2012	3A 2012-03-26	C1/SIAT1	80	160	160	640	320	320	160	1280	640	640
A/Belgrade/1420/2012	3C 2012-03-26	C1/SIAT1	80	160	160	640	320	320	160	2560	640	640
A/Belgrade/1470/2012	6 2012-03-27	C1/SIAT2	80	160	320	640	320	320	160	2560	640	640
A/Ireland/12V2794/2012	3B 2012-03-28	C1/SIAT1	40	40	40	320	160	80	160	640	160	640
A/Scotland/1/2012	2012-03-30	SIAT3/SIAT2	<	40	40	160	160	40	80	320	80	160
A/Belgrade/1545/2012	3B 2012-03-30	C1/SIAT1	80	80	80	320	160	160	160	1280	320	640
A/Beograd/1549/2012	6 2012-04-02	C1/SIAT1	80	160	160	640	160	320	160	1280	640	640
A/England/509/2012	2012-04-04	SIAT1/SIAT2	40	80	80	320	160	160	160	640	640	640
A/Beograd/1607/2012	6 2012-04-04	C1/SIAT1	160	320	320	1280	320	640	320	2560	1280	640
A/Leskovac/1701/2012	5 2012-04-05	C1/SIAT1	160	320	320	1280	320	640	320	2560	640	640
A/Belgrade/1741/2012	3C 2012-04-11	C1/SIAT1	160	320	320	1280	640	640	640	2560	1280	1280
A/England/524/2012	2012-04-12	SIAT1/SIAT3	40	80	160	160	320	160	80	640	320	320
A/Sabac/1930/2012	2012-04-24	C1/SIAT1	160	160	320	1280	320	320	320	2560	1280	640

1. <= <40

Sequences in phylogenetic trees

Vaccine
2011-12
2012

Vaccine

Vaccine

Table 13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir)

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
			Post infection ferret sera									
			A/Perth 16/09 F30/09	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10	A/HK 3969/11 F27/11	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 Egg F05/12	A/Berlin 93/11 T/C F17/12	A/Vic 361/11 T/C F15/12	A/Athens GR112/12 F16/12
Genetic group			group 5	group 3C	group 3A	group 6	group 3C	group 3C	group 3C	group 3C	group 3B	
REFERENCE VIRUSES												
A/Perth/16/2009	2009-07-04	E3/E2	640	80	160	640	160	160	160	640	320	640
A/Victoria/208/2009	2009-06-02	E3/E1	1280	5120	1280	5120	2560	5120	2560	5120	2560	5120
A/Alabama/5/2010	2010-07-13	MK1/C2/SIAT2	40	40	160	320	160	160	40	640	160	320
A/Hong Kong/3969/2011	2011-05-19	MDCK2/SIAT4	160	160	160	640	320	320	320	1280	640	640
A/Stockholm/18/2011	2011-03-28	MDCK2/SIAT4	80	80	40	320	320	160	160	640	160	320
A/Iowa/19/2010	2010-12-30	E3/E1	1280	5120	2560	5120	2560	5120	2560	5120	5120	5120
A/Victoria/361/2011	2011-10-24	E3/E2	320	1280	320	1280	160	640	5120	1280	320	160
A/Berlin/93/2011	2011-12-07	NVD3/S2	80	160	160	640	160	320	320	1280	640	640
A/Victoria/361/2011	2011-10-24	M2/S2	80	160	160	640	320	160	160	1280	320	640
A/Athens/GR112/2012	2012-02-01	SIAT4	80	160	160	640	320	160	320	1280	320	640
TEST VIRUSES												
A/Ukraine/5338/2012	2012-02-22	C2/SIAT1	160	320	320	1280	640	640	320	1280	1280	1280
A/Ukraine/5339/2012	2012-02-22	C2/SIAT1	160	320	320	1280	320	320	320	1280	640	640
A/Ukraine/5329/2012	2012-03-01	C1/SIAT1	160	320	320	1280	320	320	160	1280	640	640
A/Ukraine/5330/2012	2012-03-01	C1/SIAT1	80	160	160	640	160	320	160	640	640	640
A/Ukraine/5333/2012	2012-03-03	C1/SIAT1	160	320	320	1280	320	320	160	1280	640	1280
A/Ukraine/261/2012	2012-03-04	C1/SIAT1	80	80	160	320	160	160	160	320	320	320
A/Ukraine/5294/2012	2012-03-05	C1/SIAT1	160	320	320	1280	320	640	320	1280	1280	1280
A/Ukraine/5334/2012	2012-03-06	C1/SIAT1	80	160	320	640	160	320	160	640	640	640
A/Ukraine/5359/2012	2012-03-07	C1/SIAT1	160	320	640	1280	640	640	320	2560	1280	1280
A/Ukraine/5365/2012	2012-03-07	C2/SIAT1	80	160	320	640	320	320	160	1280	640	640
A/Ukraine/5366/2012	2012-03-07	C2/SIAT1	160	320	320	640	320	640	160	1280	640	1280
A/Ukraine/189/2012	2012-03-09	C1/SIAT1	40	80	160	320	160	160	80	640	320	320
A/Odessa/303/2012	2012-03-11	C1/SIAT1	40	80	160	320	160	160	80	640	320	320
A/Ukraine/5331/2012	2012-03-11	C1/SIAT1	80	160	320	640	160	320	160	640	640	640
A/Ukraine/184/2012	2012-03-12	C1/SIAT1	160	160	320	1280	320	320	320	2560	640	1280
A/Ukraine/269/2012	2012-03-12	C1/SIAT1	40	160	160	320	320	160	160	640	320	640
A/Ukraine/5300/2012	2012-03-12	C1/SIAT1	80	160	320	640	320	320	160	1280	640	640
A/Ukraine/5301/2012	2012-03-12	C1/SIAT1	80	160	160	640	160	160	80	640	640	640
A/Ukraine/5355/2012	2012-03-12	SIAT1	40	160	160	640	320	160	160	640	320	640
A/Odessa/309/2012	2012-03-14	C1/SIAT1	<	40	40	160	80	80	80	320	160	320
A/Dnipro/153/2012	2012-03-14	C2/SIAT1	320	320	640	1280	320	640	320	2560	1280	1280
A/Dnipro/155/2012	2012-03-14	C2/SIAT1	160	320	640	1280	640	640	320	2560	1280	1280
A/Ukraine/5358/2012	2012-03-14	C1/SIAT1	160	320	320	1280	320	640	320	1280	640	1280
A/Ukraine/5357/2012	2012-03-15	C1/SIAT1	160	320	320	1280	320	640	320	1280	640	1280
A/Ukraine/5380/2012	2012-03-15	C2/SIAT1	80	160	160	640	320	160	160	640	320	640
A/Ukraine/5381/2012	2012-03-15	C2/SIAT1	320	640	640	2560	640	1280	320	2560	1280	1280
A/Ukraine/279/2012	2012-03-16	C1/SIAT1	80	160	160	320	160	320	160	640	320	320
A/Ukraine/280/2012	2012-03-16	C1/SIAT1	80	160	320	640	320	320	160	1280	320	640
A/Odessa/288/2012	2012-03-16	C1/SIAT1	40	80	160	320	160	160	80	640	320	320
A/Ukraine/202/2012	2012-03-18	C1/SIAT1	40	80	160	320	160	160	80	640	320	320
A/Ukraine/5340/2012	2012-03-18	C1/SIAT1	80	160	160	640	320	160	160	1280	320	640
A/Ukraine/5341/2012	2012-03-18	C1/SIAT1	80	320	160	640	640	320	320	1280	640	1280
A/Ukraine/5383/2012	2012-03-18	C2/SIAT1	160	80	160	640	160	320	160	640	160	640
A/Odessa/289/2012	2012-03-19	C1/SIAT1	40	80	80	320	80	160	80	640	160	320
A/Ukraine/5342/2012	2012-03-20	C1/SIAT1	80	160	320	640	320	320	160	1280	640	640
A/Odessa/292/2012	2012-03-21	C1/SIAT1	40	40	80	320	80	160	80	640	160	320
A/Dnipro/182/2012	2012-03-22	C2/SIAT1	80	160	160	640	320	320	160	1280	640	640
A/S. Mitrovica/1387/2012	2012-03-22	C1/SIAT1	160	160	320	1280	320	320	160	1280	640	640
A/Ukraine/236/2012	2012-03-23	C1/SIAT1	80	160	160	640	320	640	160	640	640	640
A/Dnipro/194/2012	2012-03-23	C2/SIAT1	160	320	320	1280	320	640	320	1280	640	640
A/Ukraine/5375/2012	2012-03-23	C1/SIAT1	160	320	320	1280	320	640	320	1280	640	1280
A/Ukraine/5385/2012	2012-03-23	C1/SIAT1	160	320	320	1280	320	640	320	1280	640	640
A/Ukraine/223/2011	2012-03-24	C1/SIAT1	40	80	160	320	160	160	160	640	320	320
A/Ukraine/5382/2012	2012-03-25	C2/SIAT1	80	160	160	640	160	320	160	640	320	640
A/Ukraine/5384/2012	2012-03-25	C2/SIAT1	160	320	320	640	320	640	320	1280	1280	1280
A/Ukraine/225/2012	2012-03-26	C1/SIAT1	80	160	320	640	320	640	160	1280	640	640
A/Ukraine/228/2012	2012-03-26	C1/SIAT1	40	80	320	320	160	320	160	640	320	320
A/Odessa/300/2012	2012-03-28	C1/SIAT1	80	160	80	640	320	160	160	1280	320	640
A/Odessa/311/2012	2012-03-29	C1/SIAT1	40	40	80	160	80	80	80	320	160	160
A/Ukraine/5378/2012	2012-03-29	C1/SIAT1	320	320	640	1280	640	640	320	2560	1280	1280
A/Odessa/312/2012	2012-03-30	C1/SIAT1	40	80	160	320	160	160	80	640	320	320
A/Belgrade/1544/2012	2012-03-30	C1/SIAT1	160	320	320	1280	320	640	320	2560	640	1280
A/Ukraine/246/2012	2012-03-31	C1/SIAT1	40	80	160	320	160	160	80	640	320	320
A/Ukraine/239/2012	2012-04-01	C1/SIAT1	80	160	320	640	320	640	160	2560	1280	640
A/Ukraine/259/2012	2012-04-03	C1/SIAT1	40	40	40	160	80	40	40	320	160	160
A/Ireland/12M28563/2012	2012-04-10	SIAT4	160	160	320	1280	640	320	320	2560	640	1280
A/Ukraine/250/2012	2012-04-10	C1/SIAT1	40	80	160	320	160	160	80	640	320	320

1. <= <40

Sequences in phylogenetic trees

Vaccine
2011-12
2012

Vaccine

Vaccine

Table 14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir)

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
			Post infection ferret sera									
			A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/HK 3969/11	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens GR112/12
			F30/09	F7/10	F27/10	F27/11	F28/11	F15/11	Egg F05/12	T/C F17/12	T/C F15/12	F16/12
Genetic group				group 5	group 3C	group 3A	group 6	group 3C	group 3C	group 3C	group 3B	
REFERENCE VIRUSES												
A/Perth/16/2009	2009-07-04	E3/E2	640	80	320	640	160	320	160	640	640	640
A/Victoria/208/2009	2009-06-02	E3/E1	1280	2560	1280	5120	2560	5120	1280	5120	2560	5120
A/Alabama/5/2010	2010-07-13	MK1/C2/SIAT2	40	<	160	320	80	160	40	640	160	320
A/Hong Kong/3969/2011	2011-05-19	MDCK2/SIAT4	160	160	320	1280	640	320	320	2560	1280	1280
A/Stockholm/18/2011	2011-03-28	MDCK2/SIAT4	40	40	80	320	160	80	160	640	160	320
A/Iowa/19/2010	2010-12-30	E3/E1	1280	2560	2560	5120	1280	5120	1280	5120	5120	5120
A/Victoria/361/2011	2011-10-24	E3/E2	640	1280	640	1280	160	1280	2560	2560	640	640
A/Berlin/93/2011	2011-12-07	NVD3/S2	160	320	320	1280	320	640	640	2560	640	1280
A/Victoria/361/2011	2011-10-24	M2/S2	160	160	160	640	320	320	320	1280	640	640
A/Athens/GR112/2012	2012-02-01	SIAT4	160	160	160	640	320	320	320	1280	640	1280
TEST VIRUSES												
A/Denmark/3/2012	3B 2012-02-06	SIAT1/MDCK2/SIAT1	40	160	80	640	320	160	160	1280	320	640
A/Denmark/6/2012	3C 2012-02-07	SIAT1/MDCK2/SIAT1	40	80	80	320	160	160	160	640	320	320
A/Denmark/5/2012	2012-02-07	SIAT1/MDCK2/SIAT1	160	320	640	1280	320	640	320	2560	640	1280
A/Denmark/12/2012	2012-02-14	SIAT1/MDCK2/SIAT1	40	80	80	160	160	80	80	640	160	160
A/Ukraine/5332/2012	2012-03-02	C1/SIAT2	320	640	640	1280	640	1280	640	5120	1280	2560
A/Ukraine/5393/2012	2012-03-06	C2/SIAT1	160	160	320	1280	320	640	320	1280	640	1280
A/Ukraine/5401/2012	2012-03-06	C1/SIAT1	160	160	320	640	320	640	320	1280	640	1280
A/Ukraine/5402/2012	2012-03-07	C1/SIAT1	160	160	320	640	320	640	320	2560	640	1280
A/Denmark/24/2012	2012-03-11	SIAT1/MDCK1/SIAT1	80	320	160	640	640	320	320	1280	640	1280
A/Denmark/5422/2012	3A 2012-03-13	C3/SIAT1	40	80	80	320	160	160	160	640	320	320
A/Denmark/25/2012	2012-03-13	SIAT1/MDCK2/SIAT1	80	160	160	640	320	160	160	1280	640	640
A/Denmark/28/2012	2012-03-14	SIAT1/MDCK1/SIAT1	80	160	160	640	160	160	160	1280	320	640
A/Ukraine/5403/2012	6 2012-03-16	C1/SIAT1	40	80	160	320	160	160	160	640	320	320
A/Ukraine/5397/2012	2012-03-18	C2/SIAT1	160	320	640	1280	320	640	320	1280	1280	1280
A/Denmark/26/2012	2012-03-18	SIAT1/MDCK3/SIAT1	80	160	160	640	160	160	160	1280	640	640
A/Ukraine/5404/2012	2012-03-19	C1/SIAT1	40	80	160	320	160	160	160	640	320	320
A/Ukraine/5405/2012	2012-03-19	C1/SIAT1	80	160	320	640	320	320	320	1280	640	1280
A/Ukraine/5406/2012	2012-03-19	C1/SIAT1	40	80	160	320	160	160	160	640	320	640
A/Ukraine/5394/2012	2012-03-20	C2/SIAT1	160	320	640	1280	320	640	320	2560	1280	1280
A/Ukraine/5387/2012	2012-03-21	C2/SIAT1	160	160	320	640	160	320	160	1280	640	640
A/Ukraine/5389/2012	2012-03-21	C2/SIAT1	80	80	160	320	80	160	80	640	640	320
A/Ukraine/5391/2012	2012-03-21	C2/SIAT1	80	80	160	640	160	320	160	1280	640	640
A/Denmark/31/2012	2012-03-21	SIAT1/MDCK2/SIAT1	160	320	320	640	640	320	320	2560	640	1280
A/Ukraine/5388/2012	2012-03-22	C2/SIAT1	160	160	320	1280	160	640	320	1280	640	1280
A/Ukraine/5392/2012	6 2012-03-22	C2/SIAT1	320	640	640	2560	640	1280	640	5120	2560	2560
A/Ukraine/5399/2012	2012-03-22	C2/SIAT1	80	160	320	640	320	320	320	1280	640	640
A/Ukraine/5419/2012	2012-03-22	C1/SIAT1	160	160	160	640	640	320	320	2560	640	1280
A/Ukraine/5390/2012	2012-03-23	C2/SIAT1	160	160	320	1280	160	640	320	2560	640	1280
A/Ukraine/5379/2012	2012-03-25	SIAT2	160	320	320	1280	320	640	320	2560	640	1280
A/Ukraine/5386/2012	2012-03-25	C2/SIAT1	160	160	320	1280	320	640	320	2560	640	1280
A/Denmark/36/2012	3A 2012-03-26	SIAT1/MDCK2/SIAT1	320	640	640	1280	1280	640	640	5120	1280	2560
A/Ukraine/5417/2012	2012-03-27	C3/SIAT1	160	160	320	640	160	320	160	1280	640	640
A/Ukraine/5418/2012	2012-03-27	C3/SIAT1	80	80	160	320	160	160	80	640	320	320
A/Ukraine/5377/2012	2012-03-28	C1/SIAT2	40	320	320	1280	320	640	320	2560	640	1280
A/Ukraine/5421/2012	2012-03-29	C3/SIAT1	80	160	160	640	160	320	160	1280	640	640
A/Ukraine/5398/2012	6 2012-04-01	C1/SIAT1	80	160	320	640	320	320	320	1280	640	640
A/Ukraine/5400/2012	2012-04-01	C1/SIAT1	80	160	320	640	160	320	160	1280	640	640
A/Ukraine/5410/2012	2012-04-01	C1/SIAT1	2560	320	320	640	320	640	320	2560	1280	1280
A/Ukraine/5408/2012	6 2012-04-03	C1/SIAT1	320	320	640	1280	1280	1280	640	2560	1280	2560
A/Ukraine/5409/2012	2012-04-03	C1/SIAT1	80	320	640	640	320	640	320	2560	1280	1280
A/Ukraine/5395/2012	2012-04-04	C1/SIAT1	80	160	320	640	320	320	160	1280	640	640
A/Ukraine/5411/2012	2012-04-04	C1/SIAT1	640	320	320	640	320	640	320	2560	640	1280
A/Ukraine/5420/2012	2012-04-04	C2/SIAT1	80	160	160	640	160	320	320	1280	640	640
A/Ukraine/5396/2012	2012-04-05	C1/SIAT1	160	320	320	2560	320	640	320	2560	1280	1280
A/Ukraine/5429/2012	2012-04-10	C1/SIAT1	80	160	80	640	320	160	160	1280	320	640
A/Denmark/41/2012	2012-04-10	SIAT1/MDCK2/SIAT1	80	320	320	1280	640	320	320	2560	640	640
A/Ukraine/5425/2012	2012-04-12	C1/SIAT1	160	160	320	640	320	640	320	1280	640	640
A/Ukraine/5426/2012	5 2012-04-16	C1/SIAT1	40	40	80	320	80	160	80	320	160	320
A/Ukraine/5427/2012	2012-04-18	SIAT2	80	40	80	320	160	160	80	640	160	320
A/Ukraine/5428/2012	3A 2012-04-18	SIAT2	80	160	80	640	320	320	160	1280	320	6340
A/Denmark/42/2012	6 2012-04-21	SIAT1/MDCK1/SIAT1	160	160	320	640	320	320	320	2560	640	640
A/Denmark/45/2012	3C 2012-04-24	SIAT1/MDCK2/SIAT1	640	1280	1280	2560	1280	1280	1280	5120	2560	2560

1. < = <40

Vaccine

2011-12

2012

Sequences in phylogenetic trees

Vaccine

Vaccine

Table 15. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir)

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
				Post infection ferret sera									
				A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/HK 3969/11	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens GR112/12
				F35/11	F7/10	F27/10	F27/11	F28/11	F15/11	Egg F05/12	T/C F11/12	T/C F15/12	F16/12
					group 5	group 3C	group 3A	group 6	group 3C	group 3C	group 3C	group 3B	
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E2	1280	80	320	1280	320	320	320	640	640	1280
A/Victoria/208/2009		2009-06-02	E3/E1	1280	5120	2560	5120	2560	5120	5120	5120	5120	5120
A/Alabama/5/2010		2010-07-13	MK1/C2/SIAT2	40	40	160	640	160	160	80	320	160	640
A/Hong Kong/3969/2011		2011-05-19	MDCK2/SIAT4	80	160	320	1280	320	320	640	1280	2560	1280
A/Stockholm/18/2011		2011-03-28	MDCK2/SIAT4	40	40	80	640	160	80	160	320	160	640
A/Iowa/19/2010		2010-12-30	E3/E1	640	5120	1280	5120	1280	5120	2560	5120	5120	5120
A/Victoria/361/2011		2011-10-24	E3/E2	320	640	320	1280	160	640	5120	640	640	320
A/Berlin/93/2011		2011-12-07	NVD3/SI2	160	160	320	1280	320	320	320	1280	640	1280
A/Victoria/361/2011		2011-10-24	M2/SI2	160	160	320	1280	320	320	320	1280	640	1280
A/Athens/GR112/2012		2012-02-01	SIAT4	80	160	320	1280	320	320	320	1280	640	1280
TEST VIRUSES													
A/Milano/136/2012	3A	2012-02-06	SIAT1/SIAT1	40	160	160	320	320	160	160	640	320	640
A/Lisboa/23/2012		2012-02-07	SIAT1/SIAT1	40	80	160	160	160	160	160	320	320	160
A/Iceland/10/2012		2012-02-08	MDCK2/SIAT1	40	80	160	640	160	160	160	640	320	640
A/Iceland/11/2012		2012-02-08	MDCK3/SIAT1	<	40	80	80	80	80	80	320	160	320
A/Georgia/161/2012	3B	2012-02-08	SIAT2	160	320	160	640	640	320	320	1280	640	640
A/Iceland/12/2012		2012-02-09	MDCK2/SIAT1	40	160	160	160	320	160	160	1280	640	1280
A/Iceland/13/2012	3B	2012-02-09	MDCK2/SIAT1	40	80	160	160	320	160	160	1280	320	1280
A/Iceland/14/2012		2012-02-12	MDCK2/SIAT1	40	80	80	160	320	<	160	1280	320	1280
A/Iceland/15/2012		2012-02-13	MDCK1/SIAT1	40	80	80	160	160	80	160	640	320	640
A/Iceland/16/2012		2012-02-13	MDCK2/SIAT1	40	160	80	160	320	160	160	640	320	640
A/Iceland/17/2012	6	2012-02-17	MDCK2/SIAT1	320	320	640	2560	320	1280	640	2560	640	1280
A/Parma/139/2012	6	2012-02-20	MDCK2/SIAT1	160	320	640	1280	640	640	320	1280	640	1280
A/Georgia/255/2012	3C	2012-02-22	SIAT2	80	160	160	640	320	160	160	640	320	640
A/Perugia/21/2012		2012-02-22	MDCK2/SIAT1	40	160	320	640	320	320	160	640	640	640
A/Iceland/18/2012	6	2012-02-23	MDCK3/SIAT1	<	40	80	320	80	80	80	320	160	320
A/Georgia/215/2012		2012-02-23	SIAT2	80	160	160	640	320	160	160	640	320	640
A/Iceland/19/2012		2012-02-24	MDCK2/SIAT1	40	80	160	320	160	160	160	640	160	640
A/Iceland/20/2012		2012-02-27	MDCK2/SIAT1	160	160	160	320	160	160	320	640	160	1280
A/Iceland/21/2012		2012-03-02	MDCK2/SIAT1	80	160	320	640	320	320	160	1280	320	1280
A/Denmark/38/2012		2012-03-05	SIAT2	160	320	640	1280	320	640	320	1280	1280	2560
A/Lisboa/61/2012		2012-03-05	SIAT1/SIAT1	40	160	320	640	320	320	320	640	640	640
A/Georgia/370/2012	3B	2012-03-06	SIAT2	80	320	160	640	640	320	320	640	640	1280
A/Minsk/1574/2012	3B	2012-03-12	MDCK2/SIAT1	80	320	320	1280	320	1280	320	1280	640	1280
A/Montenegro/158/2012	3C	2012-03-13	SIAT3	320	640	640	2560	1280	2560	1280	2560	2560	2560
A/Gomel/1613/2012		2012-03-14	MDCK2/SIAT1	80	160	160	320	320	320	160	640	320	320
A/Lisboa/38/2012		2012-03-15	SIAT1/SIAT1	<	80	160	320	160	160	160	640	320	320
A/Montenegro/190/2012	3C	2012-03-16	SIAT2	80	160	160	640	160	320	320	640	640	640
A/Stockholm/14/2012	3C	2012-03-16	C2/SIAT1	80	160	160	1280	320	320	320	640	640	640
A/Molodechno/1598/2012	3C	2012-03-19	MDCK2/SIAT1	80	160	160	6340	320	320	320	1280	640	640
A/Lisboa/58/2012		2012-03-20	SIAT1/SIAT1	<	80	160	320	160	160	160	320	320	320
A/Georgia/512/2012	3C	2012-03-23	SIAT2	320	320	640	1280	1280	640	640	1280	1280	1280
A/Montenegro/269/2012	3C	2012-03-24	SIAT2	40	160	160	640	320	320	160	640	320	640
A/Montenegro/272/2012	3C	2012-03-24	SIAT2	40	80	160	320	160	160	160	640	320	320
A/Iceland/23/2012		2012-03-27	MDCK3/SIAT1	80	80	160	320	160	320	160	640	160	640
A/Lisboa/59/2012	6	2012-03-27	SIAT1/SIAT1	<	80	160	320	160	160	160	320	320	320
A/Gomel/1692/2012	3B	2012-03-27	MDCK2/SIAT3	40	80	80	640	160	80	160	640	320	640
A/Georgia/554/2012	3C	2012-03-27	SIAT2	80	320	320	1280	640	640	640	1280	1280	1280
A/Minsk/1737/2012	3C	2012-03-28	MDCK2/SIAT1	80	160	160	640	320	320	320	1280	640	640
A/Montenegro/312/2012		2012-03-29	SIAT2	80	80	160	160	80	160	80	320	160	320
A/Georgia/579/2012	3C	2012-04-04	SIAT2	80	320	320	1280	640	640	640	1280	1280	1280
A/Iceland/24/2012	3B	2012-04-14	MDCK2/SIAT1	<	<	<	80	40	40	40	160	40	160
A/Stockholm/19/2012	5	2012-04-14	C1/SIAT1	80	160	320	1280	320	320	320	640	640	1280
A/Stockholm/21/2012		2012-04-21	C2/SIAT1	160	160	320	1280	320	640	320	1280	640	1280

1. <= <40

Sequences in phylogenetic trees

Vaccine
2011-12
2012

Vaccine

Vaccine

Table 16. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir)

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹											
			Post infection ferret sera											
			A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/HK 3969/11	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens GR112/12	A/Glas 407664/12	A/Glas 407664/12
			F35/11	F7/10	F27/10	F27/11	F28/11	F15/11	Egg F05/12	T/C F11/12	T/C F15/12	F16/12	T/C F18/12	Egg F19/12
Genetic group					group 5	group 3C	group 3A	group 6	group 3C	group 3C	group 3C	group 3B	group 3C	group 3C
REFERENCE VIRUSES														
A/Perth/16/2009	2009-07-04	E3/E2	1280	80	320	640	160	320	160	320	640	640	160	160
A/Victoria/208/2009	2009-06-02	E3/E1	1280	5120	2560	5120	2560	2560	5120	5120	5120	5120	1280	640
A/Alabama/5/2010	2010-07-13	MK1/C2/SIAT2	<	<	80	160	80	80	40	160	80	160	<	<
A/Hong Kong/3969/2011	2011-05-19	MDCK2/SIAT4	80	80	160	640	320	160	160	640	320	640	160	160
A/Stockholm/18/2011	2011-03-28	MDCK2/SIAT4	40	80	80	320	320	80	160	320	160	640	80	40
A/Iowa/19/2010	2010-12-30	E3/E1	320	1280	640	1280	1280	1280	640	1280	1280	640	320	320
A/Victoria/361/2011	2011-10-24	E3/E2	320	1280	320	1280	160	640	5120	640	320	320	320	320
A/Berlin/93/2011	2011-12-07	NVD3/S2	160	160	320	1280	640	320	320	640	640	1280	320	320
A/Victoria/361/2011	2011-10-24	M2/S2	80	80	160	640	320	160	320	640	640	640	160	80
A/Athens/GR112/2012	2012-02-01	SIAT4	80	160	160	640	320	160	320	640	320	1280	160	80
A/Glasgow/407664/2012	2012-04-03	NVD3/S1	160	160	320	1280	320	320	320	640	640	1280	320	320
A/Glasgow/407664/2012	2012-04-03	NVD2/E4	320	80	1280	1280	640	640	320	640	1280	1280	640	2560
TEST VIRUSES														
A/Finland/227/2012	2012-02-06	SIAT2/SIAT1	40	80	160	320	160	320	160	320	320	320	80	80
A/Finland/252/2012	2012-02-16	SIAT2/SIAT1	160	320	320	1280	640	640	320	1280	1280	1280	320	320
A/Pavia/128/2012	2012-02-17	MDCK2/SIAT1	80	160	320	640	320	160	160	640	640	640	160	160
A/Pavia/158/2012	2012-02-27	MDCK2/SIAT1	80	160	160	640	640	320	320	640	320	640	80	80
A/Perugia/44/2012	2012-03-05	MDCK2/SIAT2	40	80	80	320	320	80	160	320	320	320	80	80
A/Finland/263/2012	2012-03-07	SIAT2/SIAT1	40	80	80	320	320	160	160	320	160	640	80	80
A/Montenegro/282/2012	2012-03-24	SIAT2	160	160	320	1280	640	320	640	1280	640	1280	320	320
A/Montenegro/283/2012	2012-03-24	SIAT2	160	160	320	1280	640	320	320	1280	640	1280	320	320

1. < = 40

Sequence in phylogenetic trees

Vaccine
2011-12
2012

Vaccine

Vaccine

Table 17. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir)

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
				Post infection ferret sera									
				A/Perth 16/09 F35/11	A/Vic 208/09 F7/10	A/Ala 5/10 F27/10	A/HK 3969/11 F27/11	A/Stock 18/11 F28/11	A/Iowa 19/10 F15/11	A/Vic 361/11 Egg F05/12	A/Berlin 93/11 T/C F11/12	A/Vic 361/11 T/C F15/12	A/Athens GR112/12 F16/12
				group 5	group 3C	group 3A	group 6	group 3C	group 3C	group 3C	group 3B		
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E2	640	80	160	640	160	160	160	320	320	640
A/Victoria/208/2009		2009-06-02	E3/E1	640	2560	1280	2560	1280	2560	2560	5120	2560	2560
A/Alabama/5/2010		2010-07-13	MK1/C2/SIAT2	<	<	80	320	80	80	40	160	160	320
A/Hong Kong/3969/2011		2011-05-19	MDCK2/SIAT4	80	80	160	640	160	160	160	640	640	640
A/Stockholm/18/2011		2011-03-28	MDCK2/SIAT6	40	40	80	320	160	80	160	320	160	320
A/Iowa/19/2010		2010-12-30	E3/E2	320	1280	640	2560	1280	2560	1280	1280	1280	1280
A/Victoria/361/2011		2011-10-24	E3/E2	320	640	320	1280	160	640	5120	640	320	320
A/Berlin/93/2011		2011-12-07	NVD3/S2	80	160	160	1280	160	320	320	640	640	640
A/Victoria/361/2011		2011-10-24	M2/S2	80	80	160	640	320	160	320	640	640	640
A/Athens/GR112/2012		2012-02-01	SIAT4	80	160	160	320	320	160	320	640	640	640
TEST VIRUSES													
A/Georgia/301/2012	3B	2012-02-21		80	80	80	320	320	80	160	320	320	320
A/Centre/848/2012	3B	2012-03-01	MDCK2/SIAT1	40	80	160	320	320	160	160	320	320	640
A/Bourgogne/850/2012	3B	2012-03-01	MDCK2/SIAT1	80	160	160	320	320	160	160	320	320	640
A/Caen/900/2012	6	2012-03-01	MDCK2/SIAT1	80	160	320	320	320	320	160	320	640	640
A/Pays de Loire/882/2012	3C	2012-03-02	MDCK1/SIAT1	80	80	160	640	160	160	160	640	320	640
A/Paris/919/2012	3B	2012-03-06	MDCK2/SIAT1	40	80	80	320	160	160	160	320	320	320
A/Centre/929/2012	3C	2012-03-07	MDCK2/SIAT1	80	160	160	640	320	320	160	640	320	640
A/Paris/972/2012	3B	2012-03-08	MDCK2/SIAT1	<	<	<	80	80	40	80	160	80	160
A/Guyane/1296/2012 (A/Guadeloupe/44/2012)	6	2012-03-08	MDCK2/SIAT1	40	80	160	320	320	160	160	320	320	640
A/Guyane/1297/2012 (A/Guadeloupe/45/2012)	3C	2012-03-08	MDCK2/SIAT1	40	80	80	320	160	160	160	320	320	320
A/Madagascar/03243/2012		2012-05-07	SIAT3	<	40	40	160	80	40	80	160	160	160
A/Madagascar/03233/2012	3C	2012-05-11	MDCK1/SIAT1	160	160	320	640	320	320	320	640	320	640
A/Madagascar/03281/2012		2012-05-14	SIAT2	80	80	160	320	160	160	160	640	320	640
A/Madagascar/03294/2012	3C	2012-05-16	MDCK2/SIAT1	160	160	160	640	640	320	320	640	640	640
A/Madagascar/03396/2012	3B	2012-05-21	MDCK2/SIAT1	40	80	80	320	320	80	160	320	320	320
A/Madagascar/03401/2012	3B	2012-05-22	MDCK2/SIAT1	80	80	80	320	320	160	160	640	320	320
A/Madagascar/03408/2012	3B	2012-05-22	MDCK2/SIAT1	40	80	80	320	320	160	160	320	320	320
A/Ghana/FS0677/2012	3A	2012-05-28	C2/SIAT3	<	40	40	160	80	40	80	160	160	160
A/Madagascar/03449/2012		2012-05-29	SIAT3	80	80	1602	640	320	160	160	640	640	640
A/Madagascar/03487/2012	3B	2012-05-31	MDCK2/SIAT1	<	80	80	320	160	80	160	320	320	320
A/Madagascar/03536/2012	3C	2012-06-04	MDCK1/SIAT1	160	160	320	1280	640	320	320	1280	1280	1280
A/Madagascar/03535/2012	3C	2012-06-06	SIAT2	40	80	160	320	160	160	160	320	320	640
A/Hong Kong/8117/2012	3C	2012-06-26	MDCK2/SIAT1	40	80	80	320	320	80	160	320	320	320
A/Hong Kong/8119/2012	3C	2012-06-26	MDCK2/SIAT1	40	80	80	320	320	160	160	320	320	640
A/Hong Kong/8152/2012	3C	2012-06-26	MDCK2/MDCK1	40	80	80	320	320	80	160	320	160	320
A/Hong Kong/8150/2012	3C	2012-06-27	MDCK2/SIAT1	40	160	80	320	320	160	160	320	320	320

1. < = <40

Sequence in phylogenetic trees

Vaccine
2011-12

Vaccine

Vaccine

Table 18. Antigenic analyses of influenza A H3N2 viruses (Guinea Pig RBC with 20nM Oseltamivir)

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
				Post-infection ferret antisera									
				A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/HK 3969/11	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens GR112/12
				F35/11	F7/10	F27/10	F27/11	F28/11	F15/11	Egg F05/12	T/C F11/12	T/C F15/12	F16/12
				group 5	group 3C	group 3A	group 6	group 3C	group 3C	group 3C	group 3B		
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E2	640	40	160	640	160	160	160	320	320	640
A/Victoria/208/2009		2009-06-02	E3/E1	1280	5120	1280	5120	2560	5120	5120	5120	2560	5120
A/Alabama/5/2010		2010-07-13	MK1/C2/SIAT2	<	<	80	320	80	80	40	160	160	320
A/Hong Kong/3969/2011		2011-05-19	MDCK2/SIAT5	80	80	160	640	320	160	320	640	640	640
A/Stockholm/18/2011		2011-03-28	MDCK2/SIAT6	40	40	80	320	320	80	160	320	160	640
A/Iowa/19/2010		2010-12-30	E3/E2	320	1280	640	1280	1280	2560	1280	1280	1280	1280
A/Victoria/361/2011		2011-10-24	E3/E2	320	640	320	1280	160	640	5120	640	320	160
A/Berlin/93/2011		2011-12-07	NVD3/S3	80	80	160	640	160	160	320	640	320	640
A/Victoria/361/2011		2011-10-24	M2/S2	80	80	160	640	320	320	320	640	320	640
A/Athens/GR112/2012		2012-02-01	SIAT4	80	160	160	640	320	160	320	640	320	640
TEST VIRUSES													
A/Brisbane/1/2012	3C	2012-01-06	E3/E1	320	1280	640	1280	640	640	2560	1280	1280	1280
A/Dakar/01/2012	3B	2012-01-17	MDCK2/SIAT1	80	160	160	640	320	160	160	640	320	640
A/Maryland/02/2012	3B	2012-01-18	E5/E1	160	640	320	640	640	640	640	640	320	1280
A/Dakar/02/2012		2012-01-31	MDCK1/SIAT2	160	160	320	640	320	320	320	640	640	640
A/Lorraine/329/2012		2012-02-06	MDCK2/SIAT3	80	80	160	320	320	160	320	640	320	640
A/Paris/340/2012		2012-02-06	MDCK2/SIAT3	80	160	160	640	320	160	320	640	320	640
A/Caen/539/2012	3B	2012-02-07	MDCK2/SIAT3	80	160	160	640	320	160	320	640	320	640
A/Lorraine/883/2012	3B	2012-03-02	MDCK2/SIAT3	80	160	160	640	320	160	320	640	320	1280
A/Alsace/889/2012	3A	2012-03-04	MDCK2/SIAT3	40	80	80	320	320	160	160	320	320	640
A/Ohio/2/2012	3B	2012-03-08	E4/E1	160	80	640	640	320	320	160	320	320	640
A/Centre/1066/2012	3C	2012-03-12	MDCK2/SIAT3	80	160	160	640	320	320	320	640	640	640
A/South Australia/30/2012	3A	2012-05-10	E2/E1	160	80	320	640	320	320	160	320	320	320

1. < = <40

Sequence in phylogenetic trees

Vaccine
2011-12
2012

Vaccine

Vaccine

Table 19. Antigenic analyses of influenza A H3N2 viruses (Guinea Pig RBC with 20nM Oseltamivir)

Viruses	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
			Post-infection ferret antisera									
			A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/HK 3969/11	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens GR112/12
			F35/11	F7/10	F27/10	F27/11	F28/11	F15/11	Egg F05/12	T/C F11/12	T/C F15/12	F16/12
Genetic group					group 5	group 3C	group 3A	group 6	group 3C	group 3C	group 3C	group 3B
REFERENCE VIRUSES												
A/Perth/16/2009	2009-07-04	E3/E2	1280	40	160	640	160	160	160	320	320	640
A/Victoria/208/2009	2009-06-02	E3/E1	1280	5120	1280	5120	2560	5120	5120	5120	2560	5120
A/Alabama/5/2010	2010-07-13	MK1/C2/SIAT2	<	<	160	320	80	80	40	160	160	320
A/Hong Kong/3969/2011	2011-05-19	MDCK2/SIAT5	80	80	160	1280	320	160	320	640	640	640
A/Stockholm/18/2011	2011-03-28	MDCK2/SIAT6	40	40	80	320	320	80	160	320	160	640
A/Iowa/19/2010	2010-12-30	E3/E2	320	1280	640	1280	1280	5120	1280	1280	1280	1280
A/Victoria/361/2011	2011-10-24	E3/E2	320	640	320	1280	160	640	1280	640	320	160
A/Berlin/93/2011	2011-12-07	NVD3/S3	80	80	160	640	160	160	320	1280	320	640
A/Victoria/361/2011	2011-10-24	M2/S2	80	80	160	640	320	320	320	640	640	640
A/Athens/GR112/2012	2012-02-01	SIAT4	80	160	160	640	320	160	320	640	320	1280
TEST VIRUSES												
A/Norway/878/2012	2012-02-27	SIAT1/SIAT2	80	80	160	640	320	160	80	320	320	640
A/Norway/876/2012	2012-03-06	SIAT1/SIAT2	40	80	80	640	160	160	80	320	320	320
A/Norway/943/2012	2012-03-09	MDCK1/SIAT2	80	160	320	640	320	320	160	640	640	640
A/Norway/962/2012	2012-03-10	SIAT1/SIAT2	160	160	160	640	320	320	160	640	640	1280
A/Norway/996/2012	2012-03-15	SIAT1/SIAT1	80	160	160	640	320	320	160	640	640	1280
A/Norway/1003/2012	2012-03-22	SIAT1/SIAT1	80	80	160	640	320	160	160	640	640	640
A/Norway/1059/2012	2012-03-22	SIAT1/SIAT2	40	80	80	320	160	160	80	320	320	320
A/Norway/1196/2012	2012-04-13	SIAT1/SIAT2	<	40	80	320	160	80	80	320	160	320
A/Norway/1225/2012	2012-04-18	MDCK1/SIAT1	160	320	320	1280	640	640	320	1280	1280	1280
A/Norway/1250/2012	2012-04-20	SIAT1/SIAT1	80	320	160	1280	320	320	160	640	640	1280
A/Norway/1248/2012	2012-04-23	SIAT1/SIAT1	160	320	320	1280	640	640	160	1280	640	1280
A/Norway/1296/2012	2012-04-26	MDCK1/SIAT2	40	80	80	320	640	160	80	320	320	640
A/Norway/1293/2012	2012-04-27	MDCK1/SIAT1	160	320	320	1280	640	640	320	1280	1280	1280
A/Norway/1292/2012	2012-04-29	MDCK1/SIAT1	80	80	160	640	320	320	160	640	640	640
A/Norway/1252/2012	2012-04-30	SIAT1/SIAT1	160	160	160	1280	320	320	160	1280	640	1280
A/Norway/1295/2012	2012-05-02	MDCK1/SIAT2	160	160	320	1280	320	320	160	640	640	640
A/Norway/1359/2012	2012-05-14	MDCK1/SIAT1	160	320	320	1280	640	640	320	1280	640	1280
A/South Australia/54/2012	2012-06-26	MDCK1/SIAT2	80	160	160	640	320	160	160	640	320	2560
A/Victoria/84/2012	2012-07-04	MDCK1/SIAT1	160	160	160	640	320	320	160	640	640	640

1. < = <40

Sequence in phylogenetic trees

Vaccine
2011-12
2012

Vaccine

Vaccine

Table 20. Antigenic analyses of influenza A H3N2 viruses (Guinea Pig RBC with 20nM Oseltamivir)

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre ¹									
				Post-infection ferret antisera									
				A/Perth 16/09	A/Vic 208/09	A/Ala 5/10	A/HK 3969/11	A/Stock 18/11	A/Iowa 19/10	A/Vic 361/11	A/Berlin 93/11	A/Vic 361/11	A/Athens GR112/12
				F35/11	F7/10	F27/10	F27/11	F28/11	F15/11	Egg F05/12	T/C F11/12	T/C F15/12	F16/12
				group 5	group 3C	group 3A	group 6	group 3C	group 3C	group 3C	group 3B		
REFERENCE VIRUSES													
A/Perth/16/2009		2009-07-04	E3/E2	640	40	320	640	160	160	160	320	320	640
A/Victoria/208/2009		2009-06-02	E3/E1	1280	5120	1280	5120	2560	5120	2560	2560	2560	2560
A/Alabama/5/2010		2010-07-13	MK1/C2/SIAT2	40	<	80	320	160	80	40	160	160	320
A/Hong Kong/3969/2011		2011-05-19	MDCK2/SIAT6	80	80	160	640	320	320	320	640	640	640
A/Stockholm/18/2011		2011-03-28	MDCK2/SIAT4	<	40	40	320	320	80	160	160	160	320
A/Iowa/19/2010		2010-12-30	E3/E2	320	640	640	1280	1280	1280	640	1280	1280	1280
A/Victoria/361/2011		2011-10-24	E3/E1	160	640	320	640	160	640	2560	640	320	320
A/Berlin/93/2011		2011-12-07	NVD3/SIAT3	40	80	160	320	160	160	160	320	320	320
A/Victoria/361/2011		2011-10-24	MDCK2/SIAT2	80	80	160	640	320	320	320	640	640	640
A/Athens/GR112/2012		2012-02-01	SIAT5	80	160	160	640	320	160	320	640	320	640
TEST VIRUSES													
A/Ohio/02/2012	3B	2012-03-08	E4/E2/E1	160	40	320	640	320	320	160	320	320	640
A/South Australia/30/2012	3A	2012-05-10	E2/E1	160	40	320	640	320	320	320	320	320	320
A/Johannesburg/3224/2012		2012-06-04	SIAT2/SIAT1	80	80	160	320	160	160	160	320	320	320
A/Johannesburg/5223/2012		2012-06-07	SIAT1/SIAT1	40	80	80	320	320	160	80	320	320	320
A/Idaho/20/2012	3C	2012-06-09	MDCK1/C1/SIAT1	<	40	80	320	160	80	160	320	320	320
A/Victoria/527/2012	3C	2012-06-16	E3/E1	320	640	320	1280	640	640	1280	1280	640	1280
A/Johannesburg/3303/2012		2012-06-18	MDCK2/SIAT1	40	80	80	320	320	80	80	320	320	320
A/Johannesburg/3345/2012		2012-06-20	SIAT1	<	40	40	320	160	80	80	320	160	320
A/Johannesburg/5471/2012		2012-06-20	MDCK0/SIAT1	<	40	80	320	320	160	80	320	160	320
A/Johannesburg/5569/2012		2012-06-21	MDCK2/SIAT1	<	<	<	160	160	40	40	160	160	160
A/Johannesburg/3557/2012		2012-06-22	MDCK0/SIAT1	40	40	80	320	320	160	80	320	320	320
A/Johannesburg/3392/2012		2012-06-25	MDCK2/SIAT1	40	80	160	320	320	160	160	320	320	320
A/Johannesburg/5637/2012		2012-06-28	SIAT0/SIAT1	40	40	160	320	160	160	80	320	320	320
A/Kagoshima/3/2012	3C	2012-06-28	MDCK2/SIAT1	80	80	320	640	640	320	320	640	640	640
A/Johannesburg/3469/2012		2012-06-29	SIAT1	80	160	160	640	640	320	160	640	640	640
A/Johannesburg/5903/2012		2012-07-03	SIAT0/SIAT1	40	80	160	640	640	160	160	640	640	640
A/Johannesburg/3495/2012		2012-07-04	MDCK0/SIAT1	80	80	160	640	320	320	160	640	640	640
A/Victoria/342/2012		2012-07-12	MDCK1/SIAT2	40	80	80	320	320	160	160	320	160	320

1. < = <40

Sequence in phylogenetic trees

Vaccine
2011-12
2012

Vaccine

Vaccine

Table 21. Summary of influenza A (H3N2) samples analysed, collected between 2012-02-01 and 2012-08-31

MONTH	H3N2			
		Fold reduction in HI titre		
Country	Number propagated ¹	≤2 fold ²	4 fold ²	≥8 fold ²
UNKNOWN				
Czech Republic ³	10	10		
FEBRUARY				
Albania	1		1	
Bulgaria	8	6	2	
Denmark	4	3	1	
Estonia	2	2		
Finland	2	1	1	
France	3	3		
Georgia	4	4		
Greece	13	12	1	
Iceland	11	8	3	
Ireland	1	1		
Italy	5	4	1	
Norway	15	15		
Portugal	1	1		
Russia	3	2		1
Serbia	1	1		
Slovenia	7	7		
Ukraine	12	12		
United Kingdom	4	3	1	
MARCH				
Belarus	5	5		
Denmark	7	7		
Estonia	7	7		
Finland	1		1	
France	10	9		1
Georgia	3	3		
Guyane	2	2		
Iceland	2	2		
Ireland	4	4		
Italy	1		1	
Montenegro	7	6	1	
Norway	6	6		
Portugal	4	4		
Russia	6	6		
Serbia	13	13		
Slovenia	7	6	1	
Sweden	1	1		
Ukraine	79	79		
United Kingdom	4	3		1
APRIL				
Denmark	3	3		
Estonia	3	3		
Georgia	1	1		
Iceland	1			1
Ireland	2	2		
Norway	8	7	1	
Russia	9	9		
Serbia	5	5		
Slovenia	1	1		
Sweden	2	2		
Ukraine	17	15	2	
United Kingdom	6	6		
MAY				
Ghana	1		1	
Madagascar	9	9		
Norway	2	2		
Ukraine	1	1		
JUNE				
Hong Kong	4	3	1	
Madagascar	2	2		
JULY				
South Africa	2	2		
Total	355	331	20	4

1. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir

2. Compared to homologous titres for ferret antisera raised against tissue-culture grown A/Victoria/361/2011 (Tables 11-20) or A/Norway/1789/2011 (Tables 8-10)

3. Package received 2012-05-10

Table 22. Antigenic analysis of influenza A H3N2 viruses - Plaque Reduction Neutralisation¹ - MDCK-SIAT

Viruses	Genetic group	Collection Date	Passage History	Neutralisation titre ²				
				Post infection ferret sera				
				A/Bris 10/07 F7/07	A/Per 16/09 F35/11	A/HK 3969/11 F27/11	A/Iowa 19/10 F15/11	A/Vic 361/11 F05/12
						group 3C	group 6	group 3C
REFERENCE VIRUSES								
A/Brisbane/10/2007		2007-02-06	E2/E4	5120	80	80	80	ND
A/Perth/16/2009		2009-07-04	E3/E1	<	640	160	80	ND
A/Hong Kong/3969/2011		2011-05-19	MDCK3	<	80	160	80	ND
A/Iowa/19/2010		2010-12-30	E3/E1	80	640	2560	5120	ND
A/Victoria/361/2011		2011-10-24	E3/E1	ND	160	640	ND	2560
TEST VIRUSES								
A/Norway/399/2012	3C	2012-01-30	SIAT1/SIAT1	<	160	1280	320	320
A/Norway/606/2012	3C	2012-02-20	SIAT1/SIAT1	20	160	1280	320	320
A/Glasgow/407585/2012	3A	2012-03-30	SIAT3	<	80	640	240	320
A/Rostov-on-Don/1/2012	3C	2012-04-19	C1/SIAT1	<	80	320	80	80
A/St.Petersburg/28/2012	3B	2012-04-19	C2/SIAT1	<	80	160	80	160
					Vaccine 2011-12 2012	Vaccine 2012-13		
1.80% reduction; 2. < = <40								
Sequences in phylogenetic trees								

1.80% reduction; 2. < = <40

Sequences in phylogenetic trees

Table 23. Antigenic analysis of influenza A H3N2 viruses - Plaque Reduction Neutralisation¹ - MDCK-SIAT

Viruses	Genetic group	Collection Date	Passage History	Neutralisation titre ²				
				Post infection ferret sera				
				A/HK 3969/11	A/Stockholm 18/11	A/Vic 361/11	A/Vic 361/11	A/Athens GR112/12
				T/C F27/11 group 3C	T/C F28/11 group 3A	Egg F05/12 group 3C	T/C F15/12 group 3C	T/C F16/12 group 3B
REFERENCE VIRUSES								
A/Hong Kong/3969/2011		2011-05-19	MDCK2/SIAT5	320	80	40	160	160
A/Stockholm/18/2011		2011-03-28	MDCK2/SIAT4	80	160	40	40	160
A/Victoria/361/2011		2011-10-24	E3/E2	640	80	5120	320	160
A/Victoria/361/2011		2011-10-24	MDCK2/SIAT3	80	40	<	40	80
A/Athens/GR112/2012		2012-02-01	SIAT4	160	160	40	160	320
TEST VIRUSES								
A/Torino/01/2011	6	2011-11-22	I SIAT/SIAT2	640	160	160	320	320
A/Pavia/42/2012	3B	2012-01-27	II MDCK/SIAT1	80	80	80	80	160
A/Norway/428/2012		2012-01-31	SIAT1/SIAT1	320	80	80	160	160
A/Finland/252/2012	3C	2012-02-16	MDCK/SIAT2	320	160	80	160	160
A/Pavia/158/2012	3A	2012-02-27	II MDCK/SIAT1	320	160	160	80	320

1.80% reduction; 2. < = <40

Sequences in phylogenetic trees

Vaccine
2012-13

Table 24. Antigenic analysis of influenza A H3N2 viruses - Plaque Reduction Neutralisation¹ - MDCK-SIAT

Viruses	Genetic group	Collection Date	Passage History	Neutralisation titre ²				
				Post infection ferret sera				
				A/HK 3969/11 T/C F27/11 group 3C	A/Stockholm 18/11 T/C F28/11 group 3A	A/Vic 361/11 Egg F05/12 group 3C	A/Vic 361/11 T/C F15/12 group 3C	A/Athens GR112/12 T/C F16/12 group 3B
REFERENCE VIRUSES								
A/Hong Kong/3969/2011		2011-05-19	MDCK2/SIAT5	320	80	80	160	160
A/Stockholm/18/2011		2011-03-28	MDCK2/SIAT4	80	160	40	40	160
A/Victoria/361/2011		2011-10-24	E3/E2	640	80	5120	320	160
A/Victoria/361/2011		2011-10-24	MDCK2/SIAT3	80	40	<	40	80
A/Athens/GR112/2012		2012-02-01	SIAT4	320	160	80	160	320
TEST VIRUSES								
A/Norway/479/2012	3C	2012-02-08	MDCK2/SIAT2	160	80	80	160	160
A/Norway/648/2012	3C	2012-02-15	SIAT1/SIAT1	640	160	160	320	320
A/Georgia/301/2012 ³	3B	2012-02-21	SIAT4	160	160	160	160	320
A/Georgia/255/2012	3C	2012-02-22	SIAT2	1280	640	640	2560	1280
A/Georgia/370/2012	3B	2012-03-06	SIAT2	640	320	320	640	640
A/Georgia/471/2012 ³	3B	2012-03-22	SIAT4	320	160	80	160	320
A/Georgia/491/2012 ³	3B	2012-03-26	SIAT4	320	160	80	160	320
A/Georgia/554/2012	3C	2012-03-27	SIAT2	1280	320	320	640	640
A/Georgia/579/2012	3C	2012-04-04	SIAT2	2560	640	320	1280	1280
A/Georgia/606/2012 ³	3C	2012-04-06	SIAT4	640	320	320	640	640
						Vaccine 2012-13		
1.80% reduction; 2. <= <40; 3. No HA titre in the presence of 20nM oseltamivir								

1.80% reduction; 2. < = <40; 3. No HA titre in the presence of 20nM oseltamivir

Table 25. Antigenic analysis of influenza A H3N2 viruses - Plaque Reduction Neutralisation¹ - MDCK-SIAT

Viruses	Genetic group	Collection Date	Passage History	Neutralisation titre ²				
				Post infection ferret sera				
				A/HK 3969/11 T/C F27/11 group 3C	A/Stockholm 18/11 T/C F28/11 group 3A	A/Vic 361/11 Egg F05/12 group 3C	A/Vic 361/11 T/C F15/12 group 3C	A/Athens GR112/12 T/C F16/12 group 3B
REFERENCE VIRUSES								
A/Hong Kong/3969/2011		2011-05-19	MDCK2/SIAT5	320	80	80	160	160
A/Stockholm/18/2011		2011-03-28	MDCK2/SIAT4	80	160	40	40	160
A/Victoria/361/2011		2011-10-24	E3/E2	640	80	5120	320	160
A/Victoria/361/2011		2011-10-24	MDCK2/SIAT3	80	40	<	40	80
A/Athens/GR112/2012		2012-02-01	SIAT4	320	160	80	160	320
TEST VIRUSES								
A/Centre/848/2012	3B	2012-03-01	MDCK2/SIAT1	320	160	160	160	320
A/Caen/900/2012	6	2012-03-01	MDCK2/SIAT1	320	160	160	320	320
A/Paris/919/2012	3B	2012-03-06	MDCK2/SIAT1	160	80	160	160	320
A/Centre/929/2012	3C	2012-03-07	MDCK2/SIAT1	160	80	160	160	160
A/Paris/972/2012	3B	2012-03-08	MDCK2/SIAT1	320	160	160	160	320
A/Guyane/1296/2012	6	2012-03-08	MDCK2/SIAT1	1280	320	320	640	1280
A/Guyane/1297/2012	3C	2012-03-08	MDCK2/SIAT1	1280	320	320	2560	1280
A/Madagascar/03536/2012	3C	2012-06-04	MDCK1/SIAT2	160	80	80	160	160
A/Madagascar/03535/2012	3C	2012-06-06	SIAT1	1280	320	320	640	1280
A/Hong Kong/8152/2012	3C	2012-06-26	MDCK2/SIAT2	40	40	80	40	80

Vaccine
2012-13

1.80% reduction; 2. < = <40

Sequences in phylogenetic trees

Figure 9. Phylogenetic comparison of influenza A(H3N2) HA genes

Vaccine viruses

Reference viruses

Collection date

Mar 2012

Apr 2012

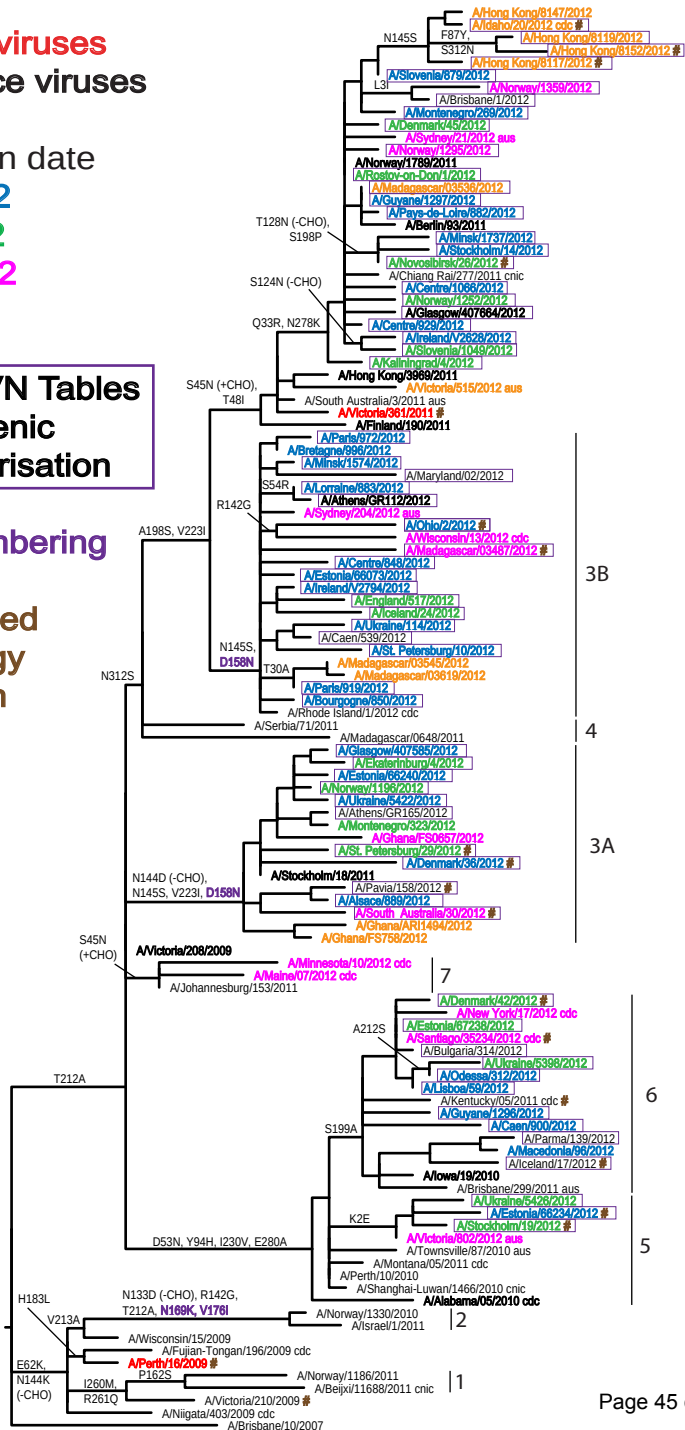
May 2012

Jun 2012

See HI/VN Tables
for antigenic
characterisation

HA2 numbering

proposed
serology
antigen



3C

3B

Vic/208
clade

4

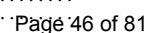
3A

6

5

Perth/16
clade

Vaccine virus: **Reference virus:** **Genetic group:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**



	310	320	330	340	350	360	370	380	390	400					
A/Brisbane/10/2007	TYGACPR	YVKQNTL	KLATGMR	NVPEKQ	TRGIFG	AIAGFI	ENGWEG	MVDGWY	GFRHQNSE	GIGQAADL	KSTQAAIDQ	INGKLN	RLIGKT	NEKFHQ	IEKEFS
A/Perth/16/2009															
A/Norway/1186/2011															
A/Israel/1/2011															
A/Victoria/208/2009															
A/Johannesburg/153/2011															
A/Minnesota/10/2012 cdc															
A/Alabama/05/2010 cdc															
A/Stockholm/19/2012															
A/Ukraine/5426/2012															
A/Iowa/19/2010															
A/Macedonia/96/2012															
A/Ukraine/5398/2012															
A/Denmark/42/2012															
A/Ghana/ARI1494/2012															
A/Stockholm/18/2011															
A/Ghana/FS0657/2012															
A/Ekaterinburg/4/2012															
A/Madagascar/0648/2011															
A/Serbia/71/2011															
A/Madagascar/03619/2012															
A/England/517/2012															
A/Athens/GRI12/2012															
A/Minsk/1574/2012															
A/Finland/190/2011															
A/Victoria/361/2011															
A/Hong Kong/3969/2011															
A/Glasgow/407664/2012															
A/Stockholm/14/2012															
A/Berlin/93/2011															
A/Norway/1789/2011															
A/Norway/1359/2012															
A/Hong Kong/8147/2012															
A/Hong Kong/8152/2012															

	410	420	430	440	450	460	470	480	490	500					
A/Brisbane/10/2007	EVEGRIQ	DLKYVED	TKIDLWS	YNAELIV	ALENQHT	IDLTDS	EMNKLFE	KTKKQL	RENAED	MGNGCF	KIYHKCD	NACIGS	IRNGTY	DHDDVY	DEALNNRF
A/Perth/16/2009															
A/Norway/1186/2011															
A/Israel/1/2011															K.
A/Victoria/208/2009															
A/Johannesburg/153/2011															
A/Minnesota/10/2012 cdc															
A/Alabama/05/2010 cdc															
A/Stockholm/19/2012															
A/Ukraine/5426/2012															
A/Iowa/19/2010															
A/Macedonia/96/2012															
A/Ukraine/5398/2012															
A/Denmark/42/2012															
A/Ghana/ARI1494/2012															N.
A/Stockholm/18/2011															N.
A/Ghana/FS0657/2012															N.
A/Ekaterinburg/4/2012															N.
A/Madagascar/0648/2011															
A/Serbia/71/2011															
A/Madagascar/03619/2012															N.
A/England/517/2012															N.
A/Athens/GRI12/2012															N.
A/Minsk/1574/2012															N.
A/Finland/190/2011															
A/Victoria/361/2011															
A/Hong Kong/3969/2011															
A/Glasgow/407664/2012															
A/Stockholm/14/2012															
A/Berlin/93/2011															
A/Norway/1789/2011															
A/Norway/1359/2012															
A/Hong Kong/8147/2012															
A/Hong Kong/8152/2012															

	510	520	530	540	550
A/Brisbane/10/2007	QIKGV	ELKSGY	KDWILW	SFAISCF	LLCVALLG
A/Perth/16/2009					
A/Norway/1186/2011					
A/Israel/1/2011					
A/Victoria/208/2009					
A/Johannesburg/153/2011					
A/Minnesota/10/2012 cdc					
A/Alabama/05/2010 cdc					
A/Stockholm/19/2012					
A/Ukraine/5426/2012					
A/Iowa/19/2010					
A/Macedonia/96/2012					
A/Ukraine/5398/2012					
A/Denmark/42/2012					
A/Ghana/ARI1494/2012					
A/Stockholm/18/2011					
A/Ghana/FS0657/2012					
A/Ekaterinburg/4/2012					
A/Madagascar/0648/2011					
A/Serbia/71/2011					
A/Madagascar/03619/2012					
A/England/517/2012					
A/Athens/GRI12/2012					
A/Minsk/1574/2012					
A/Finland/190/2011					
A/Victoria/361/2011					
A/Hong Kong/3969/2011					
A/Glasgow/407664/2012					
A/Stockholm/14/2012					
A/Berlin/93/2011					
A/Norway/1789/2011					
A/Norway/1359/2012					
A/Hong Kong/8147/2012					
A/Hong Kong/8152/2012					

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure11a. H3N2 HA locations of genetic subgroup defining amino acid substitutions: subgroups 3A, 3B and 3C.

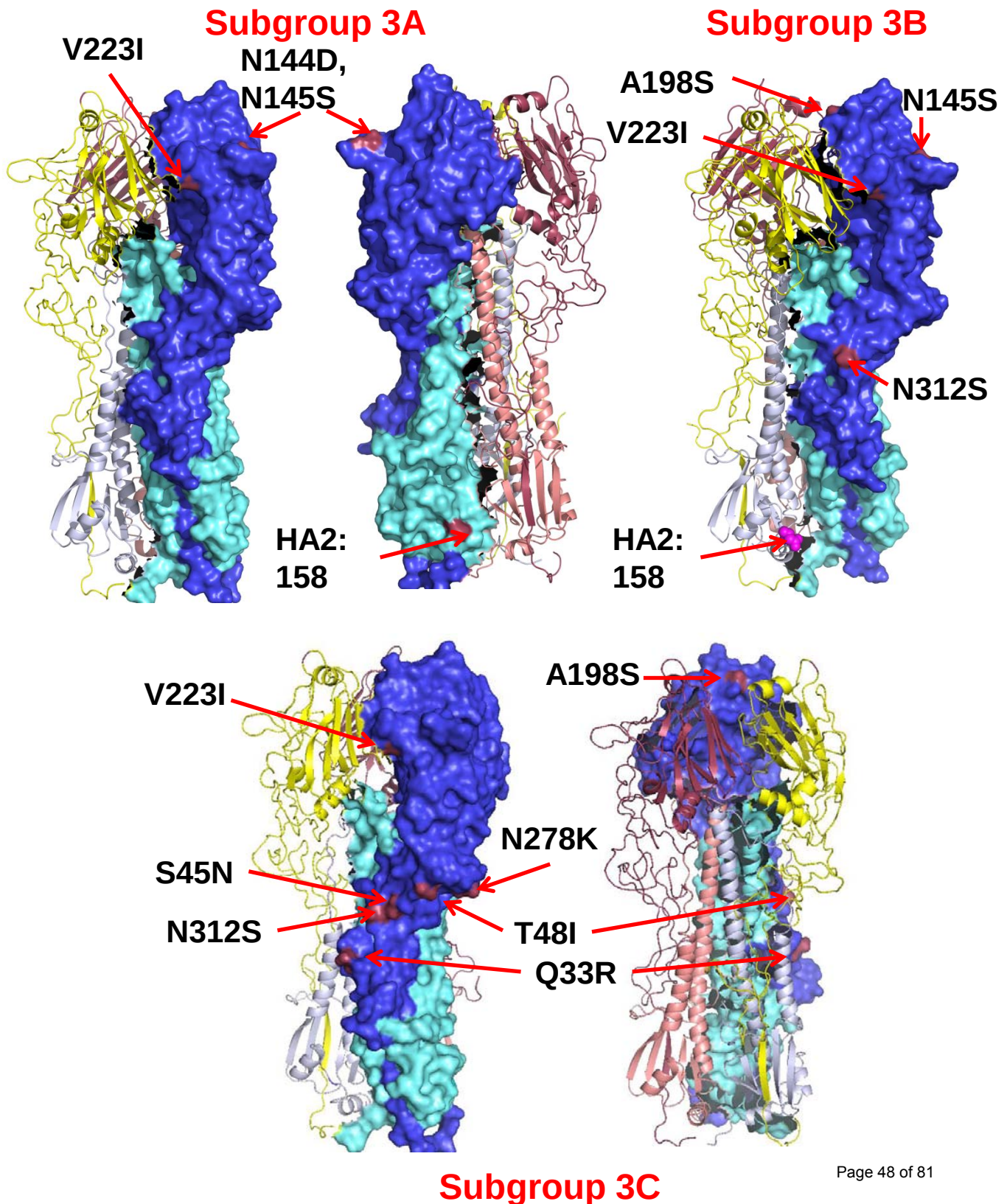
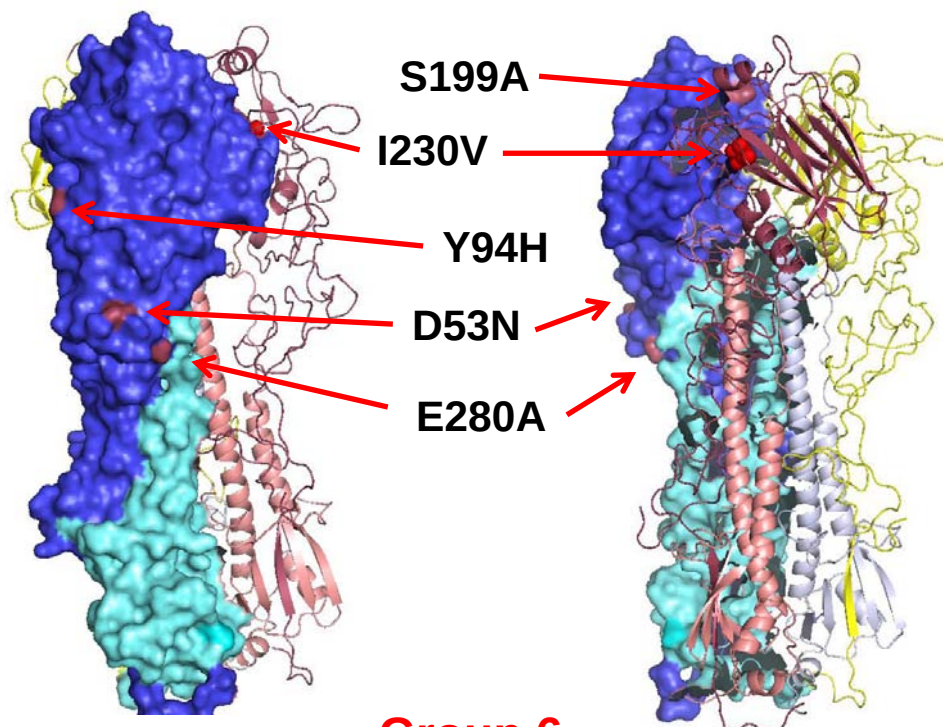
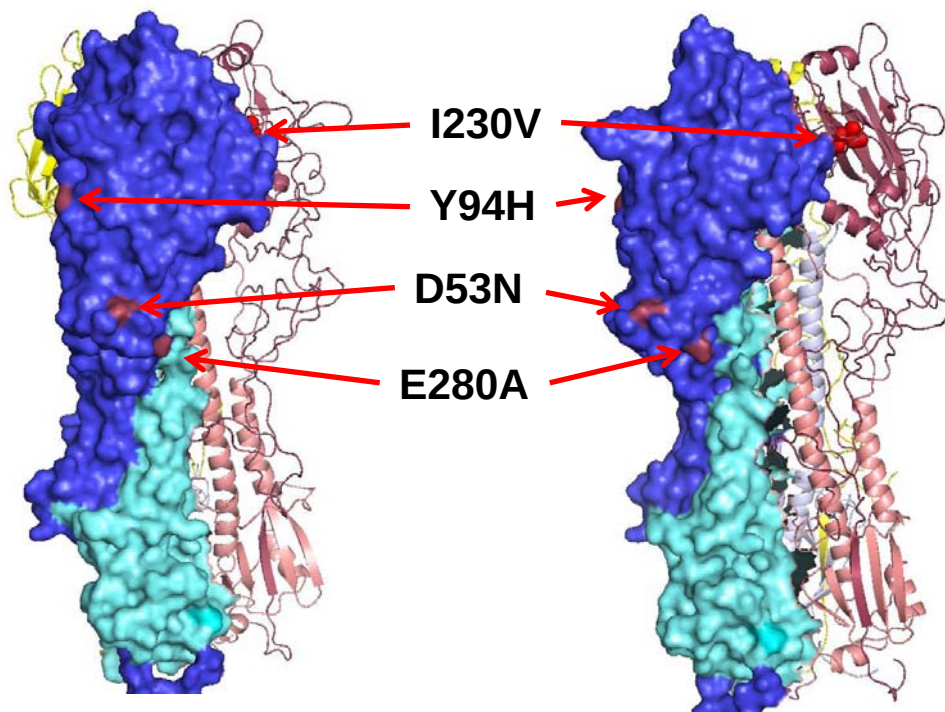


Figure 11b. H3N2 HA locations of genetic group defining amino acid substitutions: groups 5 and 6.

Group 5



Group 6

Vaccine viruses
Reference viruses

Mar 2012

Apr 2012

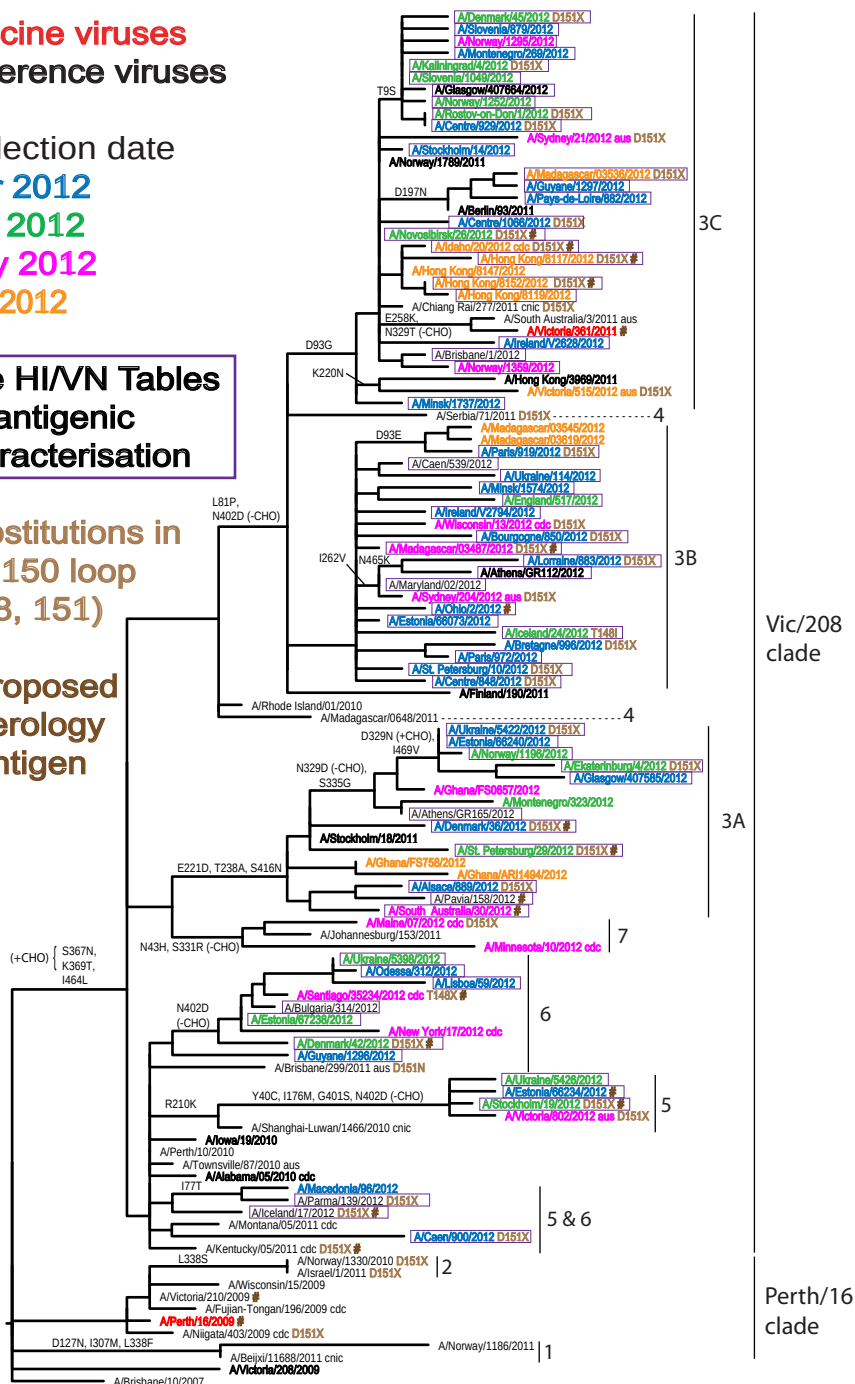
May 2012

Jun 2012

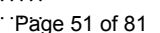
**See HI/VN Tables
for antigenic
characterisation**

Substitutions in the 150 loop (148, 151)

proposed serology antigen



Vaccine virus: Reference virus: **Genetic group:** Potential N-linked glycosylation site: NA/HA reassortants

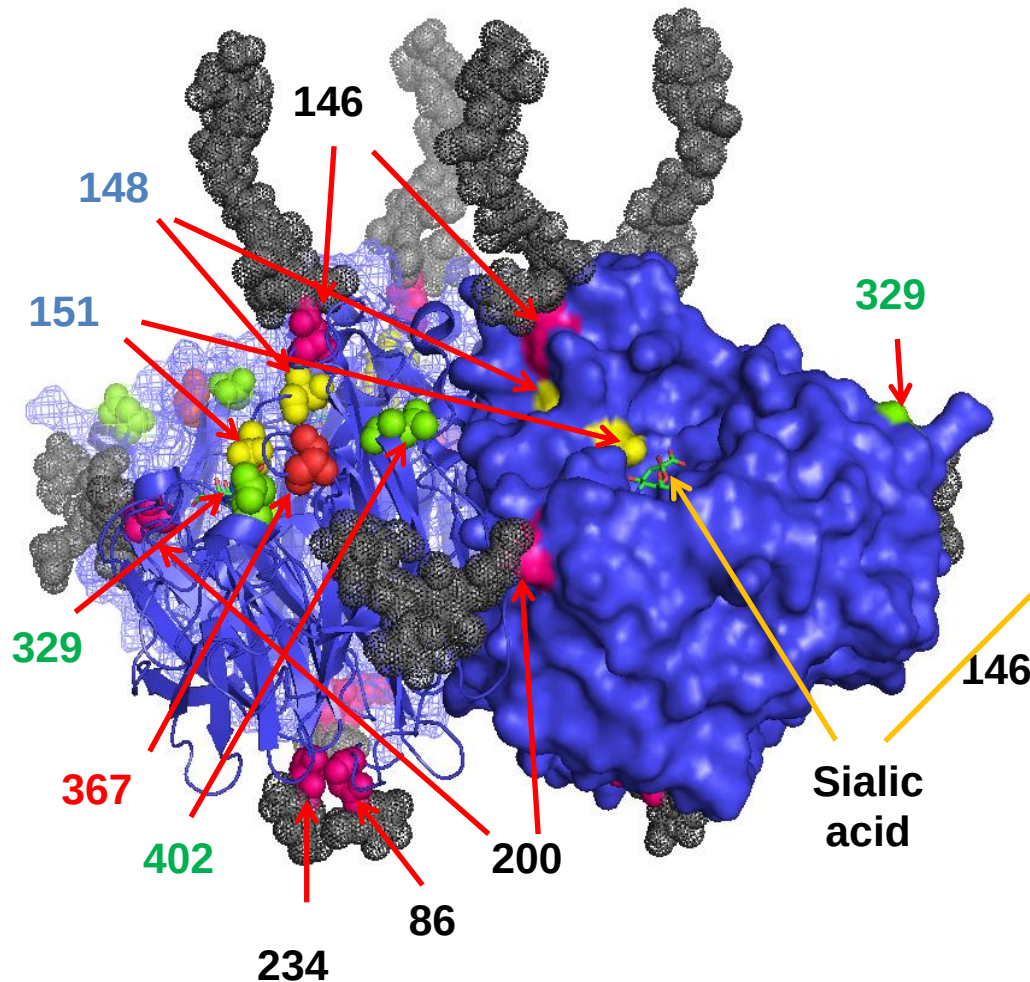


	310	320	330	340	350	360	370	380	390	400
A/Brisbane/10/2007	P I V D I N I K D H S T V S S Y V C S G L V G D T P R N D S S S S H C L D P N N E E G G H G V K G W A F D D G N D V M M G R T I S E K S R L G Y E T F K V I E G W S N P K S K L Q I N R Q V I V D R									
A/Perth/16/2009	.	I								
A/Norway/1186/2011	M	I		F	D					
A/Israel/1/2011	.	I		S						
A/Victoria/208/2009	.	I								
A/Johannesburg/153/2011	.	I	R			N	T			
A/Minnesota/10/2012 cdc	.	I	S	R		N	T			
A/Alabama/05/2010 cdc	.	I				N	T			
A/Stockholm/19/2012	.	I				N	T			
A/Ukraine/5426/2012	.	I				N	T			
A/Iowa/19/2010	.	I				N	T		M	
A/Macedonia/96/2012	.	I				N	T			
A/Ukraine/5398/2012	.	I				N	T			
A/Denmark/42/2012	.	I	D			N	T			
A/Ghana/ARI1494/2012	.	I		F		N	T			
A/Stockholm/18/2011	.	I				N	T			
A/Ghana/FSO657/2012	.	I	D	G		N	T			
A/Ekaterinburg/4/2012	.	I		G		N	T			
A/Madagascar/0648/2011	.	I				N	T			
A/Serbia/71/2011	.	I	N			N	T			
A/Madagascar/03619/2012	.	I				N	T			
A/England/517/2012	.	I				N	T			
A/Athens/GRI12/2012	.	I				N	T			
A/Minsk/1574/2012	.	I				N	T			
A/Finland/190/2011	.	I				N	T			
A/Victoria/361/2011	.	I	T			N	T			
A/Hong Kong/3969/2011	.	I				N	T			
A/Glasgow/407664/2012	.	I				N	T			
A/Stockholm/14/2012	.	I				N	T			
A/Berlin/93/2011	.	I				N	T			
A/Norway/1789/2011	.	I				N	T			
A/Norway/1359/2012	.	I	A			N	T			
A/Hong Kong/8147/2012	.	I				N	T			
A/Hong Kong/8152/2012	.	I				N	T			

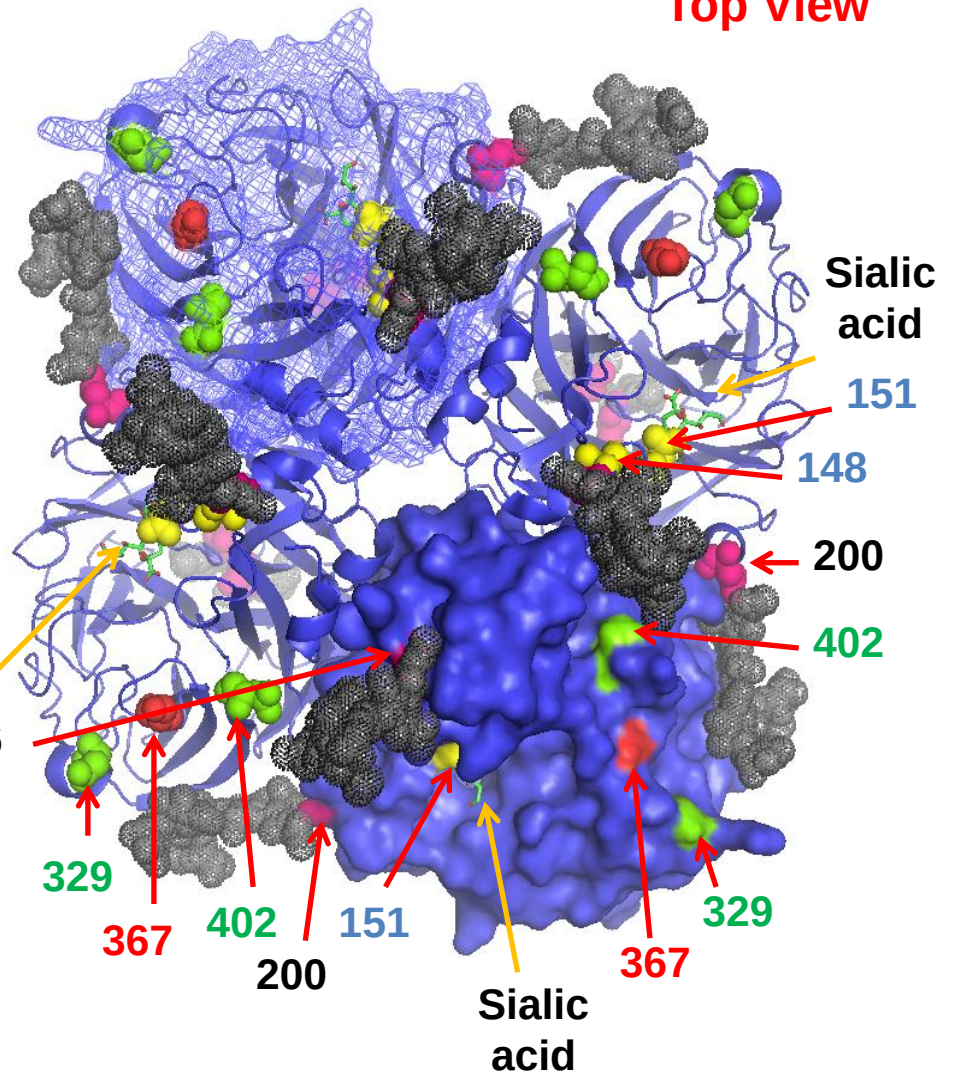
Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site: NA/HA reassortants

Figure 14. N2-NA locations of potential N-linked glycosylation sites in relation to the active site.

Side View



Top View



A/Tokyo/3/67 (H2N2) NA structure used: Conserved glycosylation: **Some lost recently:** **Most gained recently:** Substitutions in the 150-loop associated with culture on MDCK cells: Sialic acid is shown as a 'stick' representation: Speckled spheres indicate glycan structures observed in solved structure.

Table 26. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			B/Bris ² 60/08 Sh 523	B/Mal 2506/04 F28/05	B/England 393/08 F05/11	B/Bris 60/08 F06/11	B/Paris 1762/08 F07/11	B/HK 514/09 F13/10	B/Odessa 3886/10 F19/11	B/Malta 636714/11 F33/11
REFERENCE VIRUSES										
B/Malaysia/2506/2004	2004-12-06	E3/E5	1280	640	40	160	10	<	<	80
B/England/393/2008	2008-08-29	E1/E6	2560	160	320	640	80	80	40	320
B/Brisbane/60/2008	2008-08-04	E4/E3	2560	160	160	640	80	80	80	320
B/Paris/1762/2008	2009-02-09	C2/MDCK4	5120	40	40	80	160	160	80	40
B/Hong Kong/514/2009	2009-10-11	MDCK1/MDCK3	5120	40	40	80	160	160	160	40
B/Odessa/3886/2010	2010-03-19	C2/MDCK4	5120	20	40	80	160	80	160	40
B/Malta/636714/2011	2011-03-07	E4/E1	1280	160	320	640	80	80	80	320
TEST VIRUSES										
B/Finland/235/2012	2012-02-01	MDCK1/MDCK1	2560	40	40	80	160	160	160	40
B/Stockholm/5/2012	2012-02-09	C1/MDCK1	5120	20	40	80	80	80	80	40
B/Denmark/2/2012	2012-02-15	MDCK2	5120	10	20	40	80	80	160	20
B/Milano/03/2012	2012-02-16	MDCK2/MDCK1	2560	20	20	80	160	80	80	40
B/Lisboa/3/2012	2012-03-13	SIAT1/MDCK1	5120	20	40	80	160	160	80	40
B/Georgia/436/2012	2012-03-21	MDCK2	2560	10	10	80	80	80	160	40
B/Georgia/498/2012	2012-03-27	MDCK2	2560	10	10	40	80	80	160	20
B/Georgia/655/2012	2012-04-12	MDCK2	2560	<	10	40	160	80	160	20
B/Georgia/741/2012	2012-05-10	MDCK2	2560	10	10	20	80	40	80	10
B/Finland/292/2012	2012-05-14	MDCK1/MDCK1	2560	40	40	80	160	160	160	40

1. < = <10; 2. hyperimmune sheep serum

Vaccine

Sequence in phylogenetic trees

Table 27. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			B/Bris ² 60/08 Sh 523	B/Mal 2506/04 F37/11	B/England 393/08 F05/11	B/Bris 60/08 F06/11	B/Paris 1762/08 F07/11	B/HK 514/09 F13/10	B/Odessa 3886/10 F17/10	B/Malta 636714/11 F33/11
REFERENCE VIRUSES										
B/Malaysia/2506/2004	2004-12-06	E3/E3	2560	320	80	160	<	<	<	80
B/England/393/2008	2008-08-29	E1/E6	2560	160	640	1280	80	40	40	320
B/Brisbane/60/2008	2008-08-04	E4/E4	5120	160	640	1280	40	40	40	320
B/Paris/1762/2008	2009-02-09	C2/MDCK4	2560	<	10	40	160	80	80	10
B/Hong Kong/514/2009	2009-10-11	MDCK1/MDCK1	2560	<	10	40	80	160	80	10
B/Odessa/3886/2010	2010-03-19	C2/MDCK3	2560	<	10	40	80	80	160	10
B/Malta/636714/2011	2011-03-07	E4/E1	2560	80	320	640	40	40	80	320
TEST VIRUSES										
B/Pardubice/21/2012	unknown	MDCK4/MDCK1	1280	<	10	10	40	40	10	10
B/Karlovy Vary/24/2012	unknown	MDCK3/MDCK2	5120	<	10	40	80	80	40	20
B/Vsetin/51/2012	unknown	MDCK3/MDCK1	2560	<	10	20	80	80	20	20
B/Praha/27/2012	unknown	MDCK3/MDCK1	2560	<	10	20	80	80	20	10
B/Praha/52/2012	unknown	MDCK4/MDCK1	2560	<	10	20	80	80	20	20
B/Praha/90/2012	unknown	MDCK3/MDCK1	2560	<	10	20	80	80	40	20
B/Karlovy Vary/93/2012	unknown	MDCK3/MDCK1	2560	<	20	20	80	80	40	20
B/Karlovy Vary/23/2012	unknown	MDCK3/MDCK2	2560	<	20	40	40	80	20	20
B/Prerov/55/2012	unknown	MDCK3/MDCK2	2560	<	20	20	80	80	40	20
B/Athens/GR147/2012	2012-02-01	MDCK2	5120	20	40	80	160	160	160	40
B/Athens/GR152/2012	2012-02-03	MDCK2	5120	20	40	80	160	160	160	40
B/Norway/710/2012	2012-02-13	MDCK1/MDCK1	2560	<	10	40	40	20	40	<
B/England/322/2012	2012-02-29	SIAT1/MDCK1	2560	<	10	20	40	40	20	10
B/Ukraine/104/2012	2012-03-03	C1/MDCK1	2560	<	10	20	40	80	20	20
B/Slovenia/955/2012	2012-03-26	MDCKx/MDCK1	5120	10	10	20	80	80	80	20
B/England/507/2012	2012-04-04	SIAT1/MDCK1	2560	<	10	20	80	80	20	20
B/England/537/2012	2012-04-17	SIAT2/MDCK1	2560	<	10	20	80	80	20	20

1. < = <10; 2. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 28. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			B/Bris ²	B/Mal	B/England	B/Bris	B/Paris	B/HK	B/Odessa	B/Malta
			60/08 Sh 523	2506/04 F37/11	393/08 F05/11	60/08 F06/11	1762/08 F07/11	514/09 F13/10	3886/10 F19/11	636714/11 F33/11
REFERENCE VIRUSES										
B/Malaysia/2506/2004	2004-12-06	E3/E5	1280	320	40	80	<	<	<	80
B/England/393/2008	2008-08-29	E1/E1	1280	80	320	640	40	80	80	640
B/Brisbane/60/2008	2008-08-04	E4/E3	1280	40	160	320	40	40	40	320
B/Paris/1762/2008	2009-02-09	MDCK6	5120	20	40	40	80	160	160	40
B/Hong Kong/514/2009	2009-10-11	MDCK3	5120	20	40	40	80	160	320	20
B/Odessa/3886/2010	2010-03-19	C2/MDCK4	5120	10	20	20	80	80	160	20
B/Malta/636714/2011	2011-03-07	E4/E1	1280	40	320	320	40	40	40	320
TEST VIRUSES										
B/Prerov/25/2012	unknown	MDCK3/MDCK3	5120	10	40	40	80	80	160	40
B/Denmark/5/2012	2012-02-27	MDCK2/MDCK1	2560	10	20	20	80	40	80	20
B/Ireland/M17522/2012	2012-02-29	C1/SIAT1	5120	<	20	40	40	40	80	20
B/England/345/2012	2012-03-08	SIAT1/MDCK2	5120	10	40	80	80	160	160	40
B/Ukraine/5374/2012	2012-03-12	C1/MDCK1	2560	<	20	20	80	80	80	10
B/Ukraine/5376/2012	2012-03-26	C1/MDCK1	2560	<	20	20	80	80	80	20
B/St. Petersburg/18/2012	2012-04-13	C3/MDCK1	2560	<	20	40	80	80	20	20
B/St. Petersburg/27/2012	2012-04-13	C1/MDCK1	2560	20	20	80	160	80	40	40
B/Denmark/15/2012	2012-04-16	MDCK2/MDCK1	2560	20	40	40	40	40	40	40

1. < = <10; 2. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 29. Antigenic analyses of influenza B viruses (Victoria lineage)

			Haemagglutination inhibition titre ¹							
Viruses	Collection date	Passage History	Post infection ferret sera							
			B/Bris ²	B/Mal	B/England	B/Bris	B/Paris	B/HK	B/Odessa	B/Malta
			60/08 Sh 523	2506/04 F28/05	393/08 F05/11	60/08 F06/11	1762/08 F17/11	514/09 F13/10	3886/10 F19/11	636714/11 F33/11
REFERENCE VIRUSES										
B/Malaysia/2506/2004	2004-12-06	E7	1280	640	80	160	10	<	<	160
B/England/393/2008	2008-08-29	E1/E1	1280	160	320	640	80	40	160	640
B/Brisbane/60/2008	2008-08-04	E4/E3	2560	160	320	640	80	40	160	640
B/Paris/1762/2008	2009-02-09	C2/MDCK1	2560	20	10	40	80	80	160	40
B/Hong Kong/514/2009	2009-10-11	MDCK4	2560	20	10	40	80	80	320	40
B/Odessa/3886/2010	2010-03-19	MDCK2/MDCK4	5120	40	20	80	80	80	320	40
B/Malta/636714/2011	2011-03-07	E4	1280	80	160	320	40	20	80	320
TEST VIRUSES										
B/Madagascar/01551/2012	2012-02-15	MDCKI/MDCK1	2560	20	10	40	40	80	160	40
B/Madagascar/01591/2012	2012-02-21	MDCKI/MDCK1	2560	10	10	80	80	80	160	40
B/Madagascar/01593/2012	2012-02-27	MDCKI/MDCK1	2560	10	10	40	80	80	160	40
B/Madagascar/01594/2012	2012-02-27	MDCKI/MDCK1	2560	20	20	40	80	80	160	40
B/Madagascar/01706/2012	2012-03-07	MDCKI/MDCK1	2560	20	10	40	80	80	320	40
B/Madagascar/01732/2012	2012-03-07	MDCKI/MDCK1	2560	10	10	40	80	80	160	40
B/Madagascar/01731/2012	2012-03-09	MDCK2/MDCK2	5120	10	20	80	160	160	320	80
B/Madagascar/01875/2012	2012-03-27	MDCKI/MDCK1	2560	10	10	40	160	80	160	40
B/Madagascar/02853/2012	2012-03-28	MDCKI/MDCK1	2560	10	10	40	80	80	160	40
B/Madagascar/02856/2012	2012-04-02	MDCKI/MDCK1	2560	10	10	40	160	80	160	40
B/Madagascar/03142/2012	2012-05-02	MDCKI/MDCK1	2560	<	10	40	80	80	160	40
B/Madagascar/03165/2012	2012-05-03	MDCKI/MDCK1	2560	10	10	40	80	80	160	40
B/Madagascar/03283/2012	2012-05-09	MDCKI/MDCK1	5120	40	20	80	80	80	160	80

1. < = <10; 2. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 30. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			B/Bris ²	B/Mal	B/England	B/Bris	B/Paris	B/HK	B/Odessa	B/Malta
			60/08 Sh 523	2506/04 F28/05	393/08 F05/11	60/08 F22/12	1762/08 F17/11	514/09 F13/10	3886/10 F19/11	636714/11 F33/11
REFERENCE VIRUSES										
B/Malaysia/2506/2004	2004-12-06	E7	640	320	20	80	<	<	<	80
B/England/393/2008	2008-08-29	E1/E1	1280	80	160	640	40	40	80	320
B/Brisbane/60/2008	2008-08-04	E4/E3	1280	160	160	320	40	40	80	320
B/Paris/1762/2008	2009-02-09	C2/MDCK1	2560	<	<	40	40	80	80	20
B/Hong Kong/514/2009	2009-10-11	MDCK4	2560	<	<	40	80	80	320	20
B/Odessa/3886/2010	2010-03-19	MDCK2/MDCK4	2560	<	<	40	80	80	160	20
B/Malta/636714/2011	2011-03-07	E4/E1	1280	80	160	640	40	40	80	320
TEST VIRUSES										
B/Lorraine/1023/2012	2012-03-13	MDCK2/MDCK1	2560	10	10	80	80	80	80	40
B/Paris/1039/2012	2012-03-15	MDCK2/MDCK1	2560	<	10	80	160	80	160	40
B/Paris/1087/2012	2012-03-19	MDCK1/MDCK1	2560	<	10	80	80	80	160	40
B/Djibouti/IP1157/2012	2012-03-24	MDCK2/MDCK1	2560	<	10	40	80	80	160	40
B/Paris/1184/2012	2012-03-27	MDCK2/MDCK1	1280	<	<	40	80	40	80	10
B/Pays de Loire/1240/2012	2012-04-03	MDCK1/MDCK1	1280	<	10	40	80	80	80	40
B/Centre/1244/2012	2012-04-05	MDCK1/MDCK1	1280	<	<	40	80	40	80	10
B/Centre/1376/2012	2012-04-20	MDCK1/MDCK1	1280	<	<	40	40	40	80	10
B/Madagascar/03104/2012	2012-04-25	MDCK2	2560	40	160	320	40	40	80	160
B/Hong Kong/3496/2012	2012-04-30	MDCK2/MDCK1	2560	<	<	40	40	80	160	20
B/Hong Kong/3501/2012	2012-05-01	MDCK2/MDCK1	2560	40	80	160	80	80	80	160
B/Hong Kong/3508/2012	2012-05-03	MDCK2/MDCK1	2560	<	<	40	80	80	160	20
B/Hong Kong/3514/2012	2012-05-04	MDCK2/MDCK1	2560	<	<	40	40	80	160	10
B/Hong Kong/3534/2012	2012-05-08	MDCK2/MDCK1	2560	<	<	40	40	80	160	10
B/Hong Kong/3546/2012	2012-05-13	MDCK2/MDCK1	2560	<	<	40	80	40	80	10

1. < = <10; 2. hyperimmune sheep serum

Vaccine

N197S/N
N197S/N
T199N/T

Sequences in phylogenetic trees

Table 31. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹							
			B/Bris ² 60/08 Sh 523	B/Mal 2506/04 F28/05	B/England 393/08 F05/11	Post infection ferret sera				
						B/Bris 60/08 F22/12	B/Paris 1762/08 F17/11	B/HK 514/09 F13/10	B/Odessa 3886/10 F19/11	B/Malta 636714/11 F33/11
REFERENCE VIRUSES										
B/Malaysia/2506/2004	2004-12-06	E7	1280	320	80	160	10	<	<	160
B/England/393/2008	2008-08-29	E1/E1	1280	160	320	640	80	40	160	640
B/Brisbane/60/2008	2008-08-04	E4/E3	2560	160	320	320	80	40	160	640
B/Paris/1762/2008	2009-02-09	C2/MDCK1	2560	20	10	40	160	80	160	40
B/Hong Kong/514/2009	2009-10-11	MDCK4	2560	20	10	40	80	160	320	40
B/Odessa/3886/2010	2010-03-19	MDCK2/MDCK4	5120	40	20	80	80	80	160	40
B/Malta/636714/2011	2011-03-07	E4/E1	1280	80	160	320	40	20	80	320
TEST VIRUSES										
B/Dakar/11/2012	2012-03-30	MDCK3/MDCK2	1280	<	20	40	80	80	160	20
B/Dakar/12/2012	2012-04-11	MDCK1/MDCK2	1280	80	320	640	40	20	20	320
B/Dakar/13/2012	2012-04-11	MDCK2/MDCK1	1280	<	20	40	80	80	160	20
B/Dakar/14/2012	2012-04-11	MDCK1/MDCK2	1280	80	320	320	40	20	20	320
B/Dakar/01/2012	2012-04-12	MDCK1/MDCK1	1280	10	40	80	40	40	40	80
B/Dakar/02/2012	2012-04-12	MDCK4/MDCK4	1280	<	20	40	80	80	80	10
B/Dakar/03/2012	2012-04-13	MDCK3/MDCK2	1280	80	320	320	80	80	80	320
B/Dakar/04/2012	2012-04-13	MDCK1/MDCK1	1280	<	<	20	40	40	80	10
B/Dakar/05/2012	2012-04-14	MDCK1/MDCK2	1280	80	640	1280	80	40	80	640
B/Dakar/06/2012	2012-04-15	MDCK1/MDCK1	1280	<	10	40	40	80	80	20
B/Dakar/15/2012	2012-04-16	MDCK6/MDCK1	1280	<	20	40	80	80	160	20
B/Dakar/16/2012	2012-04-18	MDCK2/MDCK1	1280	<	20	40	80	80	160	20
B/Dakar/17/2012	2012-04-19	MDCK2/MDCK1	1280	<	20	40	80	80	80	20
B/Dakar/07/2012	2012-04-19	MDCK4/MDCK1	1280	<	10	20	40	40	80	10
B/Dakar/08/2012	2012-04-25	MDCK3/MDCK1	1280	<	10	20	40	40	80	10
B/Dakar/09/2012	2012-04-29	MDCK4/MDCK2	1280	<	20	40	80	80	160	20
B/Dakar/10/2012	2012-05-16	MDCK2/MDCK1	1280	<	20	40	80	80	80	40

1. < = <10; 2. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

N197T
N197S/N
N197Y/N
T199A

Table 32. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre ¹							
			Post infection ferret sera							
			B/Bris ² 60/08 Sh 523	B/Mal 2506/04 F28/05	B/England 393/08 F05/11	B/Bris 60/08 F22/12	B/Paris 1762/08 F17/11	B/HK 514/09 F13/10	B/Odessa 3886/10 F19/11	B/Malta 636714/11 F33/11
REFERENCE VIRUSES										
B/Malaysia/2506/2004	2004-12-06	E3/E5	1280	640	80	160	10	<	<	160
B/England/393/2008	2008-08-29	E1/E1	1280	160	640	640	80	40	160	640
B/Brisbane/60/2008	2008-08-04	E4/E3	2560	160	320	640	80	40	160	640
B/Paris/1762/2008	2009-02-09	C2/MDCK1	2560	20	10	40	160	80	160	40
B/Hong Kong/514/2009	2009-10-11	MDCK1/MDCK3	2560	20	10	40	80	160	320	40
B/Odessa/3886/2010	2010-03-19	MDCK2/MDCK4	5120	40	20	80	160	160	160	40
B/Malta/636714/2011	2011-03-07	E4/E1	1280	80	160	320	40	20	80	640
TEST VIRUSES										
B/Norway/1072/2012	2012-03-19	MDCK1/MDCK1	5120	40	80	80	160	160	160	80
B/Norway/1133/2012	2012-04-11	MDCK1/MDCK1	2560	<	40	40	80	80	80	40
B/Brisbane/12/2012	2012-04-16	E3/E1	2560	160	320	320	80	40	160	640
B/Brisbane/12/2012	2012-04-16	MDCK2/MDCK1	5120	40	80	80	160	160	160	80
B/New Jersey/01/2012	2012-04-26	C2/MDCK1	5120	20	40	80	80	80	80	80
B/New Jersey/01/2012	2012-04-26	E4/E1	2560	80	160	320	40	40	80	320
B/Johannesburg/3005/2012	2012-05-22	MDCK1/MDCK1	5120	10	20	80	160	80	160	40
B/Johannesburg/3958/2012	2012-08-03	E1/E1	1280	<	20	40	80	40	80	40
B/Johannesburg/3964/2012	2012-08-03	E1/E1	2560	320	320	320	80	40	80	320

1. < = <10; 2. hyperimmune sheep serum

Vaccine

Table 33. Summary of influenza B (Victoria-lineage) samples analysed, collected between 2012-02-01 and 2012-08-31

MONTH	B Victoria lineage			
	Number propagated ¹	Fold reduction in HI titre		
		≤2 fold ²	4 fold ²	≥8 fold ²
UNKNOWN				
Czech Republic ³	10	1	4	5
FEBRUARY				
Denmark	2	1	1	
Finland	1	1		
Greece	2	2		
Ireland	1		1	
Italy	1	1		
Madagascar	4	4		
Norway	1			1
Sweden	1	1		
United Kingdom	1			1
MARCH				
Djibouti	1	1		
France	4	4		
Georgia	2	2		
Madagascar	5	5		
Norway	1	1		
Portugal	1	1		
Senegal	1	1		
Slovenia	1	1		
Ukraine	3	2		1
United Kingdom	1	1		
APRIL				
Denmark	1		1	
France	3	3		
Georgia	1	1		
Hong Kong	1	1		
Madagascar	2	2		
Norway	1	1		
Russia	2		1	1
Senegal	15	7	5	3
United Kingdom	2			2
MAY				
Finland	1	1		
Georgia	1		1	
Hong Kong	5	5		
Madagascar	3	3		
Senegal	1	1		
South Africa	1	1		
Total	84	56	14	14

1. Propagated to sufficient titre to perform HI assay

2. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents

3. Package received 2012-05-10

Figure 15. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

Vaccine virus

Reference viruses

Collection date

Feb-Mar 2012

Apr 2012

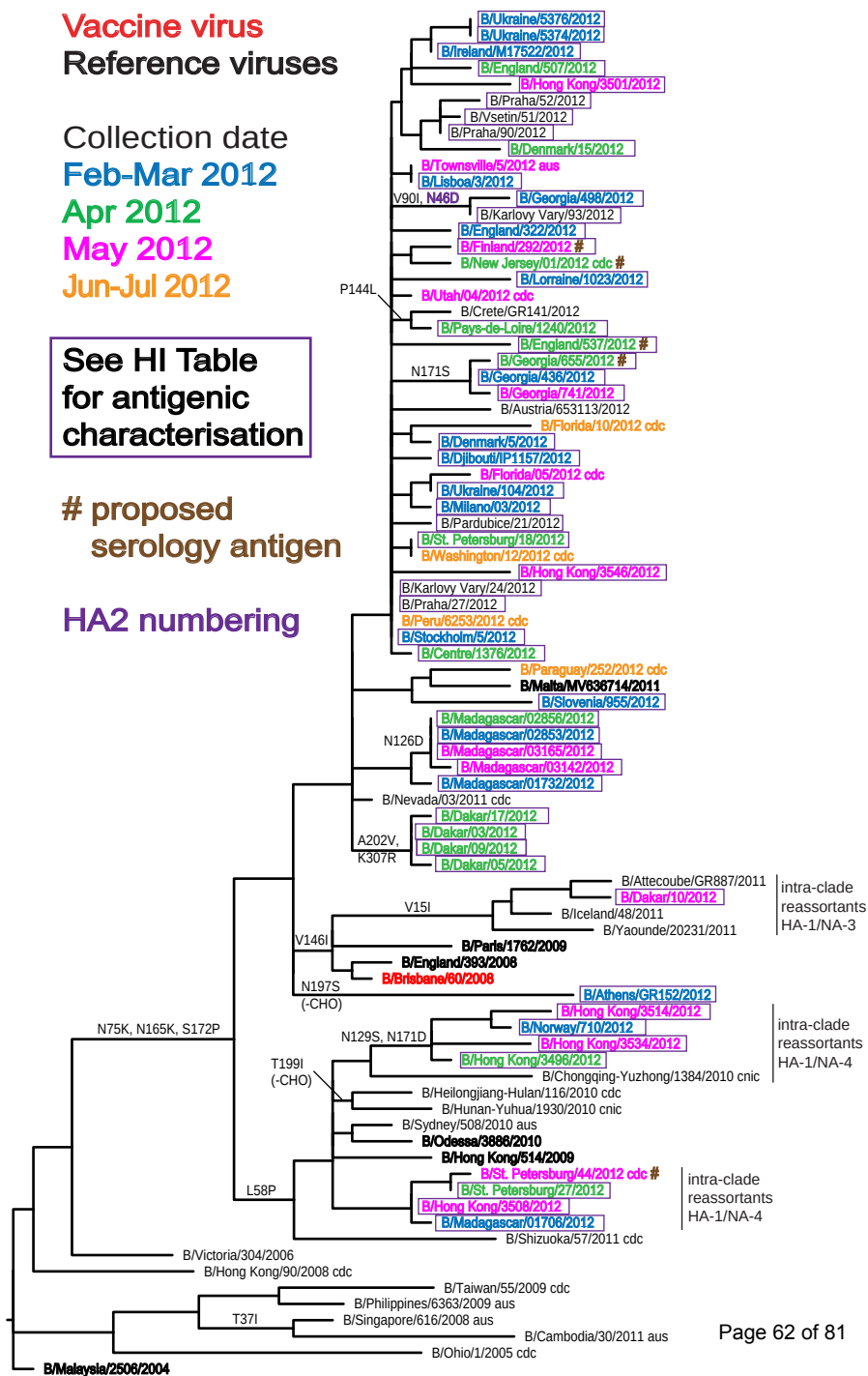
May 2012

Jun-Jul 2012

**See HI Table
for antigenic
characterisation**

proposed
serology antigen

HA2 numbering

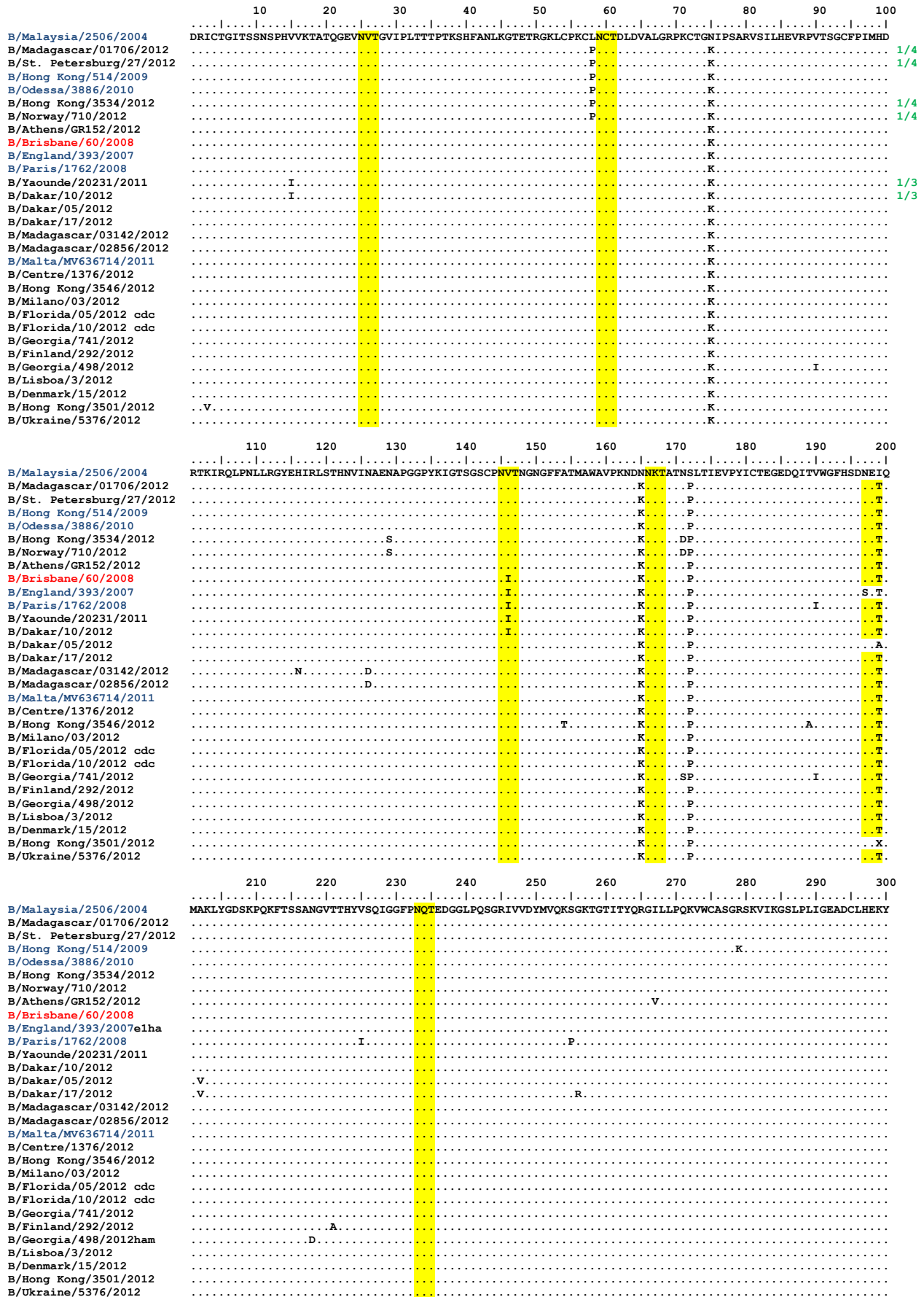


Clade 1
"Brisbane/60"

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Figure 16. HA protein alignment for B-Victoria viruses.

Vaccine virus: Reference virus: HA/NA genetic clade reassortants: HA1/HA2 boundary: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400
B/Malaysia/2506/2004	GG	LN	SK	PE	YT	GE	HA	KA	IG	NC
B/Madagascar/01706/2012		R								
B/St. Petersburg/27/2012										
B/Hong Kong/514/2009										
B/Odessa/3886/2010										
B/Hong Kong/3534/2012										
B/Norway/710/2012										
B/Athens/GR152/2012										
B/Brisbane/60/2008										
B/England/393/2007										
B/Paris/1762/2008										
B/Yaounde/20231/2011										
B/Dakar/10/2012										
B/Dakar/05/2012										
B/Dakar/17/2012										
B/Madagascar/03142/2012										
B/Madagascar/02856/2012										
B/Malta/MV636714/2011										
B/Centre/1376/2012										
B/Hong Kong/3546/2012										
B/Milano/03/2012										
B/Florida/05/2012 cdc										
B/Florida/10/2012 cdc										
B/Georgia/741/2012										
B/Finland/292/2012										
B/Georgia/498/2012									D	
B/Lisboa/3/2012										
B/Denmark/15/2012										
B/Hong Kong/3501/2012										
B/Ukraine/5376/2012										

	410	420	430	440	450	460	470	480	490	500
B/Malaysia/2506/2004	SL	SE	LV	KN	LR	LS	GA	MD	EL	HN
B/Madagascar/01706/2012										
B/St. Petersburg/27/2012										
B/Hong Kong/514/2009										
B/Odessa/3886/2010										
B/Hong Kong/3534/2012										
B/Norway/710/2012										
B/Athens/GR152/2012								I		
B/Brisbane/60/2008										
B/England/393/2007										
B/Paris/1762/2008			D							
B/Yaounde/20231/2011										
B/Dakar/10/2012										
B/Dakar/05/2012										
B/Dakar/17/2012										
B/Madagascar/03142/2012										
B/Madagascar/02856/2012										
B/Malta/MV636714/2011										
B/Centre/1376/2012										
B/Hong Kong/3546/2012										
B/Milano/03/2012										
B/Florida/05/2012 cdc										
B/Florida/10/2012 cdc										
B/Georgia/741/2012										
B/Finland/292/2012										
B/Georgia/498/2012										
B/Lisboa/3/2012										
B/Denmark/15/2012										
B/Hong Kong/3501/2012										
B/Ukraine/5376/2012										

	510	520	530	540	550	560	570
B/Malaysia/2506/2004	AG	TF	DA	GE	FS	LP	TF
B/Madagascar/01706/2012							
B/St. Petersburg/27/2012							
B/Hong Kong/514/2009							
B/Odessa/3886/2010							
B/Hong Kong/3534/2012							
B/Norway/710/2012							
B/Athens/GR152/2012							
B/Brisbane/60/2008							
B/England/393/2007							
B/Paris/1762/2008							
B/Yaounde/20231/2011							
B/Dakar/10/2012							
B/Dakar/05/2012							
B/Dakar/17/2012							
B/Madagascar/03142/2012							
B/Madagascar/02856/2012							
B/Malta/MV636714/2011							
B/Centre/1376/2012							
B/Hong Kong/3546/2012							
B/Milano/03/2012							
B/Florida/05/2012 cdc							
B/Florida/10/2012 cdc							
B/Georgia/741/2012							
B/Finland/292/2012							
B/Georgia/498/2012							
B/Lisboa/3/2012							
B/Denmark/15/2012							
B/Hong Kong/3501/2012							
B/Ukraine/5376/2012							

Vaccine virus: Reference virus: HA/NA genetic clade reassortants: HA1/HA2 boundary: Potential N-linked glycosylation site

Vaccine virus

Reference viruses

Feb-Mar 2012

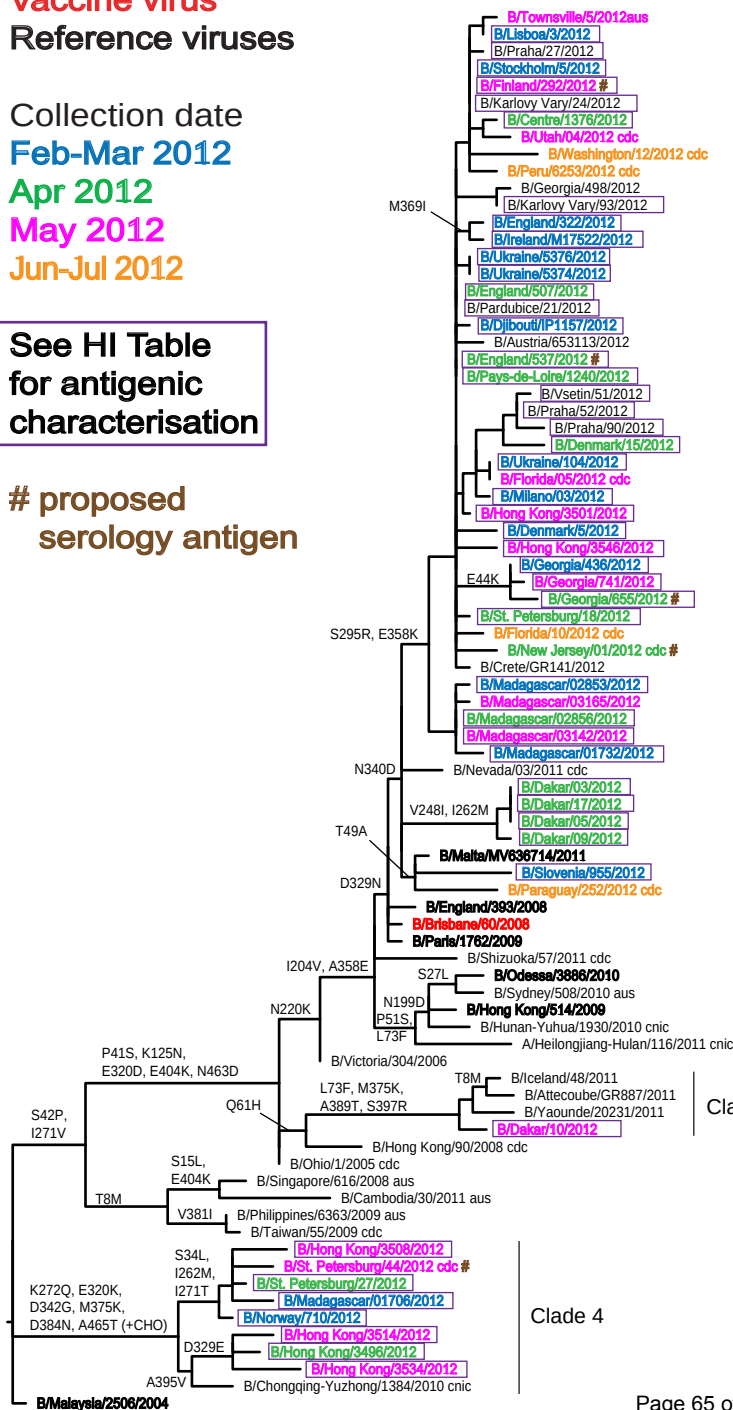
Apr 2012

May 2012

Jun-Jul 2012

**See HI Table
for antigenic
characterisation**

proposed
serology antigen



Clade 1

Clade 3

Clade 4

Figure 18. NA protein alignment for B-Victoria viruses.

Vaccine virus: Reference virus: NA genetic clade: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100		
B/Malaysia/2506/2004	M	L	P	S	T	I	Q	T	L	F	L	4
B/Madagascar/01706/2012				L								4
B/St. Petersburg/27/2012				L								1
B/Hong Kong/514/2009					S	P			S			1
B/Odessa/3886/2010			L		S	P			F			1
B/Hong Kong/3534/2012					I	S			F			4
B/Norway/710/2012			L									4
B/Brisbane/60/2008					S	P						1
B/England/393/2007					S	P						1
B/Paris/1762/2008					S	P	T					1
B/Yaounde/20231/2011					S	P		H		F		3
B/Dakar/10/2012					S	P		AH		F		3
B/Dakar/05/2012					S	P						1
B/Dakar/17/2012					S	P						1
B/Madagascar/03142/2012		X			S	P						1
B/Madagascar/02856/2012					S	P						1
B/Malta/MV636714/2011					S	P	A					1
B/Centre/1376/2012					S	P						1
B/Hong Kong/3546/2012					S	L						1
B/Milano/03/2012					S	P						1
B/Florida/05/2012 cdc					S	P						1
B/Florida/10/2012 cdc					S	P		A				1
B/Georgia/741/2012					S	P	K					1
B/Finland/292/2012					S	P						1
B/Georgia/498/2012					S	P						1
B/Lisboa/3/2012					S	P						1
B/Denmark/15/2012					S	P		A				1
B/Hong Kong/3501/2012					S	P						1
B/Ukraine/5376/2012					S	P						1
	110	120	130	140	150	160	170	180	190	200		
B/Malaysia/2506/2004	H	R	F	G	E	T	G	N	S	A	P	1
B/Madagascar/01706/2012												1
B/St. Petersburg/27/2012											V	1
B/Hong Kong/514/2009												D
B/Odessa/3886/2010												D
B/Hong Kong/3534/2012												
B/Norway/710/2012												
B/Brisbane/60/2008												
B/England/393/2007												
B/Paris/1762/2008												
B/Yaounde/20231/2011												
B/Dakar/10/2012												
B/Dakar/05/2012												
B/Dakar/17/2012												
B/Madagascar/03142/2012												
B/Madagascar/02856/2012												
B/Malta/MV636714/2011												
B/Centre/1376/2012												
B/Hong Kong/3546/2012												
B/Milano/03/2012												
B/Florida/05/2012 cdc												
B/Florida/10/2012 cdc								S				
B/Georgia/741/2012												
B/Finland/292/2012												
B/Georgia/498/2012												
B/Lisboa/3/2012												
B/Denmark/15/2012												
B/Hong Kong/3501/2012												
B/Ukraine/5376/2012												
	210	220	230	240	250	260	270	280	290	300		
B/Malaysia/2506/2004	L	L	K	I	K	Y	G	E	A	T	1	
B/Madagascar/01706/2012											1	
B/St. Petersburg/27/2012											1	
B/Hong Kong/514/2009											1	
B/Odessa/3886/2010											1	
B/Hong Kong/3534/2012											1	
B/Norway/710/2012											1	
B/Brisbane/60/2008											1	
B/England/393/2007											1	
B/Paris/1762/2008											1	
B/Yaounde/20231/2011											1	
B/Dakar/10/2012											1	
B/Dakar/05/2012											1	
B/Dakar/17/2012											1	
B/Madagascar/03142/2012											1	
B/Madagascar/02856/2012											1	
B/Malta/MV636714/2011											1	
B/Centre/1376/2012											1	
B/Hong Kong/3546/2012											1	
B/Milano/03/2012											1	
B/Florida/05/2012 cdc											1	
B/Florida/10/2012 cdc											1	
B/Georgia/741/2012											1	
B/Finland/292/2012											1	
B/Georgia/498/2012											1	
B/Lisboa/3/2012											1	
B/Denmark/15/2012											1	
B/Hong Kong/3501/2012											1	
B/Ukraine/5376/2012											1	

	310	320	330	340	350	360	370	380	390	400
B/Malaysia/2506/2004	P	F	V	K	L	N	V	E	T	D
B/Madagascar/01706/2012	.	.	.	.	.	.	.	.	.	.
B/St. Petersburg/27/2012	.	L	.	.	.	.	.	.	.	.
B/Hong Kong/514/2009	.	.	D	.	.	.	E	.	.	N
B/Odessa/3886/2010	.	.	D	.	.	.	E	.	.	N
B/Hong Kong/3534/2012	.	.	K	.	E	.	G	.	E	K
B/Norway/710/2012	.	.	K	.	.	G	.	.	E	K
B/Brisbane/60/2008	.	.	D	.	N	.	.	E	.	E
B/England/393/2007	.	.	D	.	N	.	.	E	.	E
B/Paris/1762/2008	.	.	D	.	N	.	.	E	.	E
B/Yaounde/20231/2011	.	.	D	.	.	.	.	E	K	T
B/Dakar/10/2012	.	.	D	.	.	.	.	E	K	T
B/Dakar/05/2012	.	.	D	.	N	.	D	.	E	.
B/Dakar/17/2012	.	.	D	.	N	.	D	.	E	.
B/Madagascar/03142/2012	.	.	D	.	N	.	D	.	E	.
B/Madagascar/02856/2012	.	.	D	.	N	.	D	.	E	.
B/Malta/MV636714/2011	.	.	D	.	N	.	D	.	E	.
B/Centre/1376/2012	.	.	D	.	N	.	D	.	E	.
B/Hong Kong/3546/2012	.	.	D	.	N	.	D	.	E	.
B/Milano/03/2012	.	.	D	.	N	.	D	.	E	.
B/Florida/05/2012 cdc	.	.	D	.	N	.	D	.	E	.
B/Florida/10/2012 cdc	.	.	D	.	N	.	D	.	E	.
B/Georgia/741/2012	.	.	D	.	N	.	D	.	E	.
B/Finland/292/2012	.	.	D	.	N	.	D	.	E	.
B/Georgia/498/2012	.	.	D	.	N	.	D	.	E	.
B/Lisboa/3/2012	.	.	D	.	N	.	D	.	E	.
B/Denmark/15/2012	.	.	D	.	N	.	D	.	E	.
B/Hong Kong/3501/2012	.	.	D	.	N	.	D	.	E	.
B/Ukraine/5376/2012	.	.	D	.	N	.	D	.	E	.

	410	420	430	440	450	460
B/Malaysia/2506/2004	V	S	M	E	E	P
B/Madagascar/01706/2012	.	.	.	.	.	.
B/St. Petersburg/27/2012	.	.	.	.	.	.
B/Hong Kong/514/2009	.	.	K	.	.	.
B/Odessa/3886/2010	.	.	K	.	.	.
B/Hong Kong/3534/2012	.	.	.	.	.	.
B/Norway/710/2012	.	.	.	.	.	.
B/Brisbane/60/2008	.	.	K	.	.	.
B/England/393/2007	.	.	K	.	.	.
B/Paris/1762/2008	.	.	K	.	.	.
B/Yaounde/20231/2011	.	.	K	.	.	.
B/Dakar/10/2012	.	.	K	.	.	.
B/Dakar/05/2012	.	.	K	.	.	.
B/Dakar/17/2012	.	.	K	.	.	.
B/Madagascar/03142/2012	.	.	K	.	.	.
B/Madagascar/02856/2012	.	.	K	.	.	.
B/Malta/MV636714/2011	.	.	K	.	.	.
B/Centre/1376/2012	.	.	K	.	.	.
B/Hong Kong/3546/2012	.	.	K	.	.	.
B/Milano/03/2012	.	.	K	.	.	.
B/Florida/05/2012 cdc	.	.	K	.	.	.
B/Florida/10/2012 cdc	.	.	K	.	.	.
B/Georgia/741/2012	.	.	K	.	.	.
B/Finland/292/2012	.	.	K	.	.	.
B/Georgia/498/2012	.	.	K	.	.	.
B/Lisboa/3/2012	.	.	K	.	.	.
B/Denmark/15/2012	.	.	K	.	.	.
B/Hong Kong/3501/2012	.	.	K	.	.	.
B/Ukraine/5376/2012	.	.	K	.	.	.

Vaccine virus: Reference virus: NA genetic clade: Potential N-linked glycosylation site

Table 34. Antigenic analyses of influenza B viruses (Yamagata lineage)

			Haemagglutination inhibition titre										
Viruses		Collection date	Passage History	Post infection ferret sera									
				B/FI ²	B/Eg ¹	B/FI ¹	B/Bris ¹	B/Eng ¹	B/Bang ¹	B/Wis ¹	B/Stock ¹	B/Estonia ¹	B/Serbia ¹
				4/06 SH479	144/05 F7/05	4/06 F21/07	3/07 F24/07	145/08 F9/08	3333/07 F21/08	1/10 F26/10	12/11 F12/12	55669/11 F26/11	1894/11 F25/11
	Clade					Clade 2	Clade 3	Clade 3	Clade 3	Clade 3	Clade 2	Clade 3	
REFERENCE VIRUSES													
B/Egypt/144/2005		2005-05-01	E3/E6	2560	640	640	640	40	80	160	640	320	40
B/Florida/4/2006		2006-12-15	E3/E4	5120	640	2560	640	160	160	320	640	320	40
B/Brisbane/3/2007		2007-09-03	E2/E3	5120	640	1280	640	40	80	160	320	160	10
B/England/145/2008			Ex/E5	320	40	80	<	80	10	20	320	<	<
B/Bangladesh/3333/2007		2007-08-07	E3/E4	2560	320	320	160	40	160	320	640	10	20
B/Wisconsin/1/2010		2010-02-20	E3/E1	1280	80	160	80	40	40	320	1280	20	20
B/Stockholm/12/2011		2011-03-28	E4/E1	2560	160	160	80	40	160	80	640	20	40
B/Estonia/55669/2011		2011-03-14	MDCK2/MDCK2	1280	40	40	40	10	10	<	160	1280	80
B/Serbia/1894/2011		2011-03-08	MDCK1/MDCK4	2560	80	80	80	40	20	40	320	160	160
TEST VIRUSES													
B/Athens/GR123/2012	2	2012-02-01	MDCK2	2560	80	320	320	160	160	80	320	1280	1280
B/Athens/GR132/2012	2	2012-02-01	MDCK2	1280	40	80	40	40	40	10	160	1280	80
B/Stockholm/2/2012	3	2012-02-02	C1/MDCK1	5120	160	80	40	160	160	160	640	160	320
B/Athens/GR179/2012		2012-02-06	MDCK2	5120	40	80	80	40	40	40	320	1280	40
B/Athens/GR199/2012	2	2012-02-07	MDCK2	5120	160	80	80	320	160	80	640	1280	320
B/Novosibirsk/2/2012	3	2012-02-14	C2/MDCK2	1280	<	20	40	40	80	80	320	80	320
B/Lisboa/1/2012	2	2012-02-28	SIAT2/MDCK1	2560	40	160	80	160	40	40	160	1280	80
B/Stockholm/8/2012	2	2012-03-06	C0/MDCK1	2560	160	80	80	80	40	40	160	1280	80
B/Lisboa/6/2012	3	2012-03-15	SIAT1/MDCK2	5120	160	160	80	320	320	320	640	640	640
B/Lisboa/5/2012		2012-03-19	SIAT1/MDCK1	2560	40	80	40	160	160	160	320	80	160

1. < = <10 ; 2. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 35. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre									
			Post infection ferret sera									
			B/FI ² 4/06 SH479	B/Eg ¹ 144/05 F3/07	B/FI ¹ 4/06 F21/07	B/Bris ¹ 3/07 F24/07	B/Eng ¹ 145/08 F9/08	B/Bang ¹ 3333/07 F21/08	B/Wis ¹ 1/10 F07/12	B/Stock ¹ 12/11 F34/11	B/Estonia ¹ 55669/11 F26/11	B/Serbia ¹ 1894/11 F25/11
Clade						Clade 2	Clade 3	Clade 3	Clade 3	Clade 3	Clade 2	Clade 3
REFERENCE VIRUSES												
B/Egypt/144/2005	2005-05-01	E3/E5	5120	320	640	640	80	160	160	320	160	10
B/Florida/4/2006	2006-12-15	E3/E4	5120	160	640	320	80	80	80	320	160	10
B/Brisbane/3/2007	2007-09-03	E2/E4	2560	80	640	320	40	80	80	320	160	<
B/England/145/2008		Ex/E4	320	<	40	20	80	10	20	160	<	<
B/Bangladesh/3333/2007	2007-08-07	E3/E4	1280	80	160	80	20	160	80	320	10	10
B/Wisconsin/1/2010	2010-02-20	E3/E1	2560	80	160	80	40	40	160	320	10	10
B/Stockholm/12/2011	2011-03-28	E4/E1	2560	80	160	80	40	80	80	320	10	20
B/Estonia/55669/2011	2011-03-14	MDCK2/MDCK2	1280	40	80	20	10	20	20	160	640	40
B/Serbia/1894/2011	2011-03-08	MDCK1/MDCK4	2560	40	80	20	40	40	80	320	160	160
TEST VIRUSES												
B/Hradec Kralove/26/2012		MDCK3/MDCK1	640	<	40	40	10	10	10	80	320	20
B/Hradec Kralove/29/2012	2	MDCK3/MDCK1	2560	40	160	80	80	160	80	160	640	160
B/Norway/579/2012	2	SIAT1/MDCK1	1280	40	80	20	20	20	40	160	640	80
B/Novosibirsk/3/2012	3	C2/MDCK2	1280	40	80	80	80	160	80	320	160	160
B/Novosibirsk/1/2012	3	C2/MDCK1	1280	40	80	80	40	160	80	320	160	160
B/Norway/777/2012	2	MDCK1/MDCK1	2560	40	80	40	40	40	40	160	640	160
B/Milano/05/2012	3	MDCK3/MDCK1	2560	10	40	40	80	80	160	160	80	160
B/England/267/2012	3	SIAT1/MDCK1	2560	40	40	40	40	80	40	160	80	160
B/Denmark/3/2012	3	MDCK3/MDCK1	640	40	40	20	40	40	40	160	20	40
B/Norway/773/2012	2	MDCK1/MDCK1	1280	40	80	20	20	20	40	160	640	80
B/Milano/08/2012		MDCK2/MDCK1	1280	10	20	40	80	80	160	160	80	160
B/Roma/01/2012	2	MDCK3/MDCK1	1280	<	<	40	40	40	20	40	640	40
B/Denmark/7/2012		MDCK2/MDCK1	640	<	<	10	40	20	40	160	10	40
B/Denmark/8/2012	3	MDCK2/MDCK1	1280	40	40	40	80	40	80	320	80	80
B/Milano/10/2012		MDCK2/MDCK1	1280	10	20	20	80	80	160	320	40	80
B/Milano/11/2012		MDCK2/MDCK1	1280	10	20	40	80	160	160	160	80	80
B/Milano/12/2012		MDCK2/MDCK1	1280	<	10	20	80	80	160	160	40	80
B/Milano/13/2012		MDCK2/MDCK1	2560	10	20	40	80	160	160	160	160	320
B/Milano/19/2012		MDCK2/MDCK1	1280	<	<	20	40	40	80	160	40	40
B/Milano/15/2012		MDCK2/MDCK1	1280	10	20	40	80	80	160	160	80	80
B/Milano/16/2012	3	MDCK2/MDCK1	2560	<	10	40	80	80	160	160	80	80
B/England/386/2012	2	SIAT1/MDCK1	2560	80	80	80	40	40	40	160	640	40
B/Perugia/02/2012	3	MDCK2/MDCK1	1280	10	20	40	40	40	80	160	20	40
B/Estonia/67205/2012	3	MDCK1/MDCK1	2560	80	160	160	80	160	160	640	160	160
B/England/538/2012	3	SIAT2/MDCK1	5120	80	80	80	80	160	160	320	320	320
B/St. Petersburg/24/2012	3	E2/E1	2560	40	80	320	20	160	80	320	10	10
B/St. Petersburg/24/2012	3	C2/MDCK1	2560	40	40	40	80	80	80	160	80	40
B/Stockholm/9/2012	2	C1/MDCK1	1280	10	20	40	20	20	20	40	640	40

1. < = <10 ; 2. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 36. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses		Collection date	Passage History	Haemagglutination inhibition titre										
				Post infection ferret sera										
				B/FI ³ 4/06 SH479	B/Eq ¹ 144/05 F3/07	B/FI ¹ 4/06 F01/10	B/Bris ² 3/07 F21/12 Clade 2	B/Enq ² 145/08 F20/12 Clade 3	B/Banq ² 3333/07 F25/08 Clade 3	B/Wis ² 1/10 F23/10 Clade 3	B/Stock ² 12/11 F12/12 Clade 3	B/Estonia ² 55669/11 F26/11 Clade 2	B/Serbia ² 1894/11 F25/11 Clade 3	B/Stock ² 12/11 T/C F8/12 Clade 3
Clade														
REFERENCE VIRUSES														
B/Egypt/144/2005		2005-05-01	E3/E5	5120	160	2560	1280	320	640	160	320	160	10	640
B/Florida/4/2006		2006-12-15	E3/E4	5120	160	1280	1280	160	640	320	320	160	10	160
B/Brisbane/3/2007		2007-09-03	E2/E4	1280	40	320	320	80	160	160	160	80	<	160
B/England/145/2008			Ex/E4	1280	<	160	80	160	40	20	160	<	<	80
B/Bangladesh/3333/2007		2007-08-07	E3/E5	5120	40	320	160	80	320	160	320	<	10	640
B/Wisconsin/1/2010		2010-02-20	E3/E2	1280	80	160	80	40	40	160	1280	20	20	640
B/Stockholm/12/2011		2007-08-07	E4/E1	5120	160	640	160	160	320	160	320	10	20	640
B/Estonia/55669/2011		2011-03-14	MDCK2/MDCK2	1280	<	160	80	40	20	40	160	640	20	40
B/Serbia/1894/2011		2011-03-08	MDCK1/MDCK4	2560	<	80	40	160	80	40	160	40	160	160
B/Stockholm/12/2011		2011-03-28	Cx/MDCK3	2560	<	160	80	80	80	80	160	40	320	160
TEST VIRUSES														
B/Guyane/1307/2012 (B/Guadeloupe/55/2012)	2	2012-03-08	MDCK2/MDCK1	2560	<	160	80	80	40	40	80	640	40	40
B/Bretagne/1160/2012	2	2012-03-23	MDCK2/MDCK1	2560	<	80	80	40	20	20	80	320	20	20
B/Nord Pas de Calais/1175/2012		2012-03-26	MDCK2/MDCK1	1280	<	160	80	80	10	20	80	320	10	20
B/Lorraine/1220/2012		2012-04-01	MDCK2/MDCK1	2560	<	80	40	80	40	20	160	20	40	80
B/Centre/1363/2012		2012-04-19	MDCK1/MDCK1	1280	<	80	40	40	10	10	80	320	20	20
B/Lorraine/1408/2012	2	2012-04-28	MDCK1/MDCK1	1280	<	80	40	40	10	40	80	320	20	20
B/Hong Kong/3569/2012	2	2012-06-02	MDCK2/MDCK1	2560	40	160	80	80	40	20	160	320	40	80
B/Hong Kong/3583/2012	3	2012-06-12	MDCK2/MDCK1	5120	40	160	80	160	80	80	160	80	160	160
B/Hong Kong/3577/2012	2	2012-06-13	MDCK2/MDCK1	5120	<	160	80	80	40	20	160	640	40	80
B/Hong Kong/3579/2012	2	2012-06-13	MDCK2/MDCK1	1280	<	160	80	80	20	20	80	320	20	40
B/Hong Kong/3581/2012	2	2012-06-13	MDCK2/MDCK1	5120	40	160	80	80	40	20	160	640	80	80
B/Ghana/DILI-12-068/2012		2012-06-15	C1/MDCK2	2560	<	80	40	160	40	20	160	40	80	160
B/Ghana/DILI-12-098/2012	3	2012-06-19	C1/MDCK1	640	<	40	20	40	20	20	80	10	20	40
B/Ghana/DILI-12-102/2012	3	2012-06-19	C1/MDCK1	1280	<	80	80	80	40	20	160	20	40	160
B/Hong Kong/3585/2012	3	2012-06-24	MDCK2/MDCK1	2560	<	160	80	80	80	40	160	40	80	160
B/Ghana/FS-12-808/2012	3	2012-06-25	C1/MDCK1	1280	<	80	40	80	80	40	160	20	80	160
B/Ghana/FS-12-831/2012	3	2012-06-26	C1/MDCK1	1280	<	80	40	80	40	40	160	10	20	80
A/Ghana/DILI-12-229/2012		2012-07-11	SIAT2	2560	<	160	40	160	80	40	160	80	80	80

1. < = <40; 2. < = <10 ; 3. hyperimmune sheep serum

Vaccine

Sequences in phylogenetic trees

Table 37. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre										
			Post infection ferret sera										
			B/FI ³ 4/06 SH479	B/Eg ¹ 144/05 F3/07	B/FI ¹ 4/06 F01/10	B/Bris ² 3/07 F21/12 Clade 2	B/Eng ² 145/08 F20/12 Clade 3	B/Bang ² 3333/07 F25/08 Clade 3	B/Wis ² 1/10 F23/10 Clade 3	B/Stock ² 12/11 F12/12 Clade 3	B/Estonia ² 55669/11 F26/11 Clade 2	B/Serbia ² 1894/11 F25/11 Clade 3	B/Stock ² 12/11 T/C F8/12 Clade 3
Clade													
REFERENCE VIRUSES													
B/Egypt/144/2005	2005-05-01	E3/E5	5120	320	1280	1280	320	640	320	640	160	20	640
B/Florida/4/2006	2006-12-15	E3/E3	5120	320	1280	1280	320	640	320	640	160	20	640
B/Brisbane/3/2007	2007-09-03	E2/E1	5120	160	640	640	160	320	160	640	160	10	320
B/England/145/2008		Ex/E4	1280	40	320	80	320	40	20	160	<	<	80
B/Bangladesh/3333/2007	2007-08-07	E4/E4	5120	80	640	320	80	320	80	640	<	20	640
B/Wisconsin/1/2010	2007-08-07	E3/E1	2560	40	640	320	80	320	160	640	10	20	640
B/Stockholm/12/2011	2007-08-07	E4/E1	5120	160	640	320	160	320	160	640	10	40	640
B/Estonia/55669/2011	2011-03-14	MDCK2/MDCK2	5120	10	320	160	160	80	40	160	640	80	80
B/Serbia/1894/2011	2011-03-08	MDCK1/MDCK4	5120	160	320	160	320	160	80	320	160	320	320
B/Stockholm/12/2011	2011-03-28	Cx/MDCK4	5120	80	320	160	160	160	80	320	160	320	320
TEST VIRUSES													
B/Norway/959/2012	2012-03-13	MDCK1/MDCK1	5120	80	320	160	320	160	80	320	640	160	320
B/Norway/1073/2012	2012-03-15	MDCK1/MDCK1	5120	80	320	160	320	160	80	640	1280	320	320
B/Norway/1071/2012	2012-03-16	MDCK1/MDCK1	5120	80	320	160	320	160	40	320	640	160	160
B/Norway/1165/2012	2012-04-06	MDCK1/MDCK1	5120	40	320	80	160	160	80	640	160	320	160
B/Norway/1249/2012	2012-04-26	MDCK1/MDCK1	5120	80	320	160	320	80	40	320	1280	320	160
B/Malaysia/412/2012	2012-05-25	K1	5120	40	320	160	160	80	40	160	1280	160	320
B/Victoria/525/2012	2012-07-23	MDCK1/MDCK1	2560	40	160	80	80	40	20	80	640	40	80
B/Christchurch/505/2012	2012-07-23	MDCK1/MDCK1	2560	40	160	80	80	40	10	40	640	40	80

1. < = <40; 2. < = <10 ; 3. hyperimmune sheep serum

Vaccine

Table 38. Antigenic analyses of influenza B viruses (Yamagata lineage)

			Haemagglutination inhibition titre										
			Post infection ferret sera										
Viruses	Collection date	Passage History	B/FI ³ 4/06 SH479	B/Eg ¹ 144/05 F3/07	B/FI ¹ 4/06 F01/10	B/Bris ² 3/07 F21/12 Clade 2	B/Eng ² 145/08 F20/12 Clade 3	B/Bang ² 3333/07 F25/08 Clade 3	B/Wis ² 1/10 F23/10 Clade 3	B/Stock ² 12/11 F34/11 Clade 3	B/Estonia ² 55669/11 F26/11 Clade 2	B/Serbia ² 1894/11 F25/11 Clade 3	B/Stock ² 12/11 T/C F8/12 Clade 3
Clade													
REFERENCE VIRUSES													
B/Egypt/144/2005	2005-05-01	E3/E5	5120	640	1280	1280	320	640	320	640	160	20	640
B/Florida/4/2006	2006-12-15	E3/E3	5120	320	2560	1280	640	640	320	640	160	20	640
B/Brisbane/3/2007	2007-09-03	E2/E1	5120	160	640	1280	160	320	160	640	160	10	320
B/England/145/2008		Ex/E4	1280	40	320	80	320	40	20	160	<	<	80
B/Bangladesh/3333/2007	2007-08-07	E4/E4	5120	80	640	320	80	320	80	640	<	20	640
B/Wisconsin/1/2010	2007-08-07	E3/E1	2560	40	640	320	80	320	160	640	10	20	640
B/Stockholm/12/2011	2007-08-07	E4/E1	5120	160	640	320	160	320	160	640	10	40	640
B/Estonia/55669/2011	2011-03-14	MDCK2/MDCK2	5120	10	320	160	160	80	40	160	640	80	80
B/Serbia/1894/2011	2011-03-08	MDCK1/MDCK4	5120	160	320	160	320	160	80	320	160	320	320
B/Stockholm/12/2011	2011-03-28	Cx/MDCK4	5120	80	320	160	160	160	80	640	160	320	320
TEST VIRUSES													
B/Sendai/6/2012	2	2012-02-18	MDCK1+1/MDCK1	1280	<	80	40	40	10	<	80	320	20
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2	5120	320	640	320	320	320	160	320	640	640
B/Massachusetts/02/2012	2	2012-03-13	E3/E1	2560	80	640	320	160	80	80	160	80	320
B/Johannesburg/242/2012		2012-04-07	MDCK1/MDCK1	5120	80	320	80	160	160	80	320	160	160
B/Taiwan/1367/2012	3	2012-04-27	MDCK2+1/MDCK1	1280	40	80	40	80	40	10	80	40	320
B/Linares/34634/2012	2	2012-05-07	E3/E4	2560	40	320	320	160	80	40	160	40	160
B/Johannesburg/2757/2012		2012-05-22	MDCK1/MDCK1	2560	40	160	80	160	80	40	320	80	160
B/Johannesburg/3009/2012		2012-05-23	MDCK1/MDCK1	2560	80	320	80	80	40	20	160	640	40
B/Santiago/39712/2012	2	2012-06-04	C2/C1/MDCK1	5120	80	320	160	160	80	40	160	640	80
B/Johannesburg/3137/2012		2012-06-05	MDCK0/MDCK1	2560	80	160	80	160	80	40	320	80	80
B/Johannesburg/3166/2012		2012-06-07	MDCK1/MDCK1	5120	80	640	160	160	80	40	320	640	160
B/Johannesburg/3559/2012		2012-06-22	MDCK0/MDCK1	2560	80	160	80	160	80	40	320	80	80
B/Johannesburg/5564/2012		2012-06-25	MDCK0/MDCK1	2560	80	160	80	80	80	40	320	80	80

1. <= <40; 2. <= <10; 3. hyperimmune sheep serum

Vaccine

Table 39. Summary of influenza B (Yamagata-lineage) samples analysed, collected between 2012-02-01 and 2012-08-31

MONTH	B Yamagata							
	Country	Number propagated ¹	Fold reduction in HI titre					
			B/Stockholm/12/2011 antisera Group 3			B/Estonia/55669/2011 antisera Group 2		
			≤2 fold ²	4 fold ²	≥8 fold ²	≤2 fold ³	4 fold ³	≥8 fold ³
UNKNOWN								
Czech Republic ⁴		2	1	1		2		
FEBRUARY								
Denmark		3	3					3
Greece		4	3	1		4		
Italy		5	4		1	1		4
Norway		3	3			3		
Portugal		1		1		1		
Russia		3	3				2	1
Sweden		1	1					1
United Kingdom		1	1					1
MARCH								
Denmark		0						
France		2		2		2		
Guyane		1		1		1		
Italy		6	6				1	5
Norway		3	3			3		
Portugal		2	2			1		1
Sweden		1		1		1		
United Kingdom		1	1			1		
APRIL								
Estonia		1	1				1	
France		3	1	2		2		1
Norway		2	2			1	1	
Russia		2	2					2
South Africa		1	1				1	
Sweden		1			1	1		
United Kingdom		1	1			1		
MAY								
South Africa		2	1	1		1		1
JUNE								
Ghana		5	4	1				5
Hong Kong		6	6			4		2
JULY								
Ghana		1	1					1
Total		64	51	11	2	30	6	28
Total sequenced			30	7	2	19	3	17
Group 2			10 (33.3%)	6 (85.7%)	2 (100%)	18 (94.7%)	0 (0%)	0 (0%)
Group 3			20 (66.7%)	1 (14.3%)	0 (0%)	1 (5.3%)	3 (100%)	17 (100%)

1. Propagated to sufficient titre to perform HI assay

2. Compared to homologous titres for ferret antiserum raised against tissue-culture grown B/Stockholm/12/2011 (Group 3)

3. Compared to homologous titres for ferret antiserum raised against tissue-culture grown B/Estonia/55669/2011 (Group 2)

4. Package received 2012-05-10

Figure 19. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

Vaccine virus
Reference viruses

Collection date
Feb 2012
Mar 2012
Apr-May 2012
Jun-Jul 2012

See HI Table
for antigenic
characterisation

proposed
serology antigen

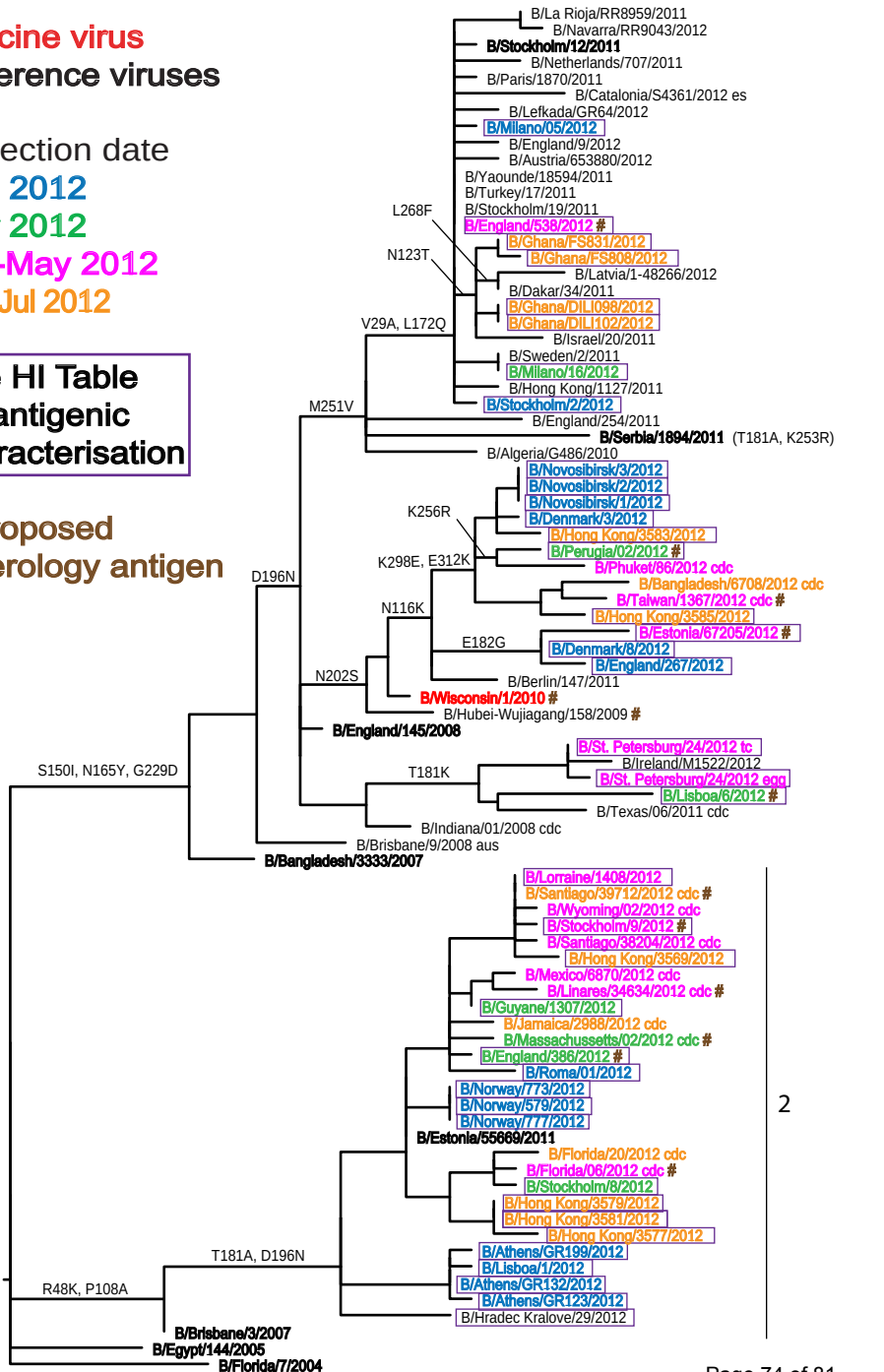


Figure 20. HA protein alignment for B-Yamagata viruses.

Vaccine virus: Reference virus: Genetic clade: HA1/HA2 boundary: Potential N-linked glycosylation site

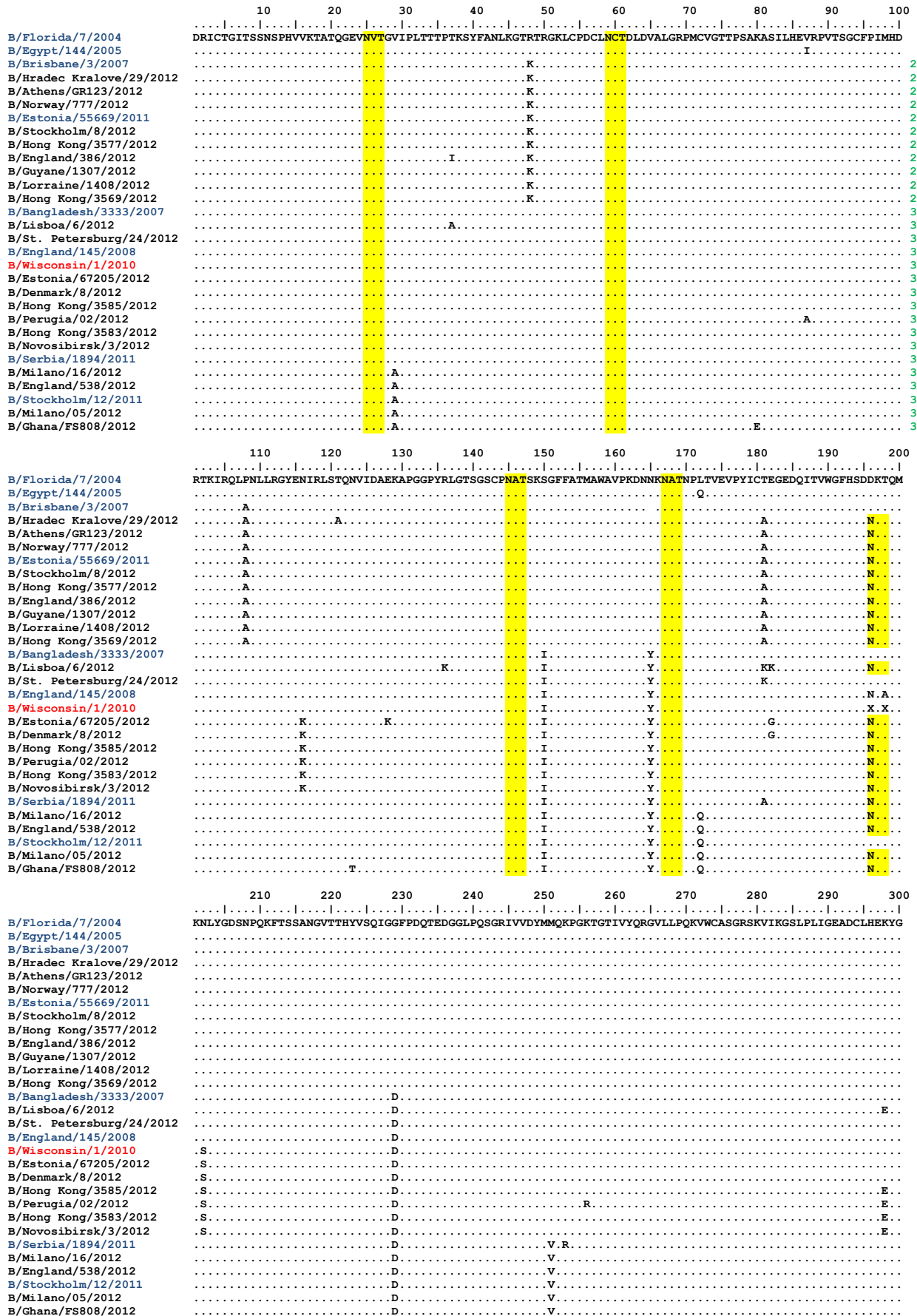


Figure 21. B/Yamagata HA locations of genetic clade defining amino acid substitutions.

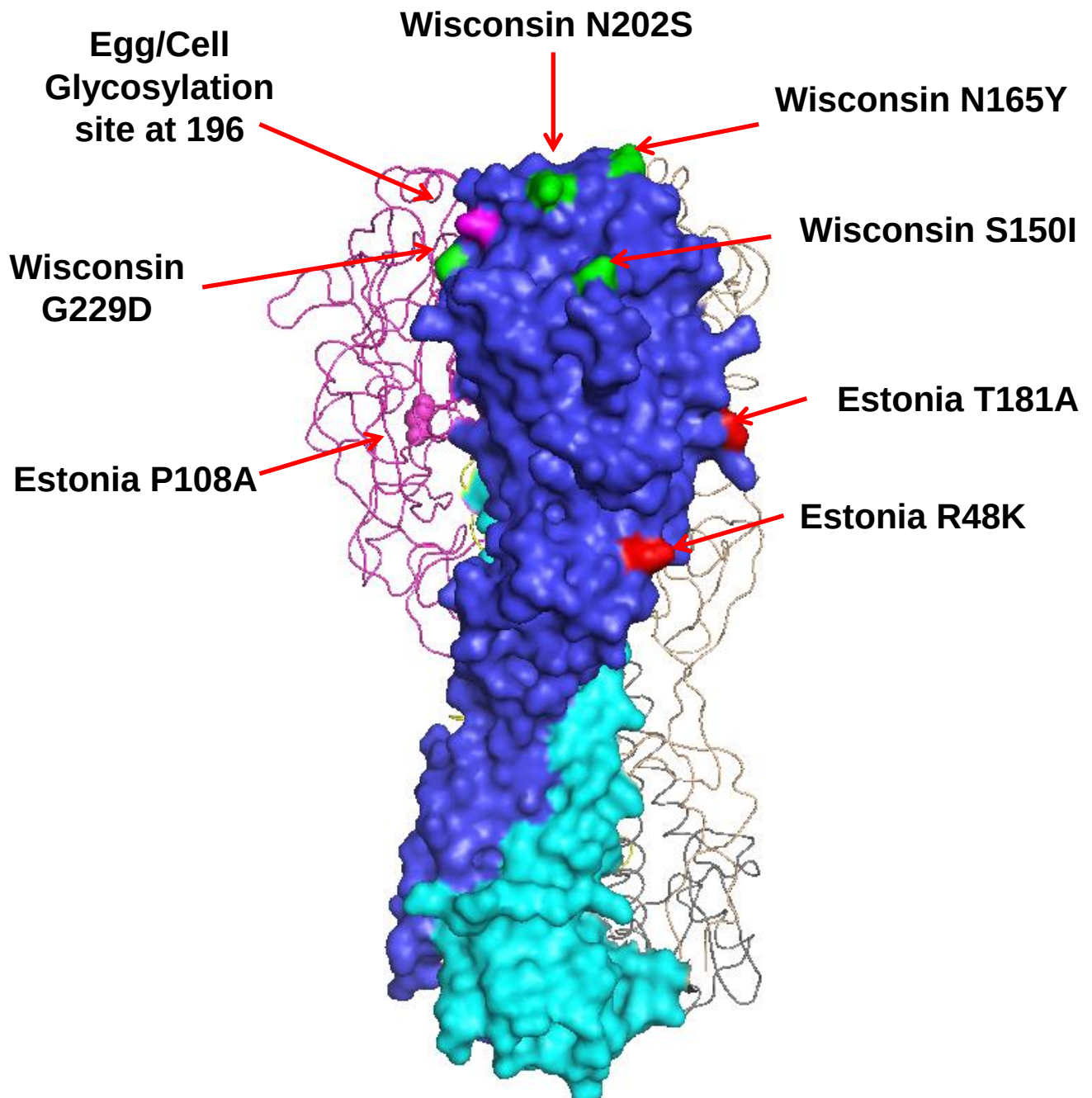


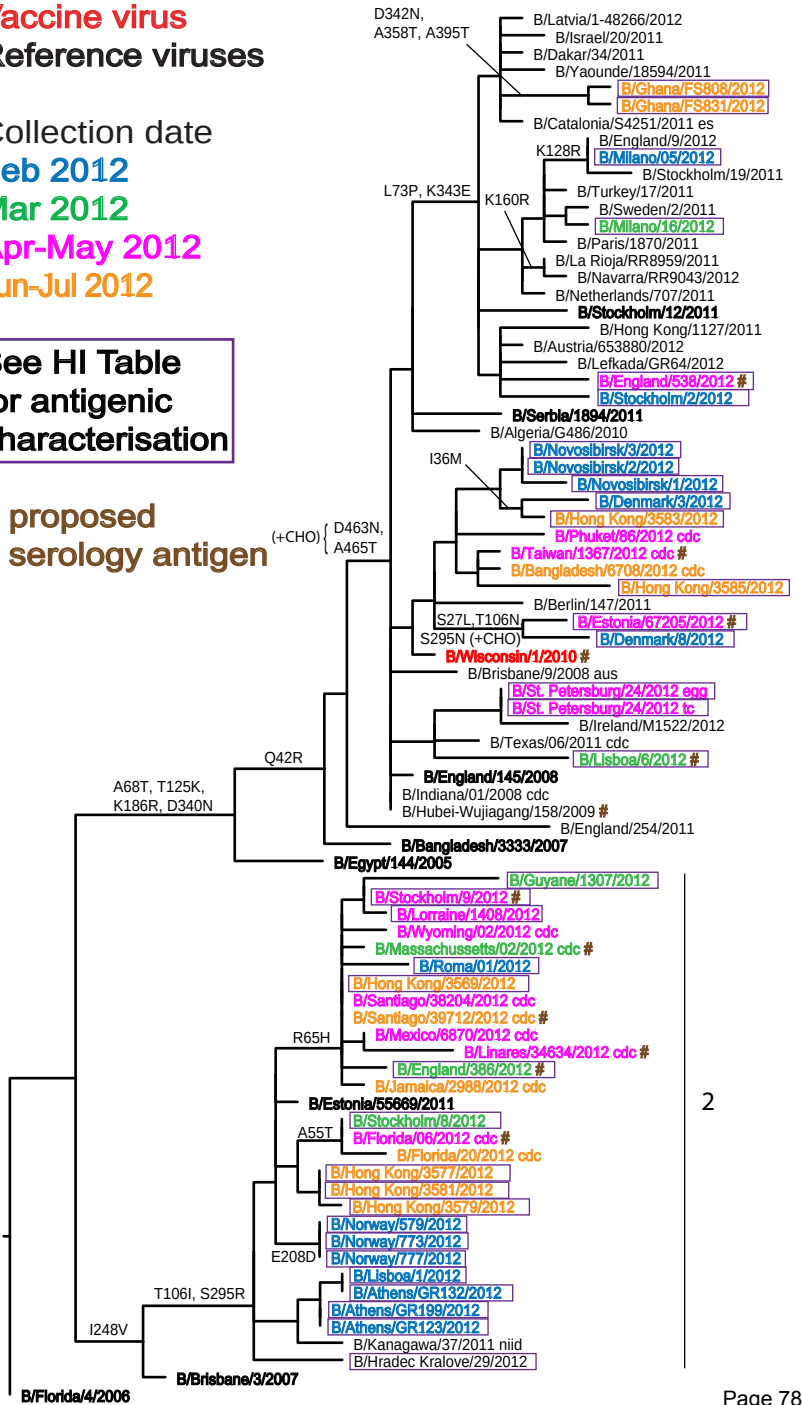
Figure 22. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

Vaccine virus
Reference viruses

Collection date
Feb 2012
Mar 2012
Apr-May 2012
Jun-Jul 2012

See HI Table
for antigenic
characterisation

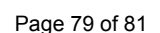
proposed
serology antigen



3

2

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400
B/Florida/7/2004	P	F	V	K	L	N	V	E	T	D
B/Egypt/144/2005	T	A	E	I	R	L	M	C	T	E
B/Brisbane/3/2007	T	E	T	Y	L	D	T	P	R	P
B/Hradec Kralove/29/2012	N	D	G	S	I	T	G	P	C	E
B/Athens/GR123/2012	S	N	G	D	K	G	S	G	G	I
B/Norway/777/2012	K	G	F	V	H	Q	R	M	A	S
B/Estonia/55669/2011	K	I	G	R	W	Y	S	R	T	M
B/Stockholm/8/2012	S	K	T	R	M	G	M	G	L	V
B/Hong Kong/3577/2012	K	D	G	P	W	T	D	S	E	A
B/England/386/2012	L	S	G	V	M					
B/Guyane/1307/2012										
B/Lorraine/1408/2012										
B/Hong Kong/3569/2012										
B/Bangladesh/3333/2007										
B/Lisboa/6/2012										
B/St. Petersburg/24/2012										
B/England/145/2008										
B/Wisconsin/1/2010										
B/Estonia/67205/2012										
B/Denmark/8/2012										
B/Hong Kong/3585/2012										
B/Hong Kong/3583/2012										
B/Novosibirsk/3/2012										
B/Serbia/1894/2011										
B/Milano/16/2012										
B/England/538/2012										
B/Stockholm/12/2011										
B/Milano/05/2012										
B/Ghana/FS808/2012										

	410	420	430	440	450	460
B/Florida/7/2004	V	S	M	E	E	P
B/Egypt/144/2005	G	W	S	F	G	F
B/Brisbane/3/2007	E	I	K	D	K	C
B/Hradec Kralove/29/2012	D	V	P	C	I	G
B/Athens/GR123/2012	I	E	M	V	H	D
B/Norway/777/2012	G	G	K	T	T	W
B/Estonia/55669/2011	S	A	A	T	A	I
B/Stockholm/8/2012	Y	C	L	M	G	S
B/Hong Kong/3577/2012	G	L	L	W	D	T
B/England/386/2012	V	T	G	V	D	M
B/Guyane/1307/2012	A	L	*			
B/Lorraine/1408/2012						
B/Hong Kong/3569/2012						
B/Bangladesh/3333/2007						
B/Lisboa/6/2012						
B/St. Petersburg/24/2012						
B/England/145/2008						
B/Wisconsin/1/2010						
B/Estonia/67205/2012						
B/Denmark/8/2012						
B/Hong Kong/3585/2012						
B/Hong Kong/3583/2012						
B/Novosibirsk/3/2012						
B/Serbia/1894/2011						
B/Milano/16/2012						
B/England/538/2012						
B/Stockholm/12/2011						
B/Milano/05/2012						
B/Ghana/FS808/2012						

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 40. Antiviral (NI) testing from 2012-02-01 to 2012-09-07

Year	Month of Collection	Type/Subtype of influenza			Total
		H3N2*	H1N1pdm*	Flu B	
2010	March	2			
	June			1	
	July		1	2	
	September			1	
	December		2	4	
2011	January		1	2	
	February		1	5	
	March		2	2	
	April			1	
	June			2	
	July		1	2	
	August			11	
	September	9	7	23	
	October	9	6	13	
	November	49	6	20	
	December	168	3	9	
2012	January	136	4	14	154
	February	94	13	30	137
	March	151	6	32	189
	April	51	2	30	83
	May	12	2	8	22
	June	10	1	11	22
	July			1	1
Total collected from 2012-02-01		318	24	112	454
Unknown collection date		11		9	
Total		702	58	233	993

All viruses tested were sensitive to Oseltamivir and Zanamivir.

* All influenza A virus M-gene sequences indicated resistance to adamantanes.